



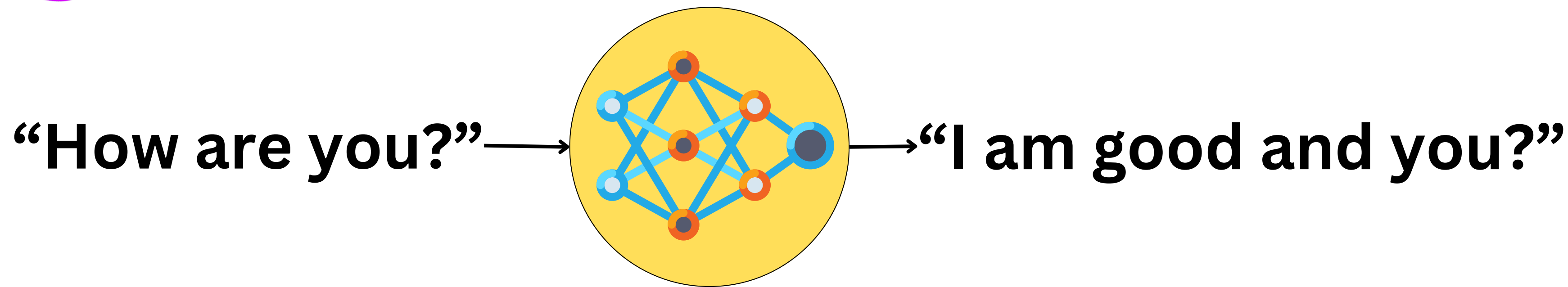
The **AiEdge** ACADEMY

The Transformer Architecture

Introduction to LLMs



The Sequence to Sequence models

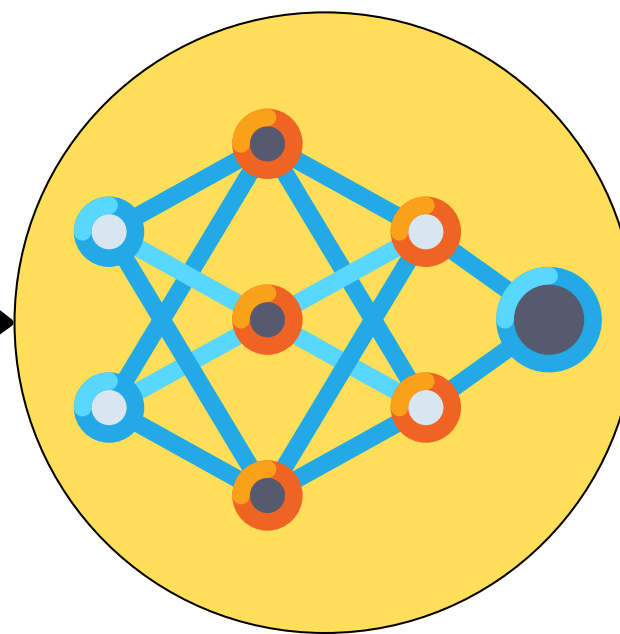


- Natural language generation
- Machine translation
- Text summarization
- Question answering
- Conversational AI



The Sequence to Sequence models

“How are you?”



“I am good and you?”

The RNN
Encoder-decoder

Bahdanau-style
Attention

Luong-style
Attention

“Attention is all
you need”
Self-Attention

Jun
2014

Sept
2014

Aug
2015

Jun
2017





Attention Is All You Need



Attention Is All You Need

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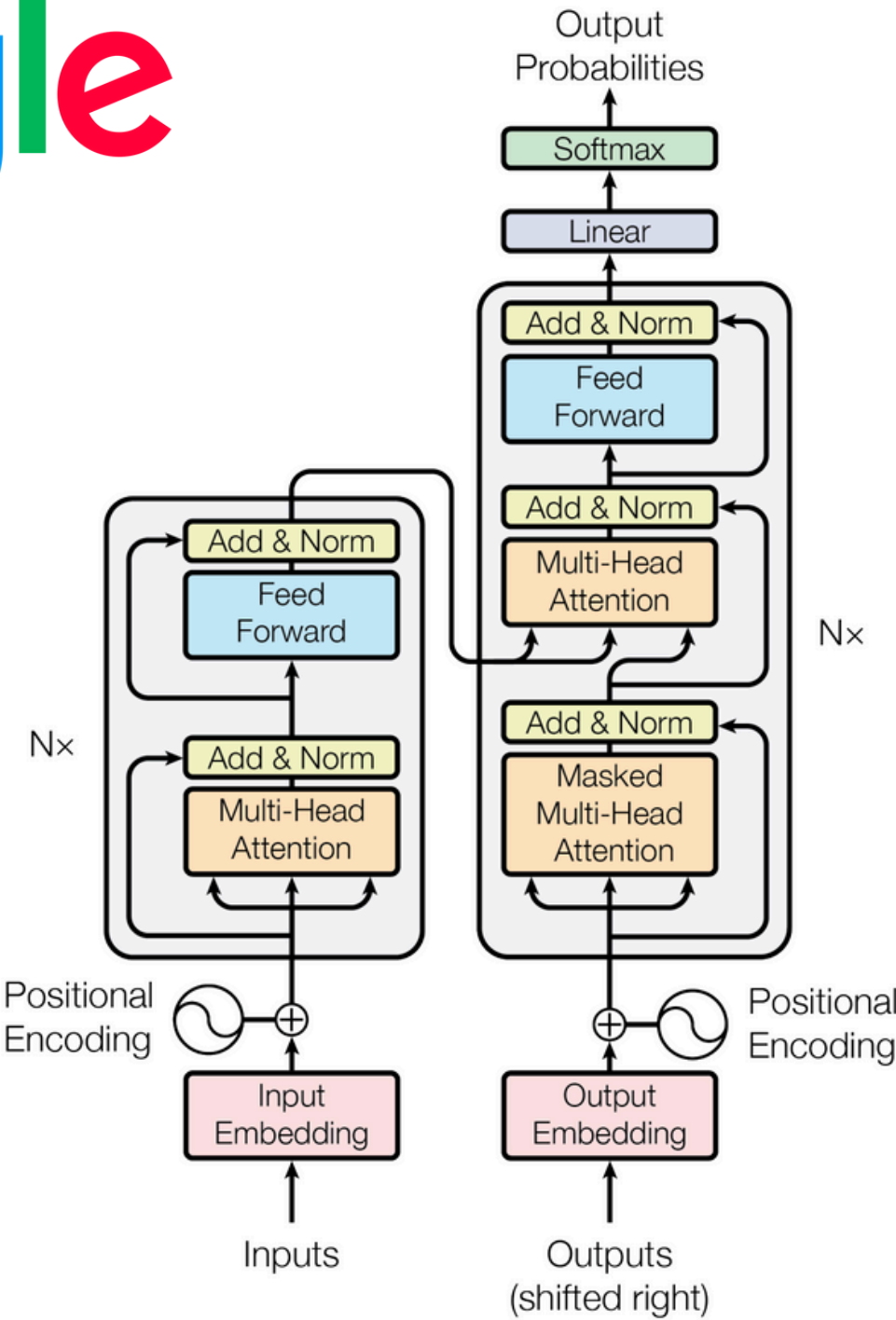
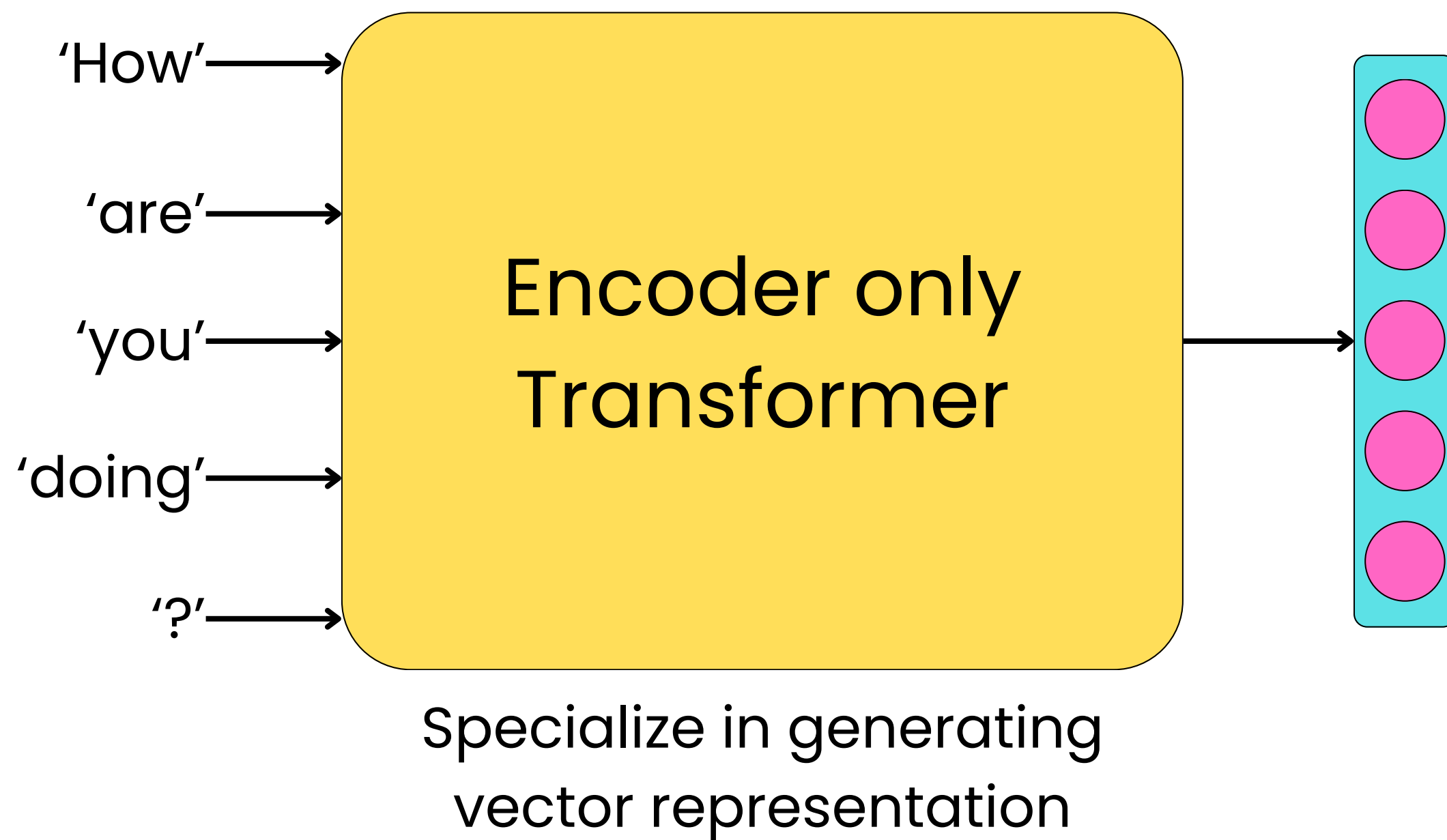


Figure 1: The Transformer - model architecture.

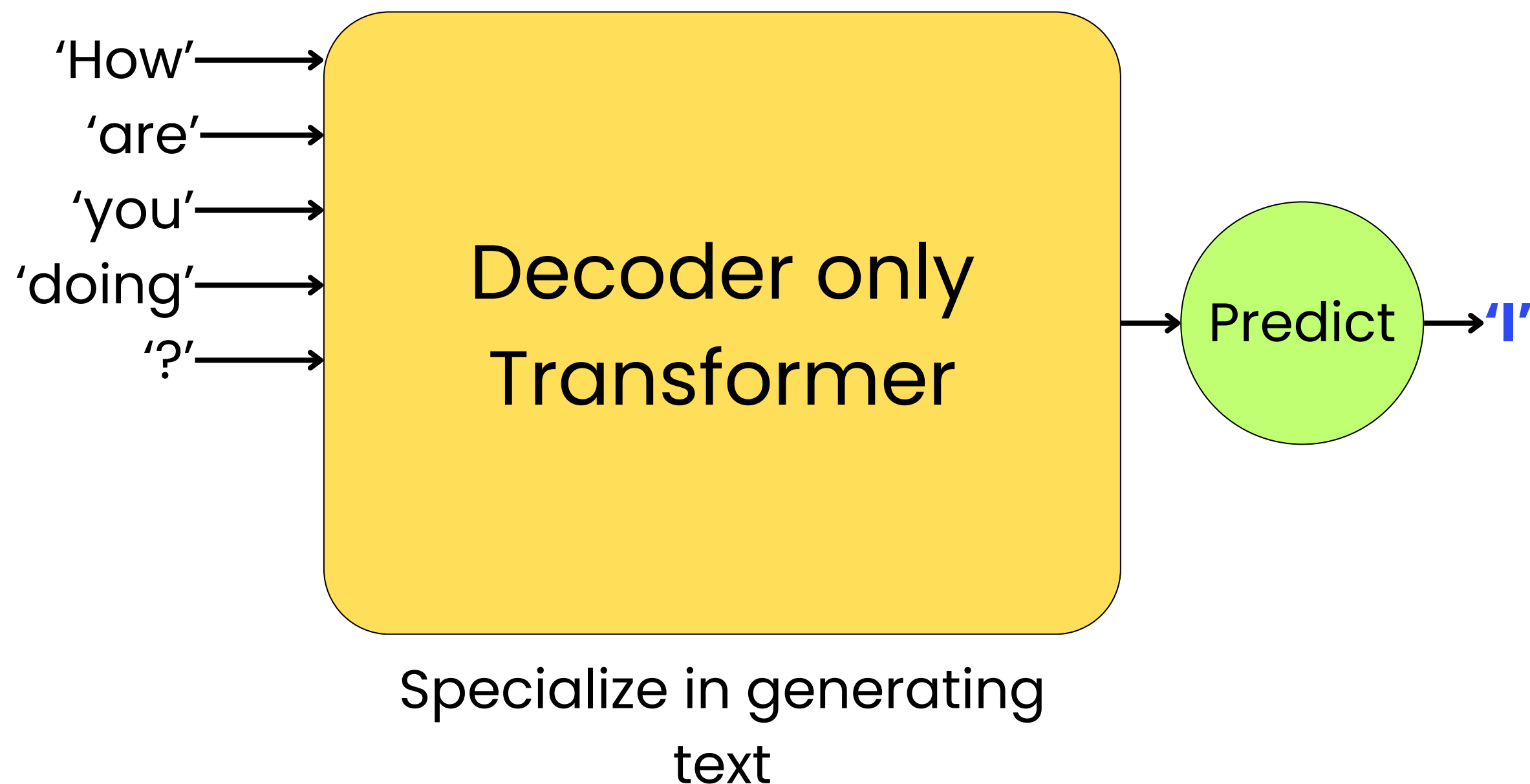


The Different Transformer Architectures



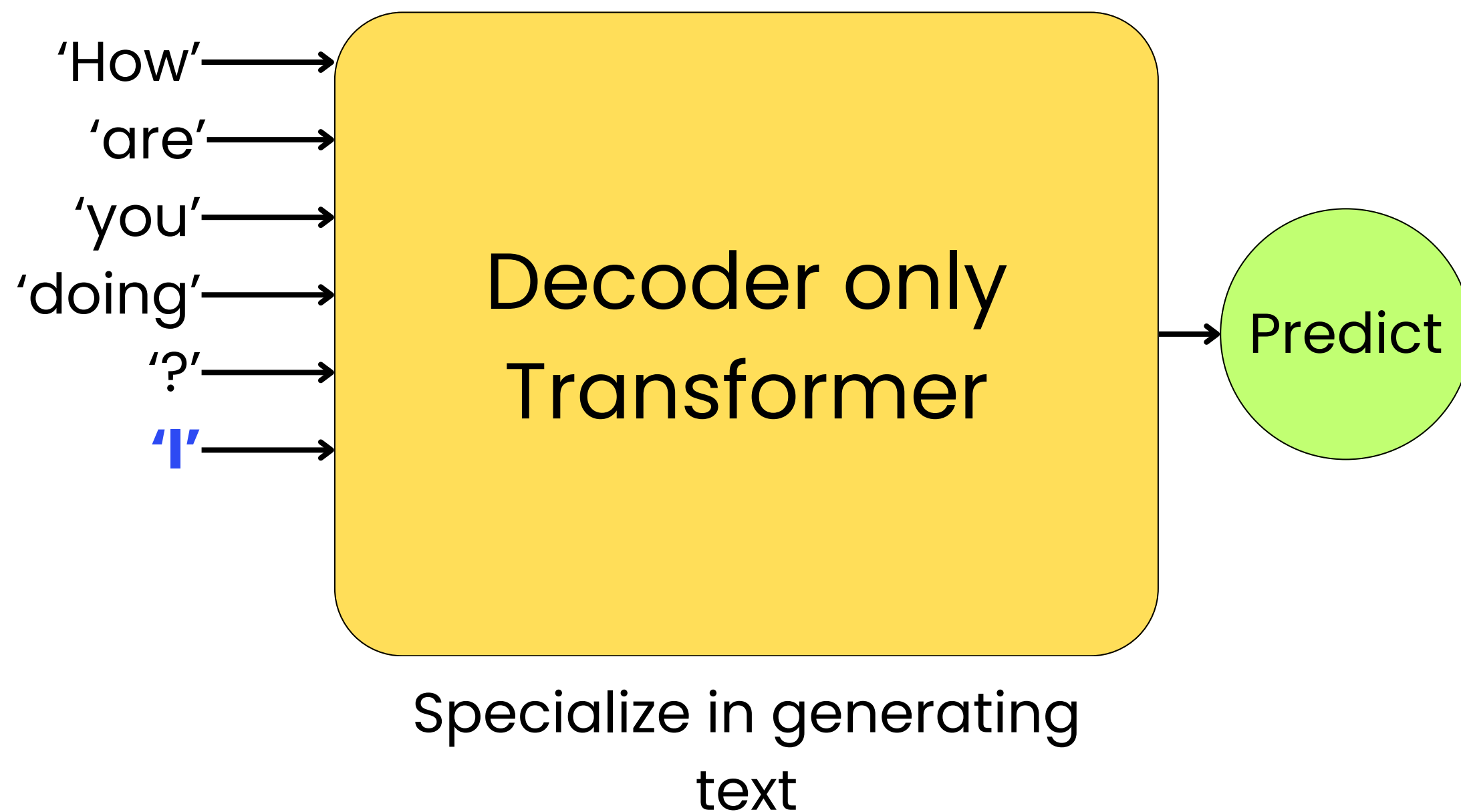


The Different Transformer Architectures



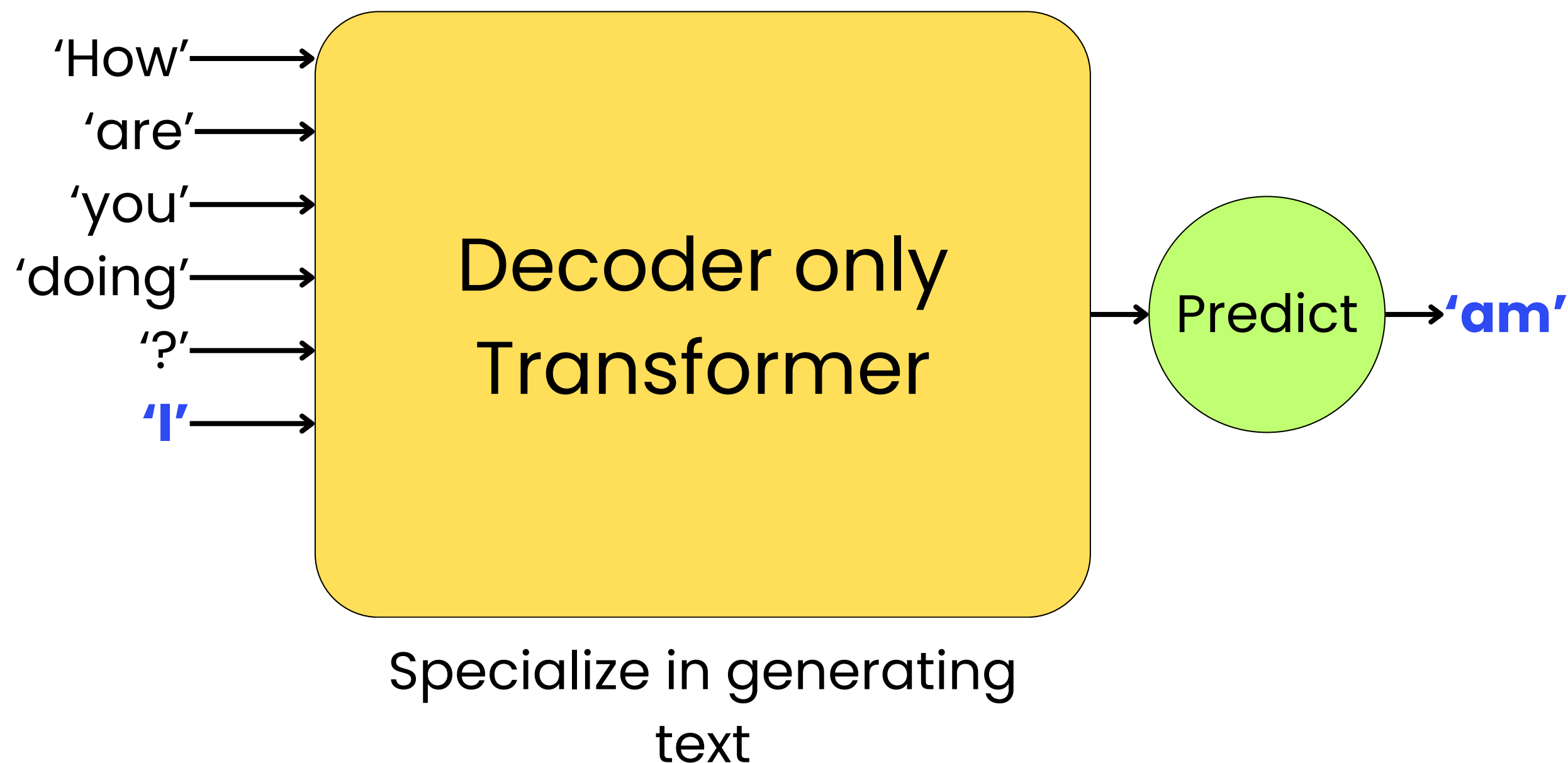


The Different Transformer Architectures



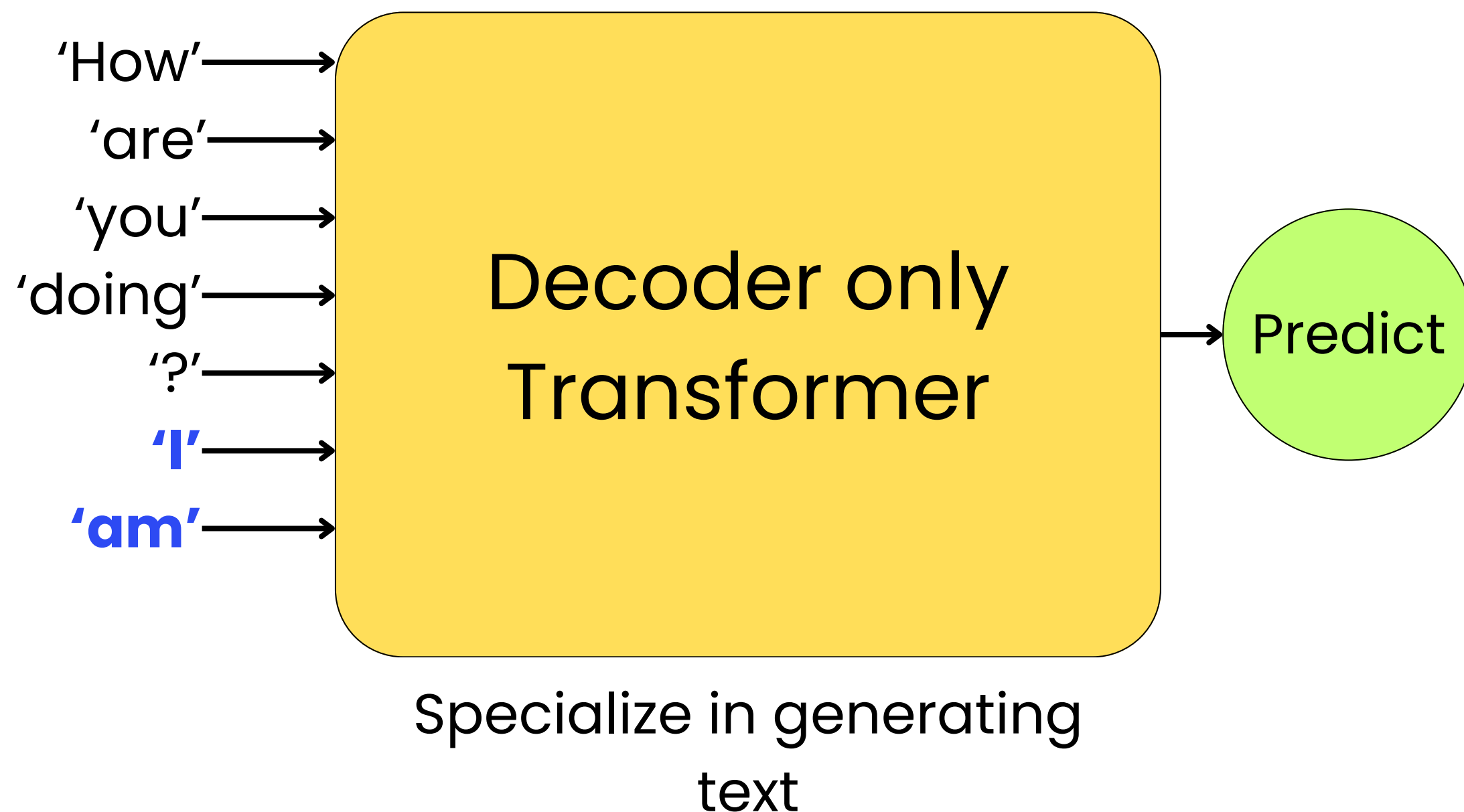


The Different Transformer Architectures



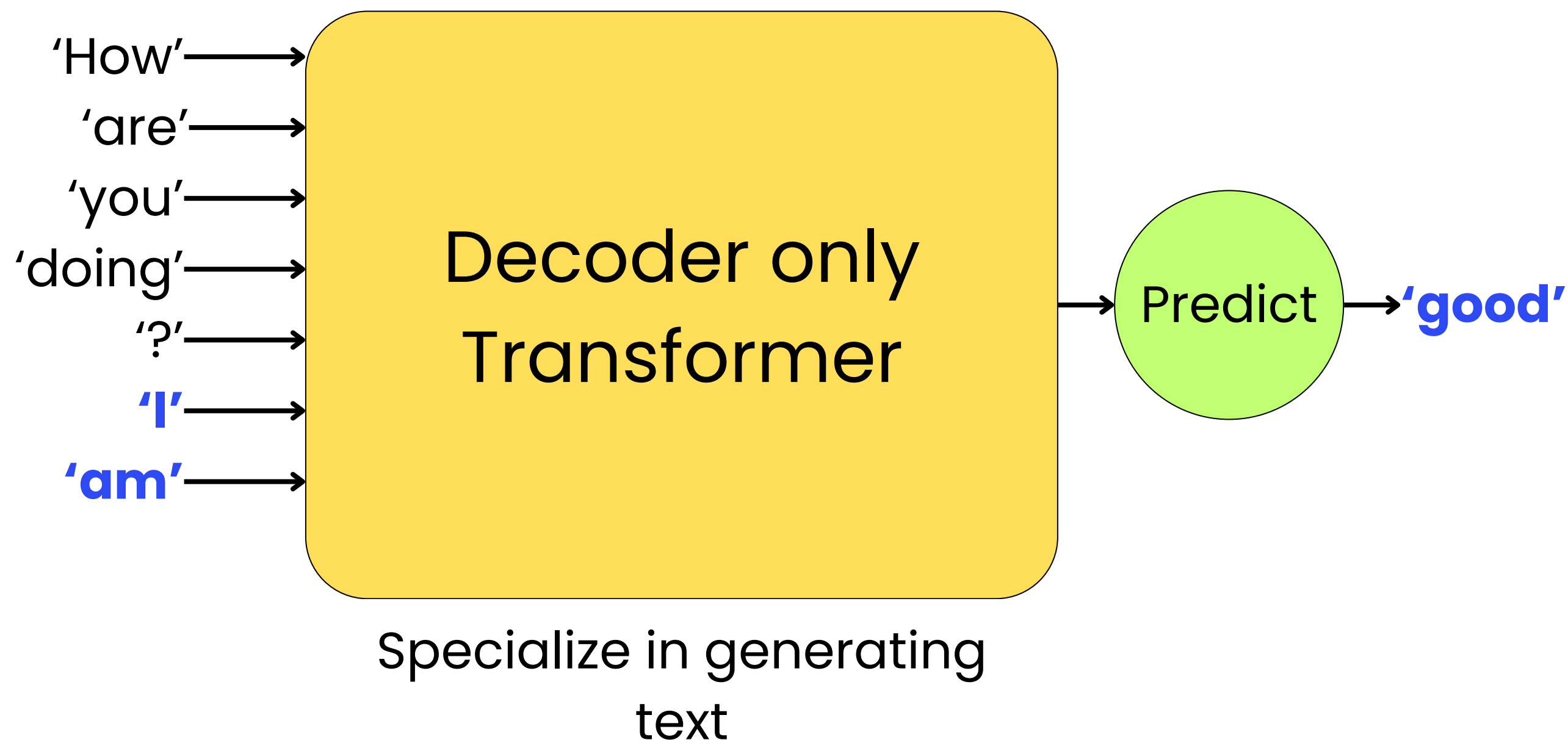


The Different Transformer Architectures



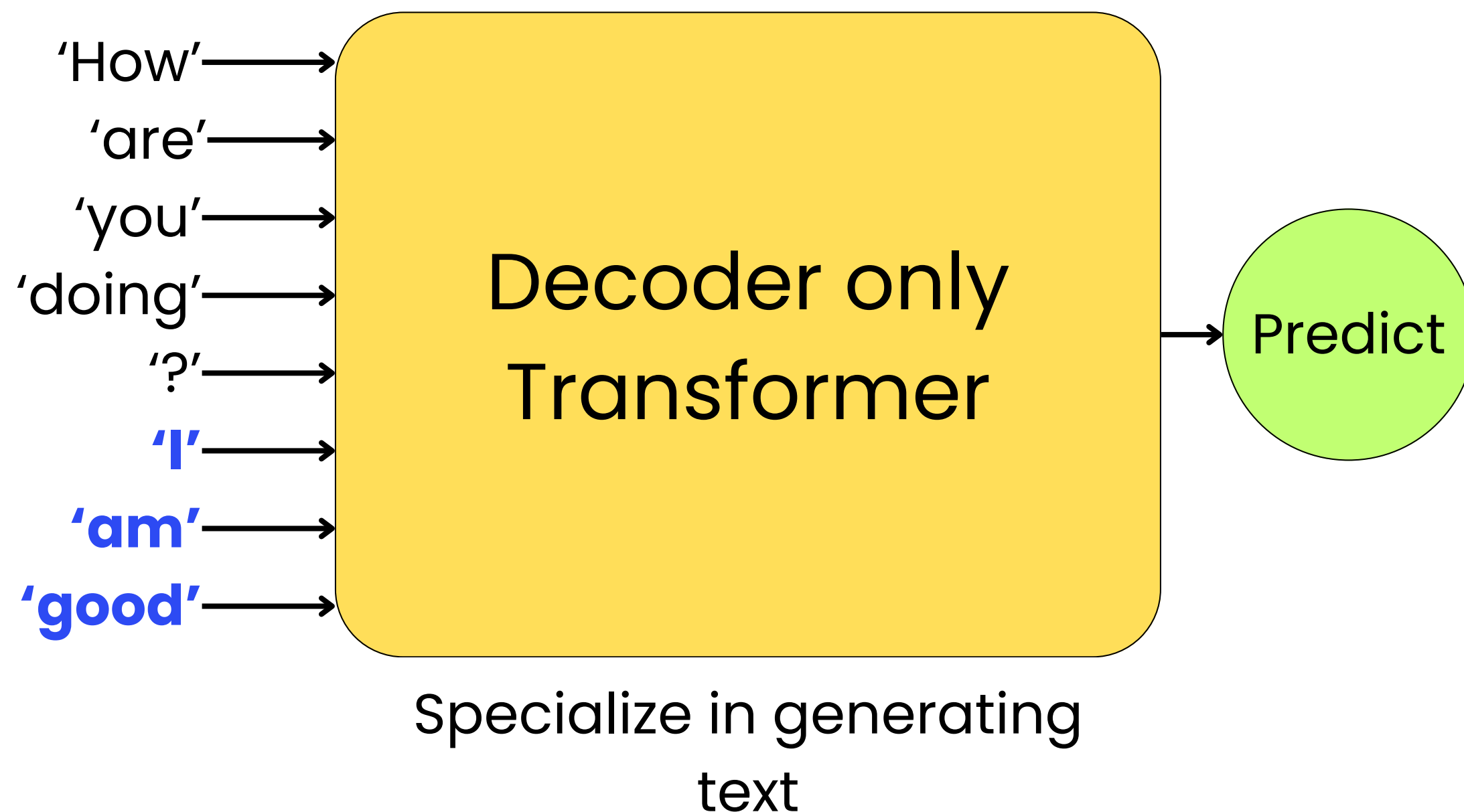


The Different Transformer Architectures



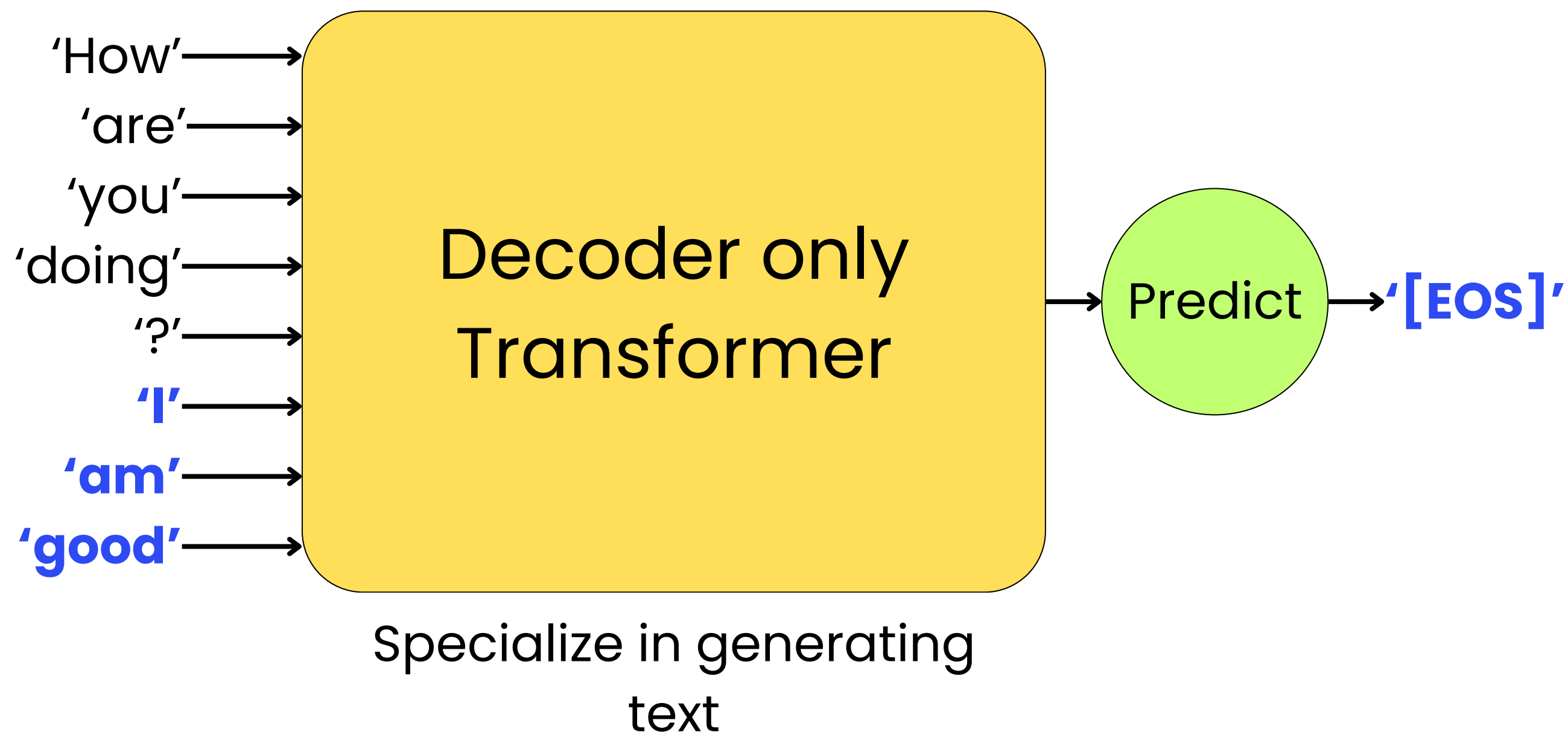


The Different Transformer Architectures



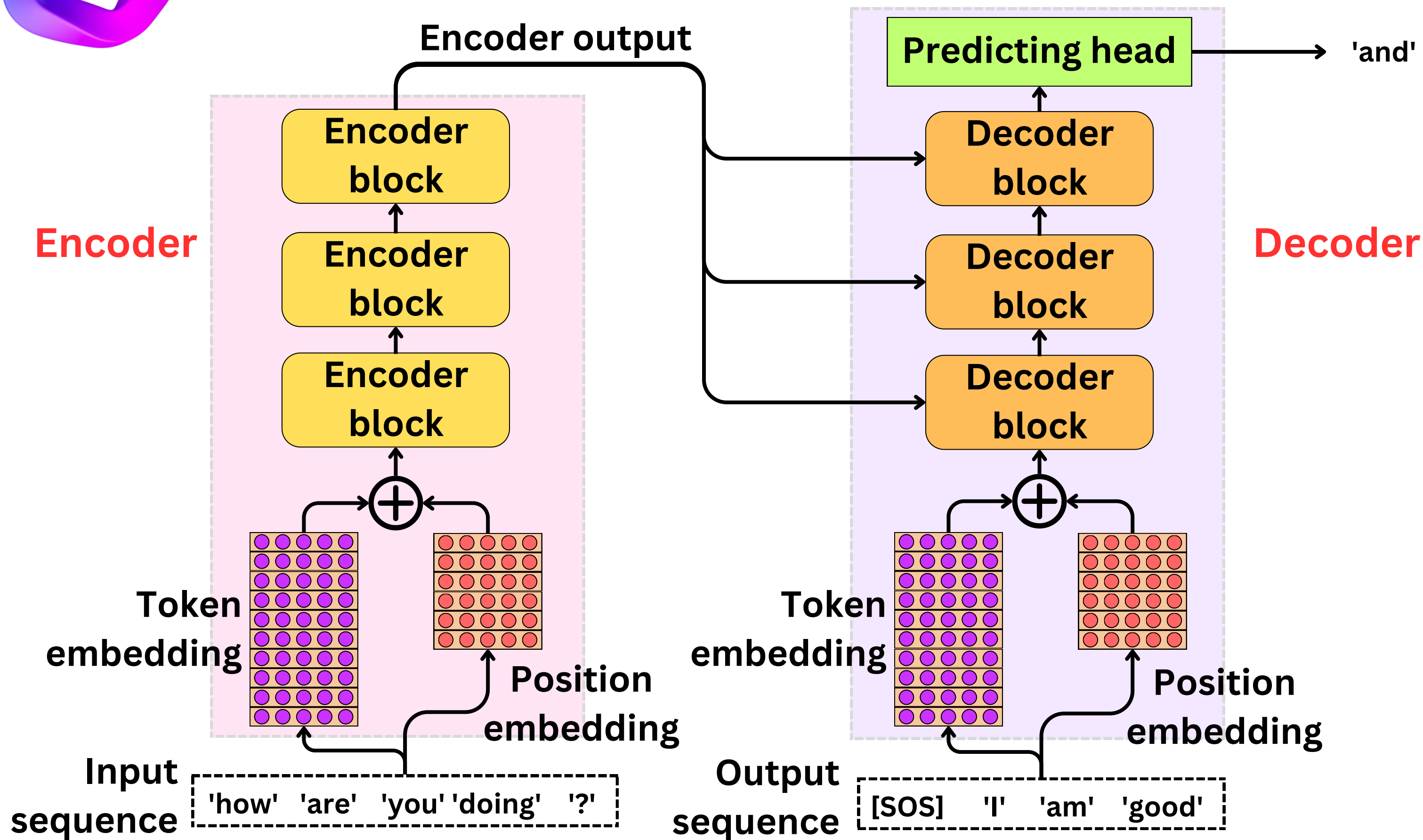


The Different Transformer Architectures



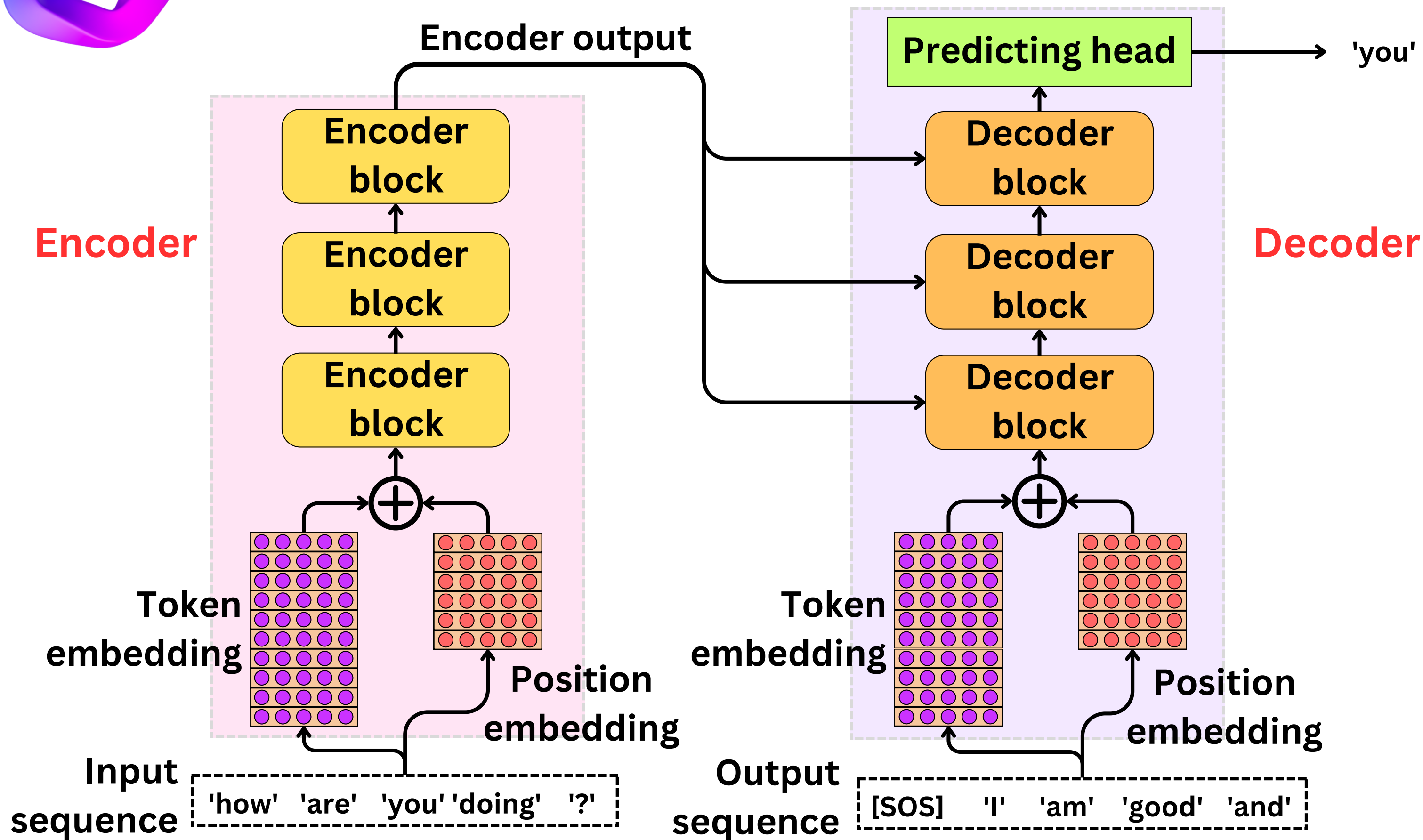


The Overall Architecture



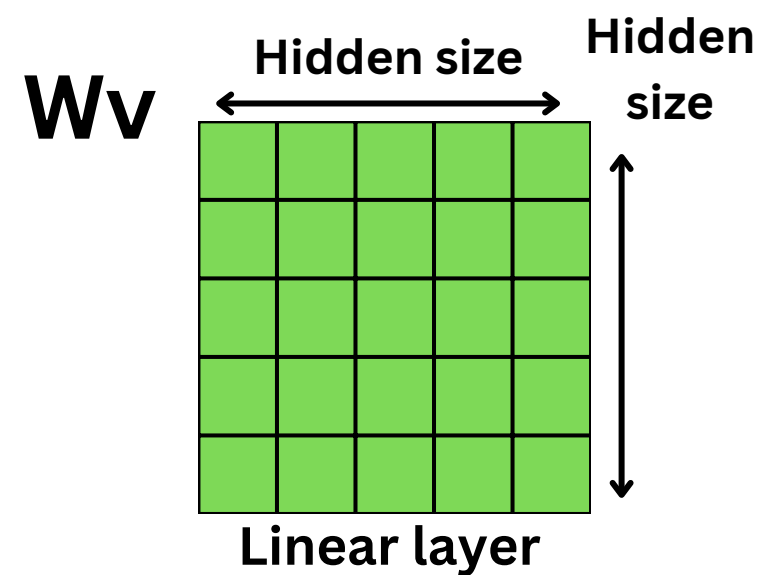
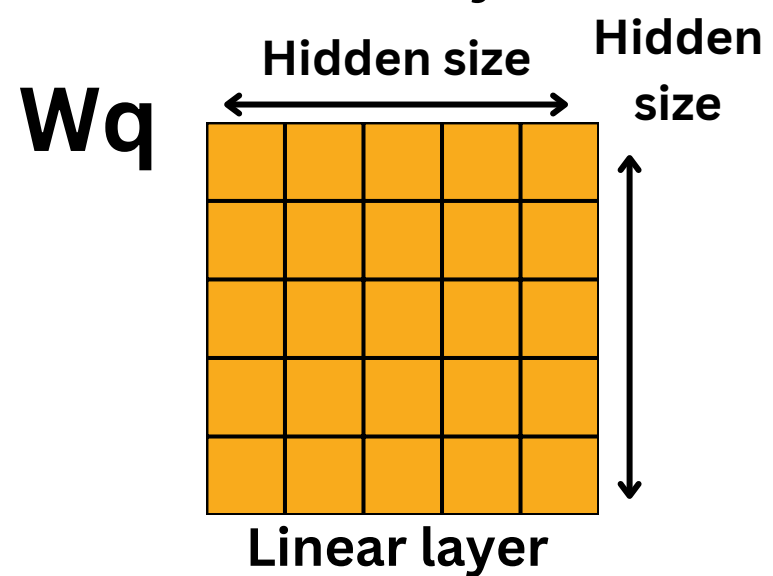
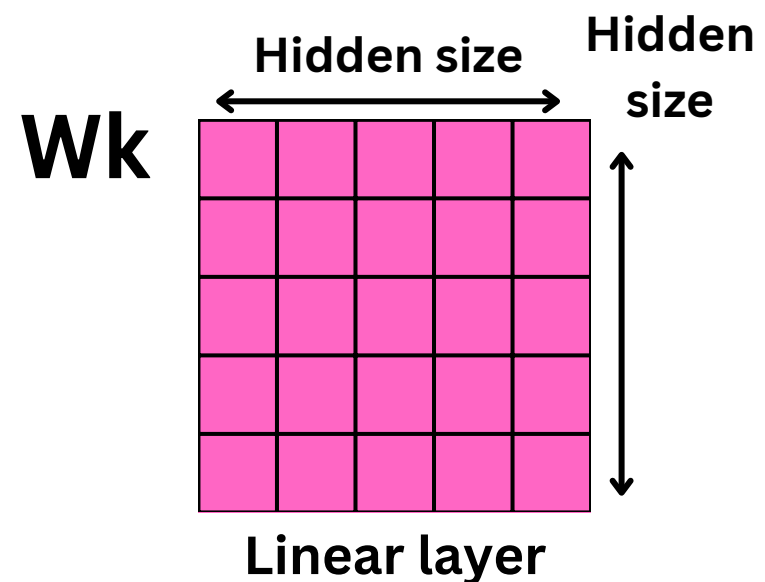
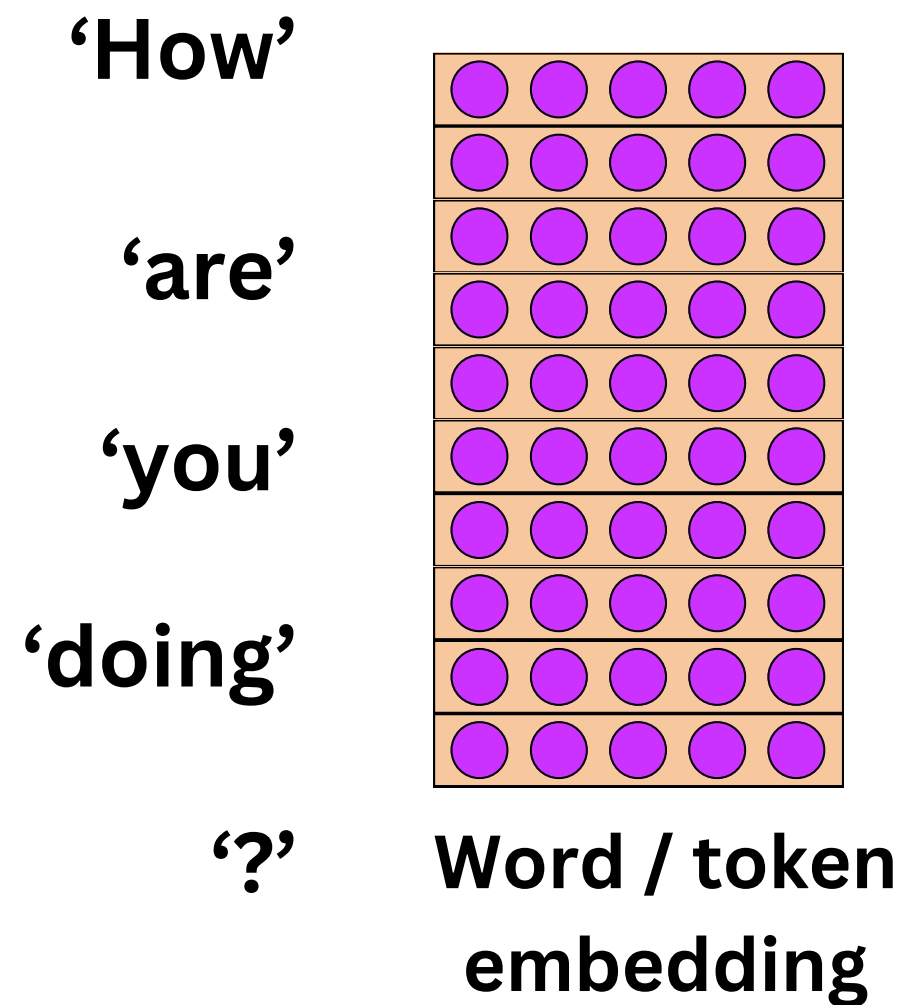


The Overall Architecture



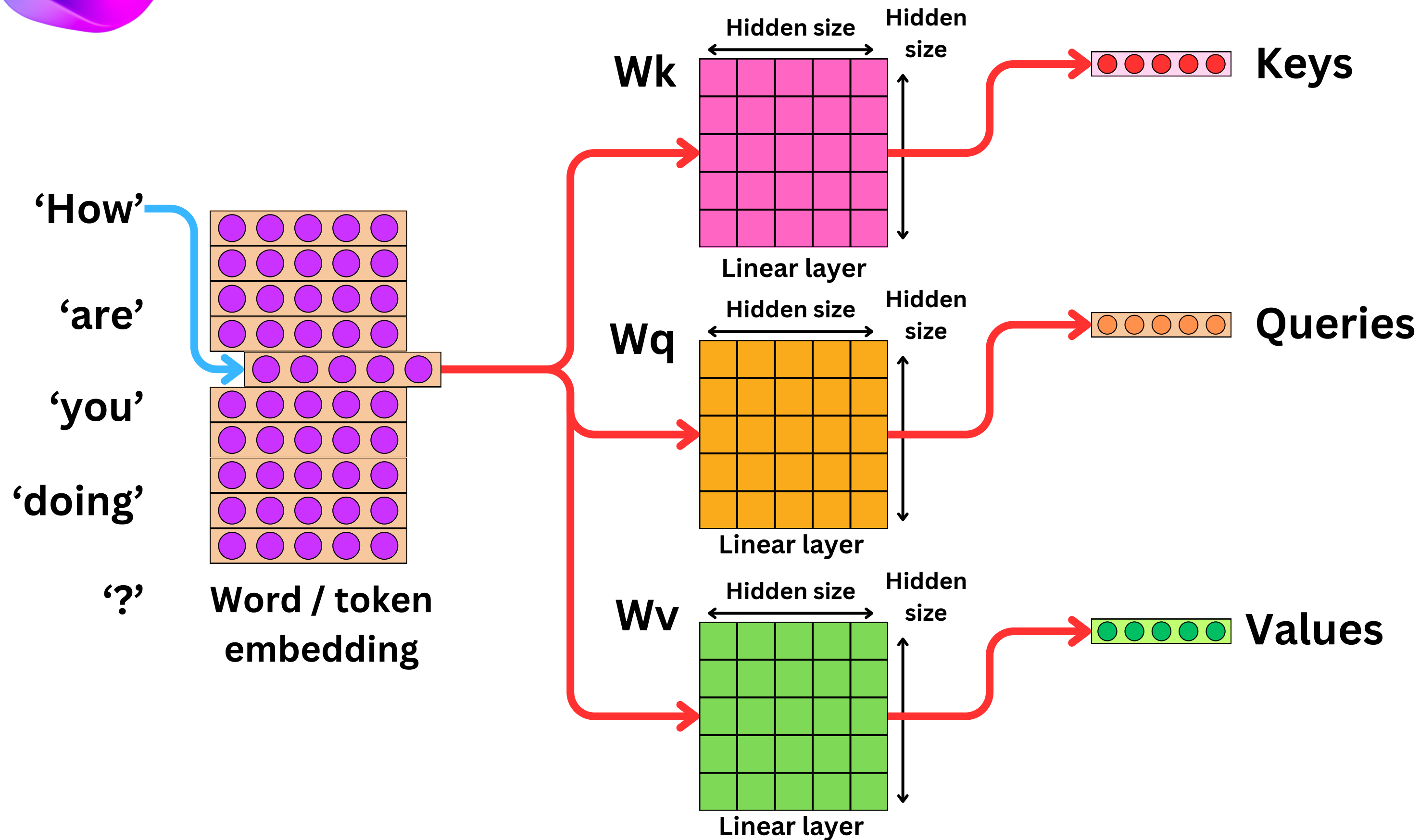


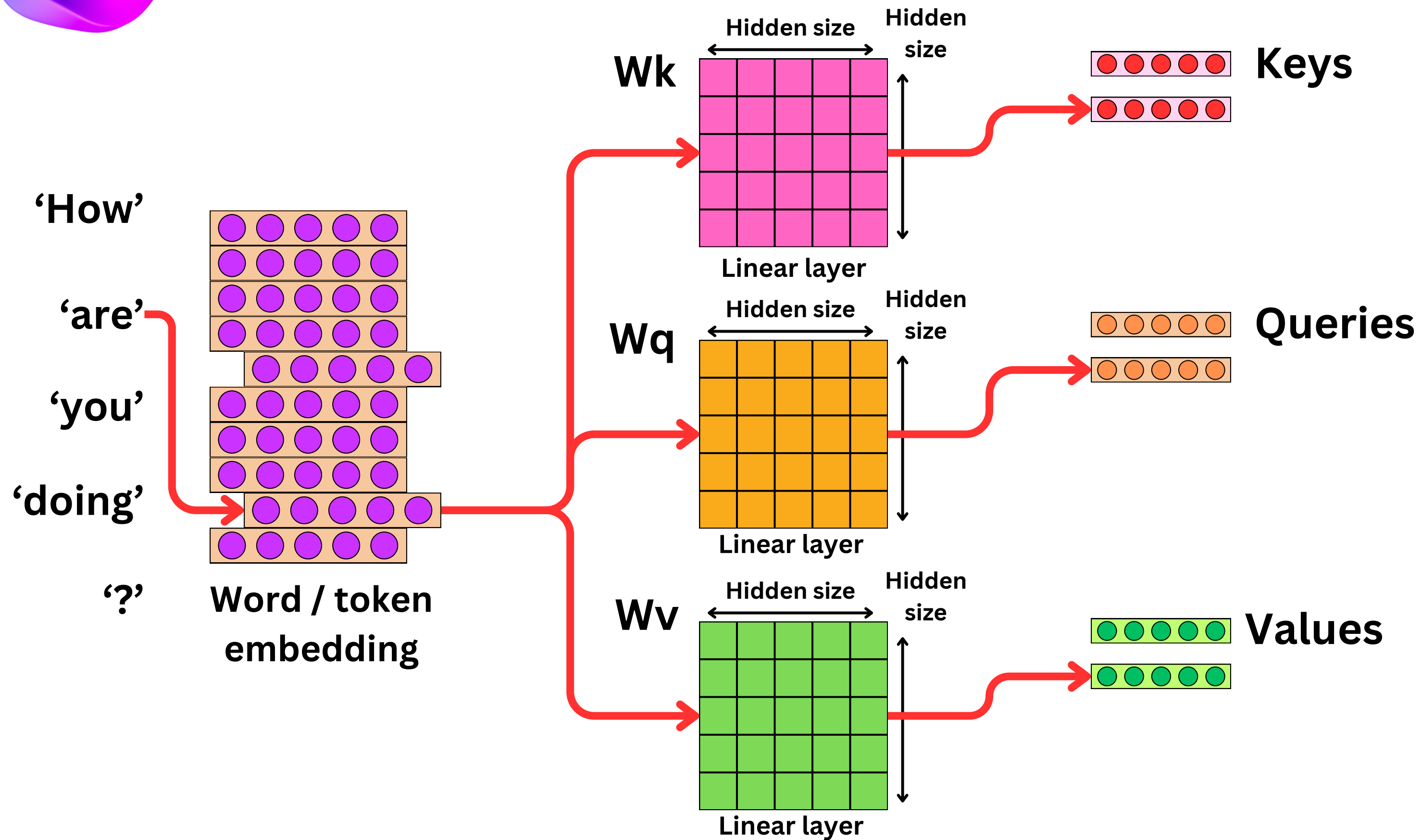
The Self-Attention Layer





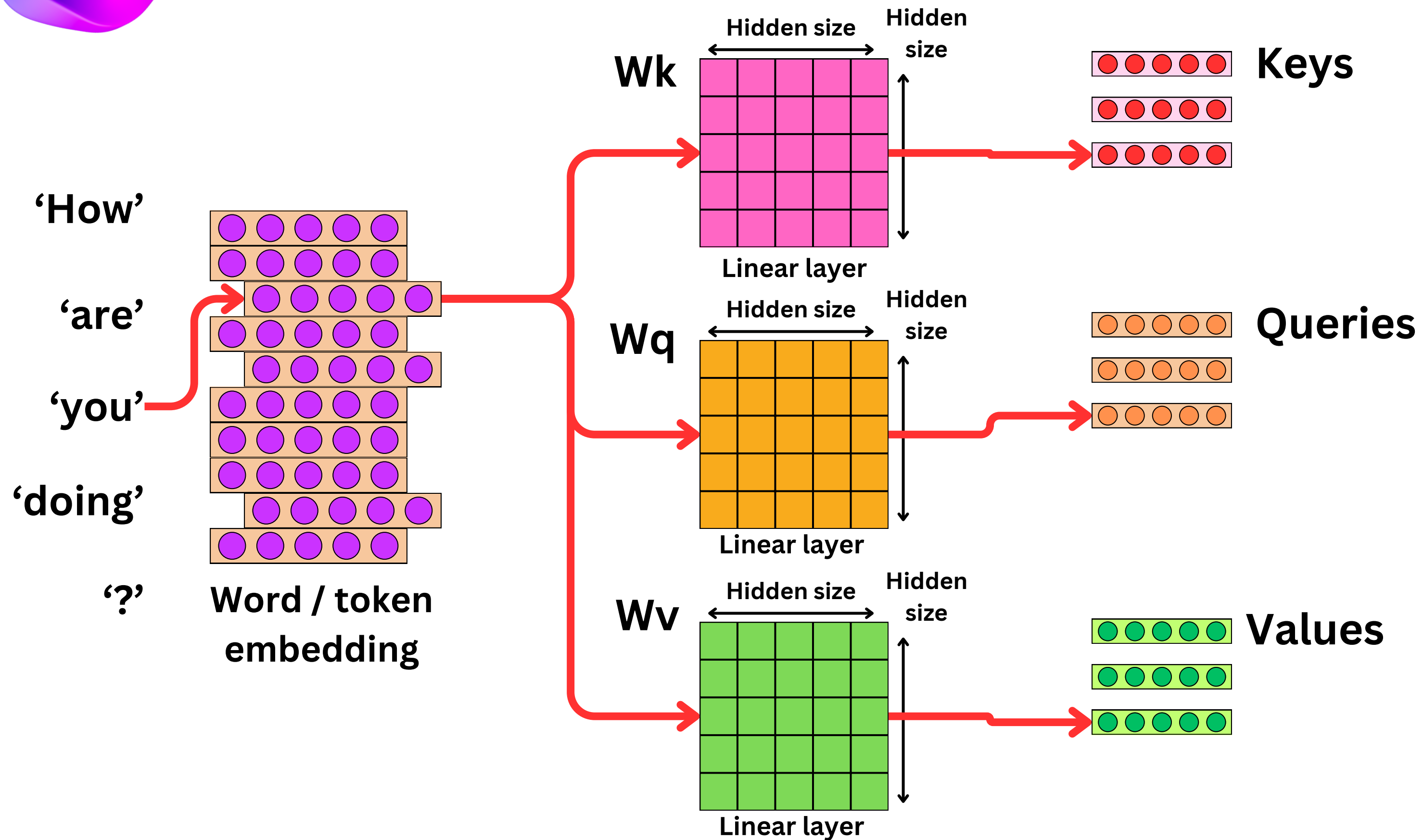
The Self-Attention Layer





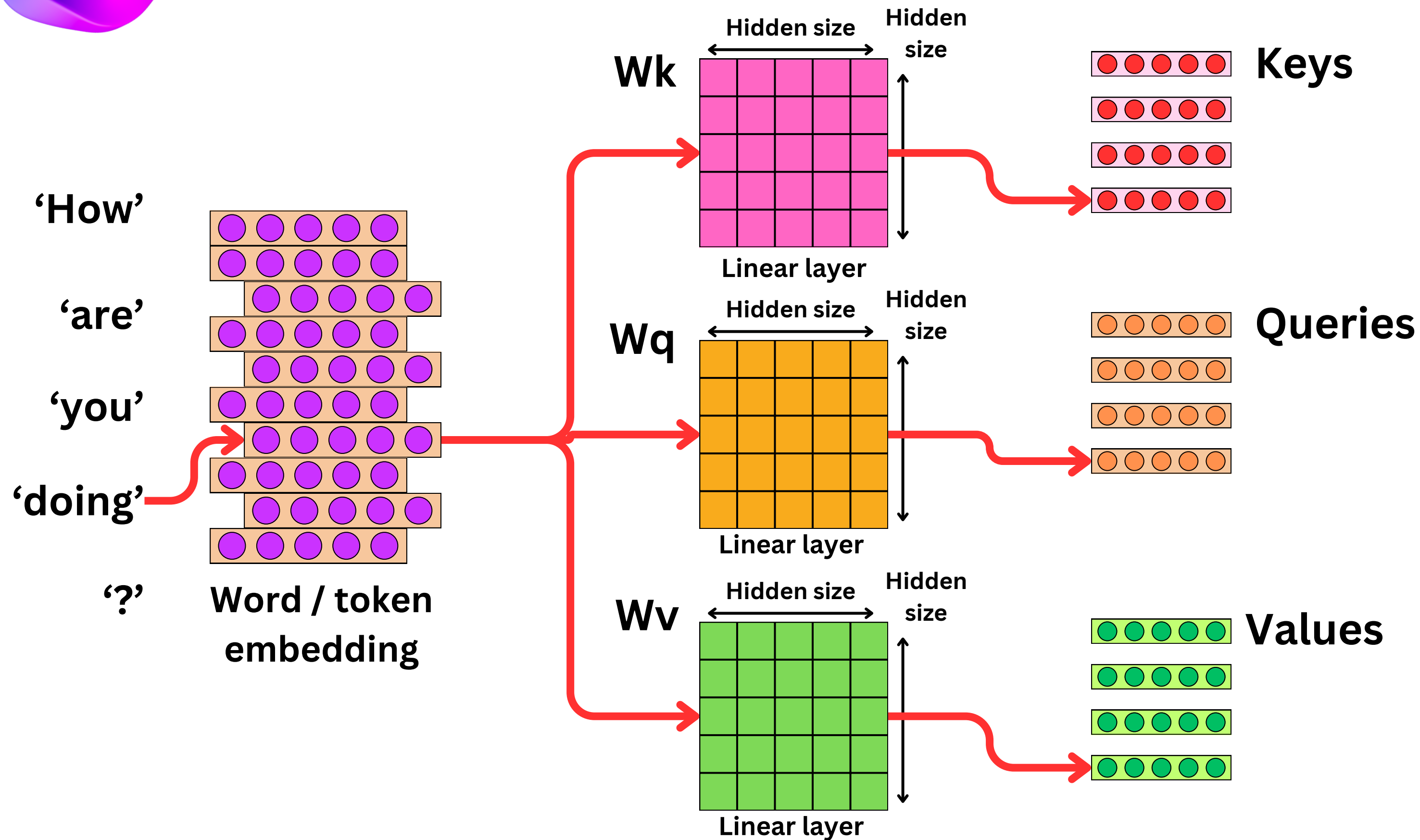


The Self-Attention Layer



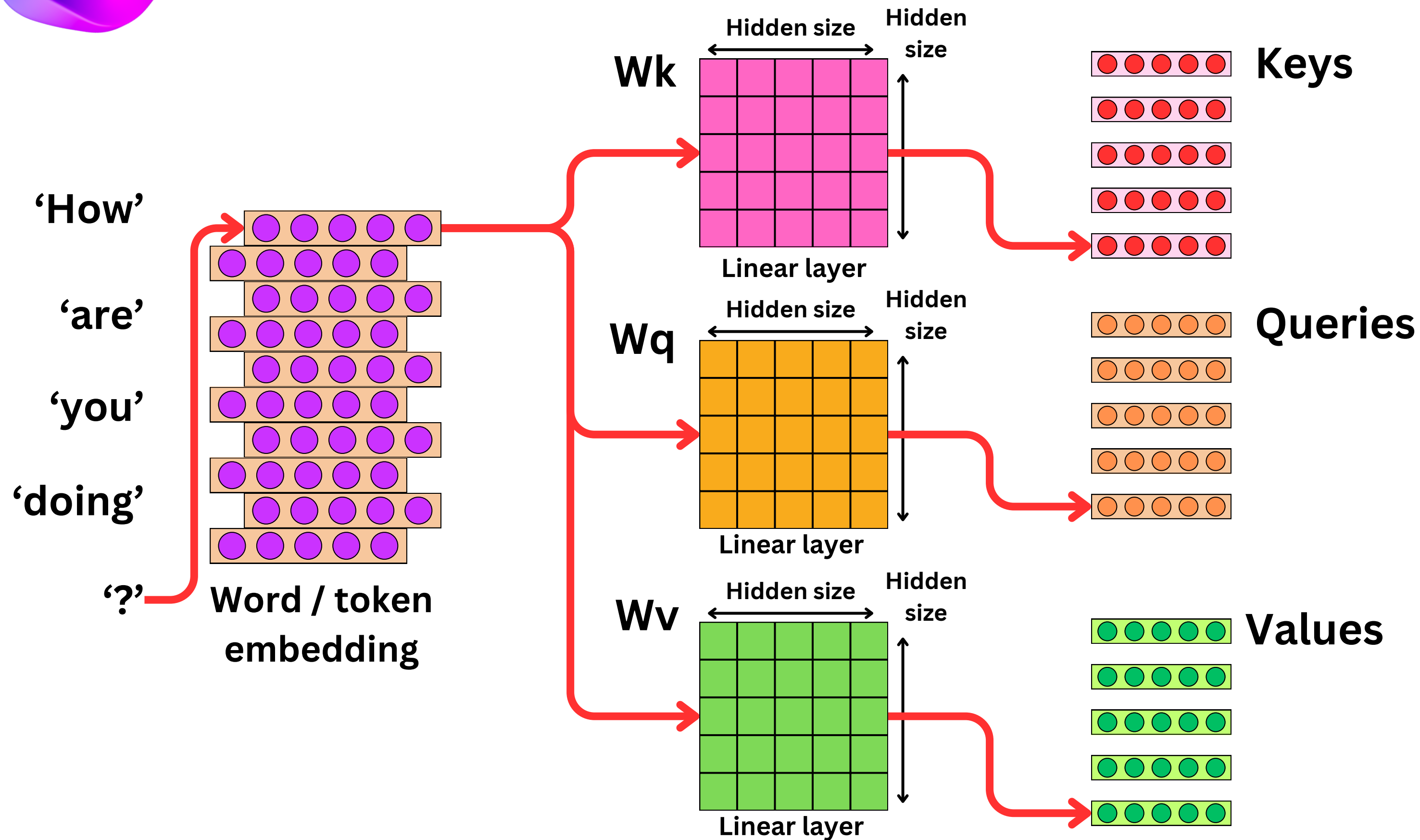


The Self-Attention Layer



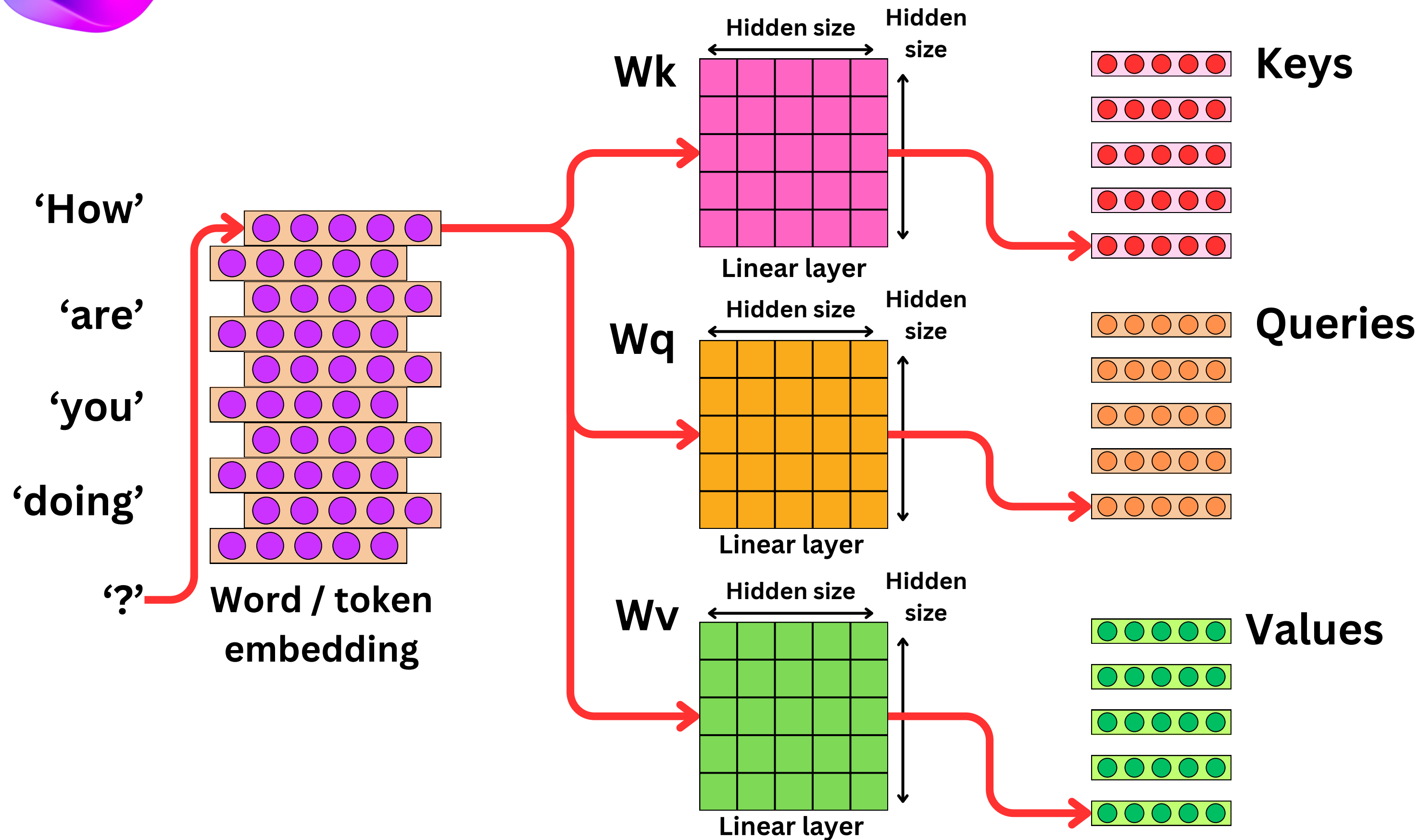


The Self-Attention Layer



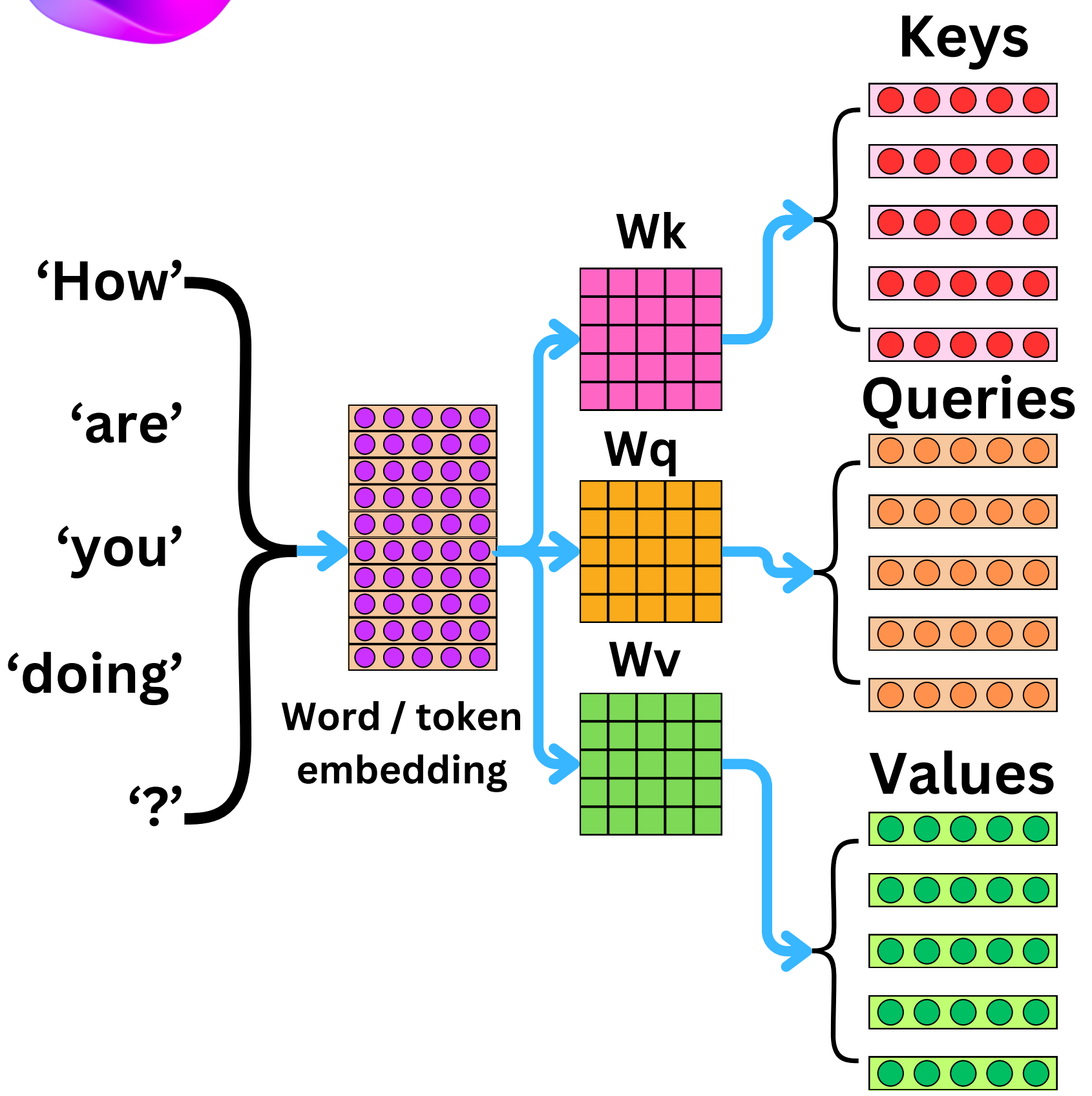


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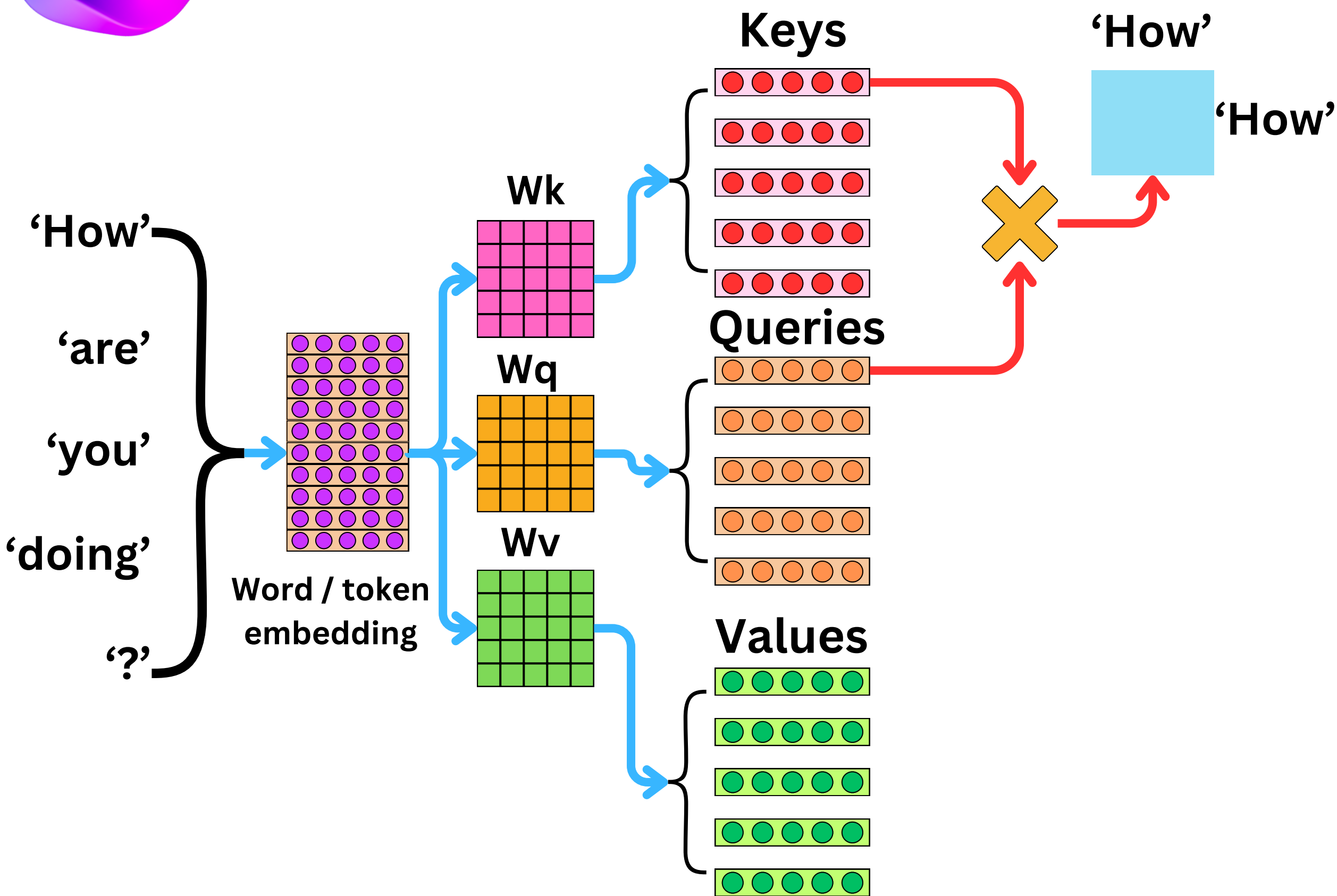


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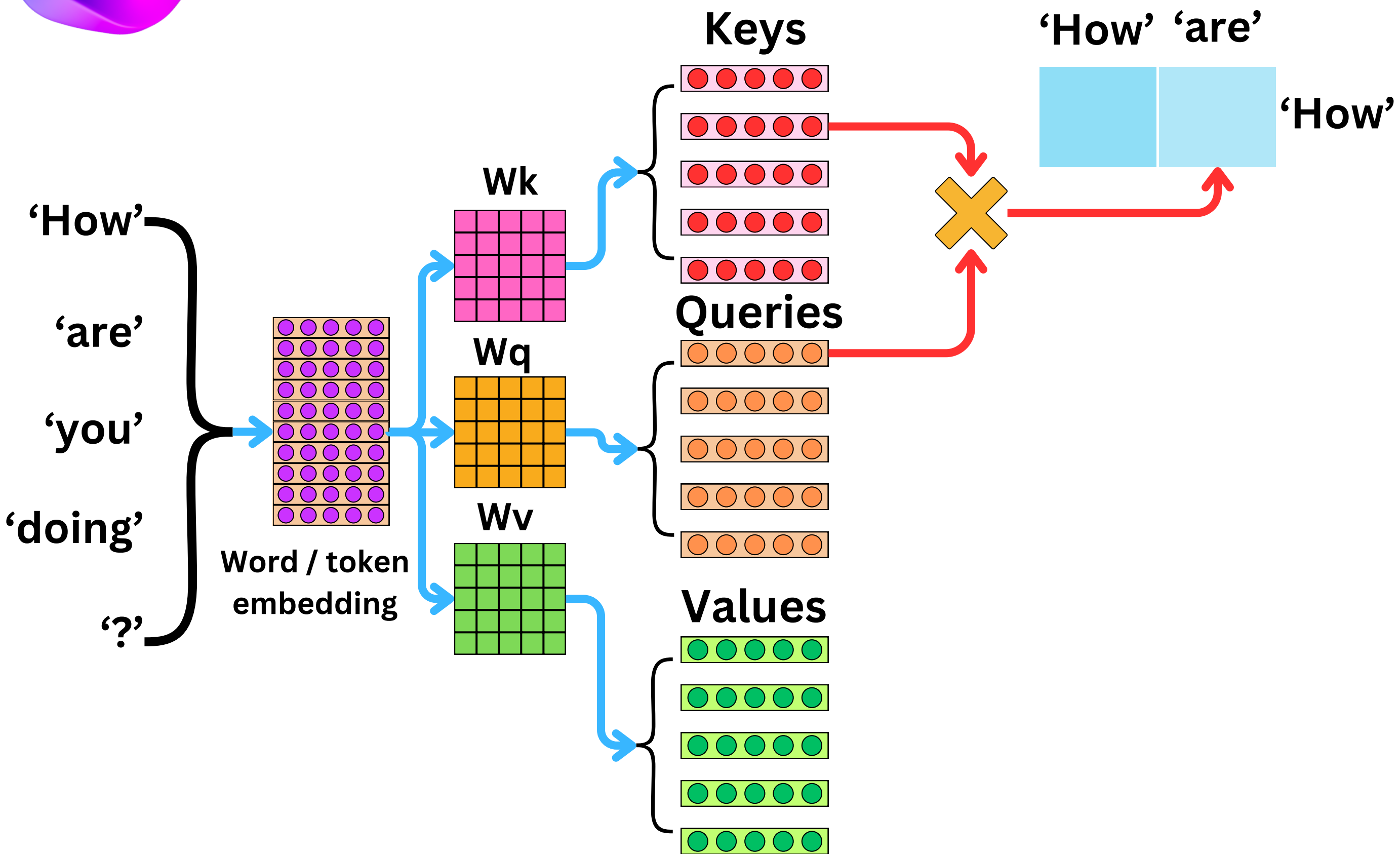


The Self-Attention Layer



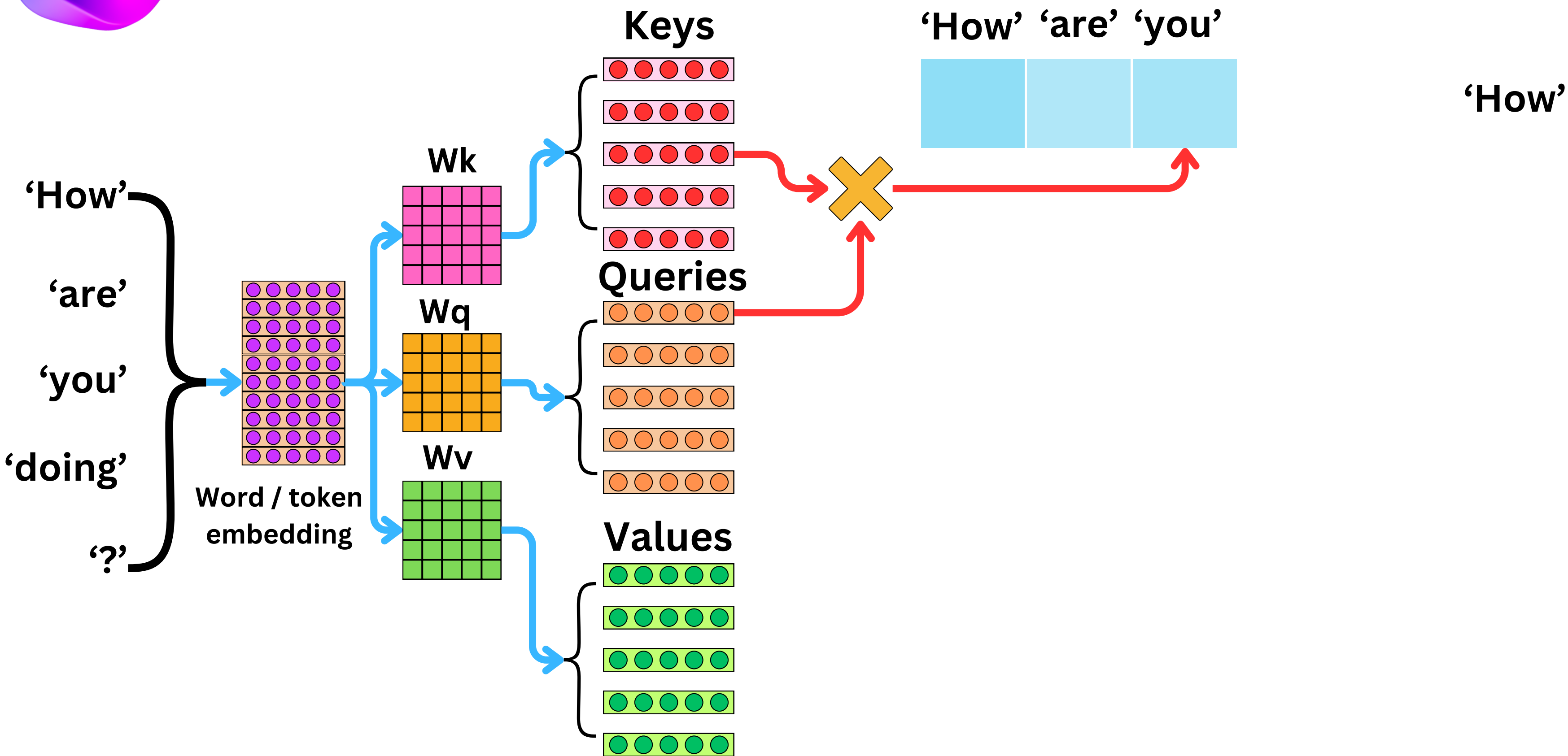


The Self-Attention Layer



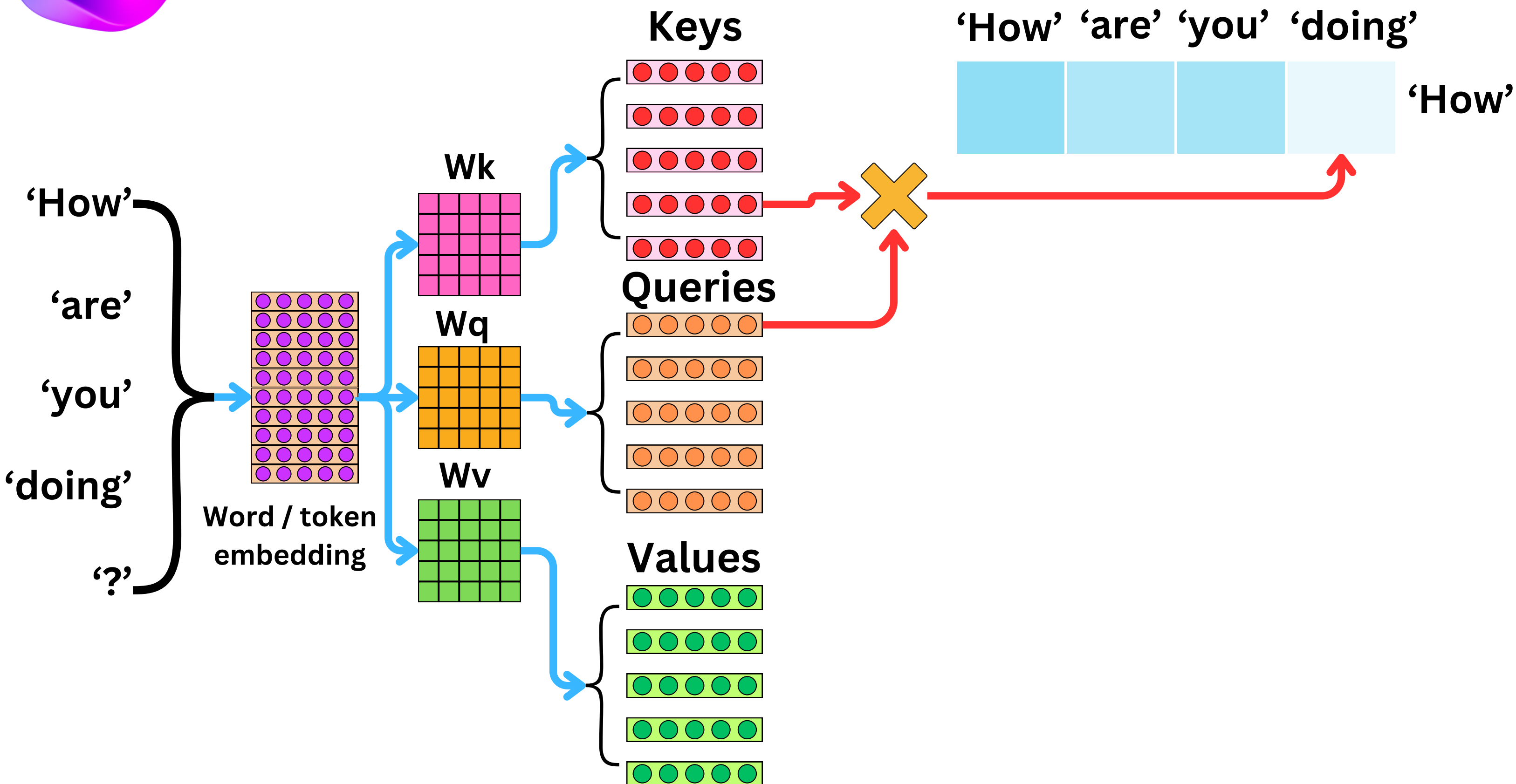


The Self-Attention Layer



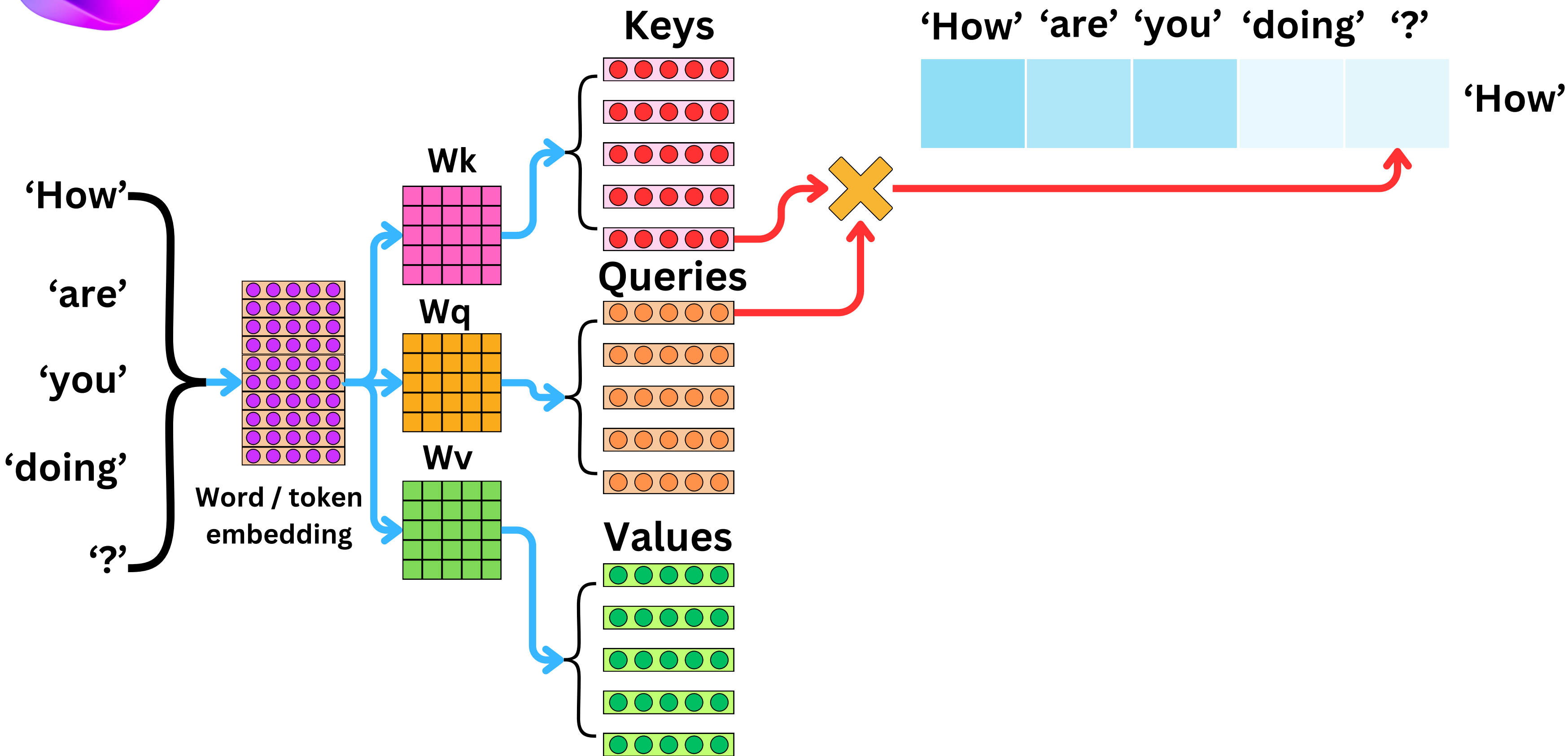


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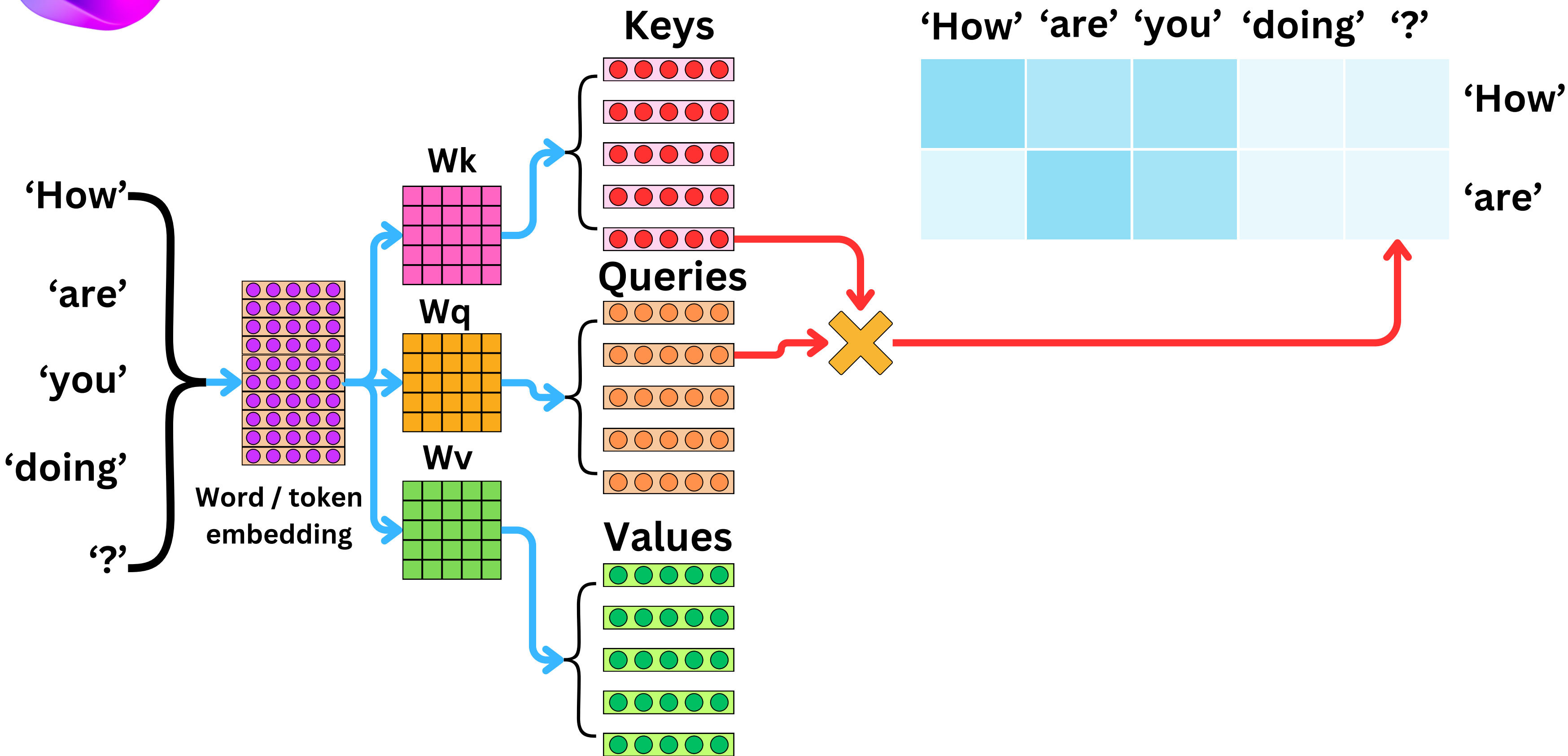


The Self-Attention Layer



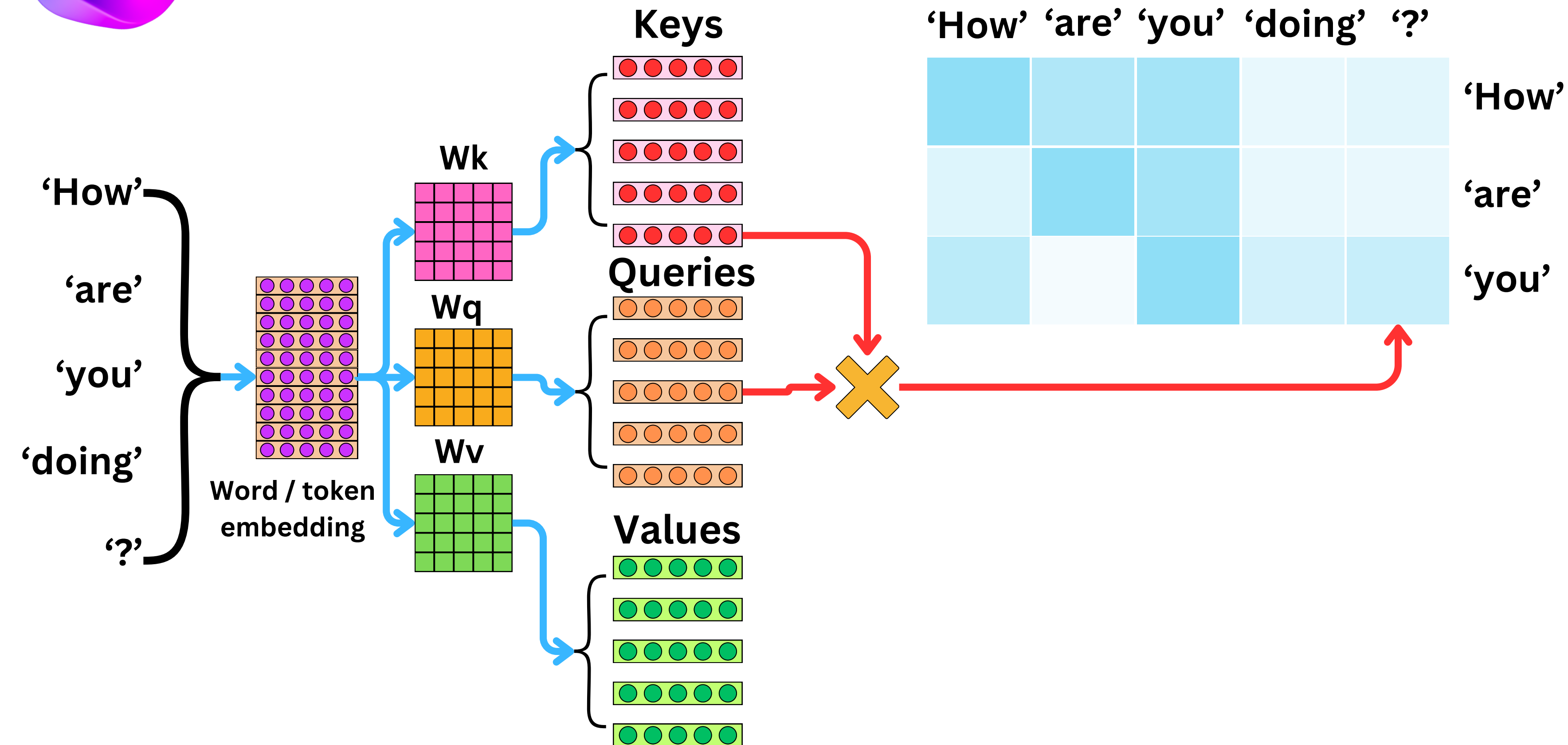


The Self-Attention Layer



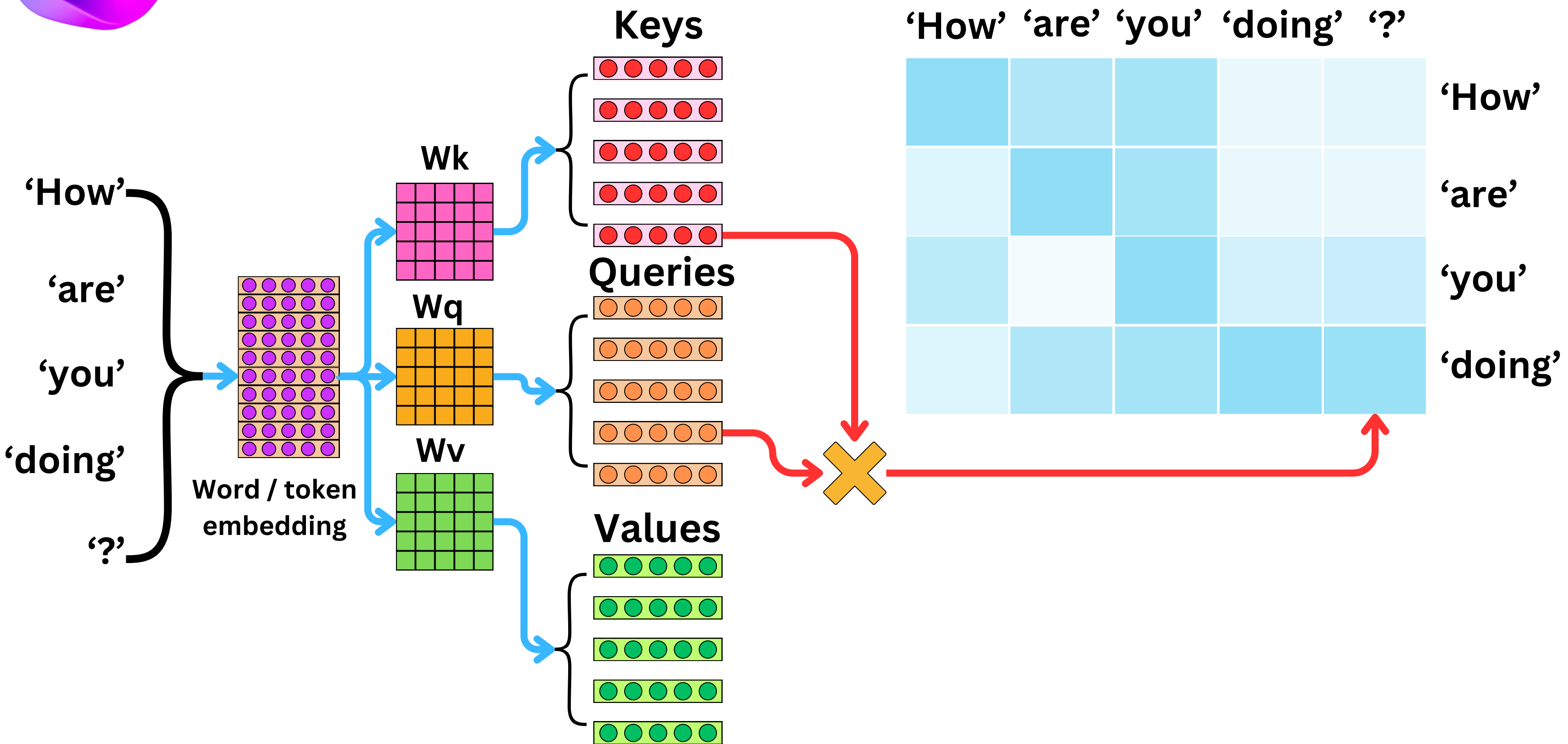


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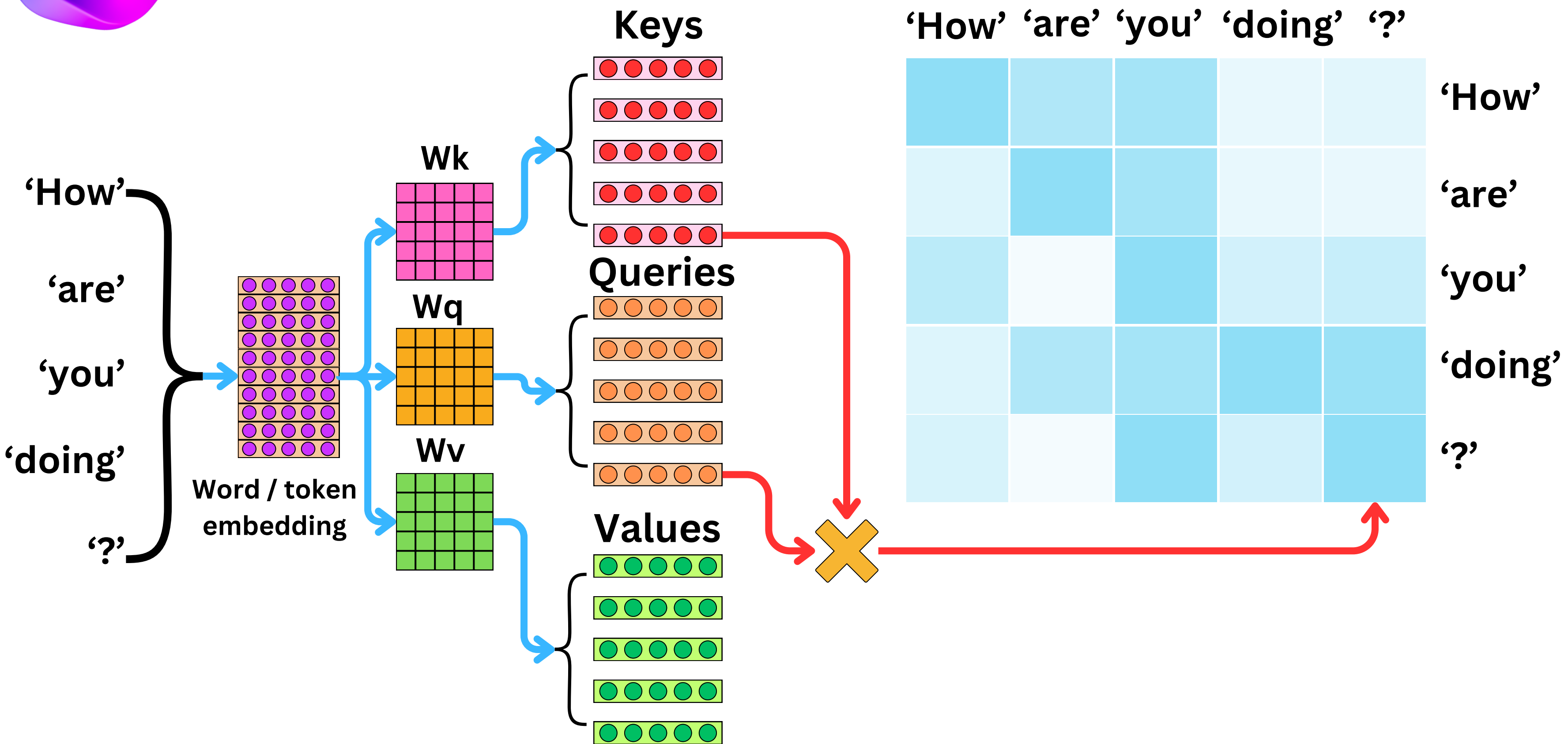


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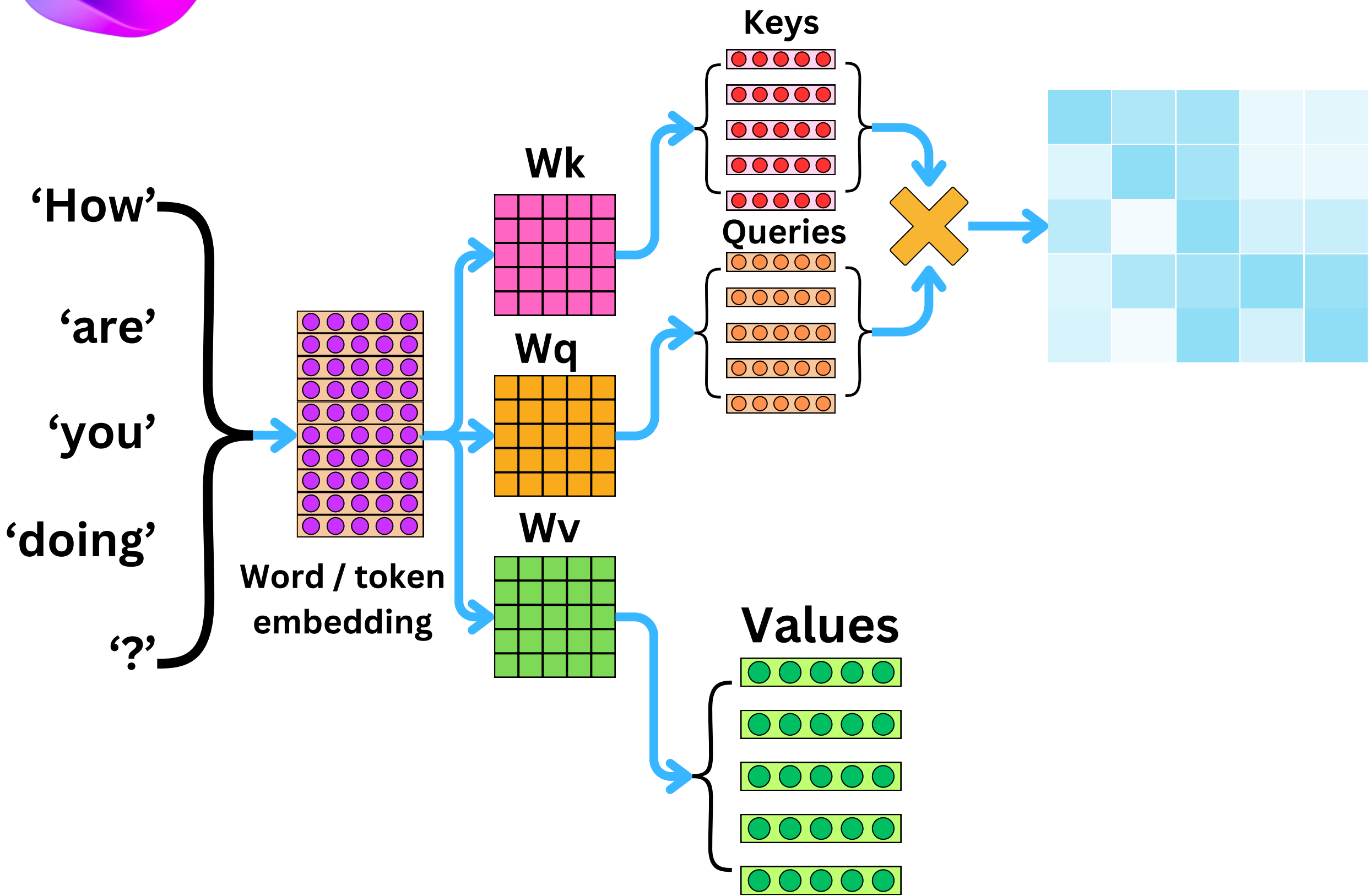


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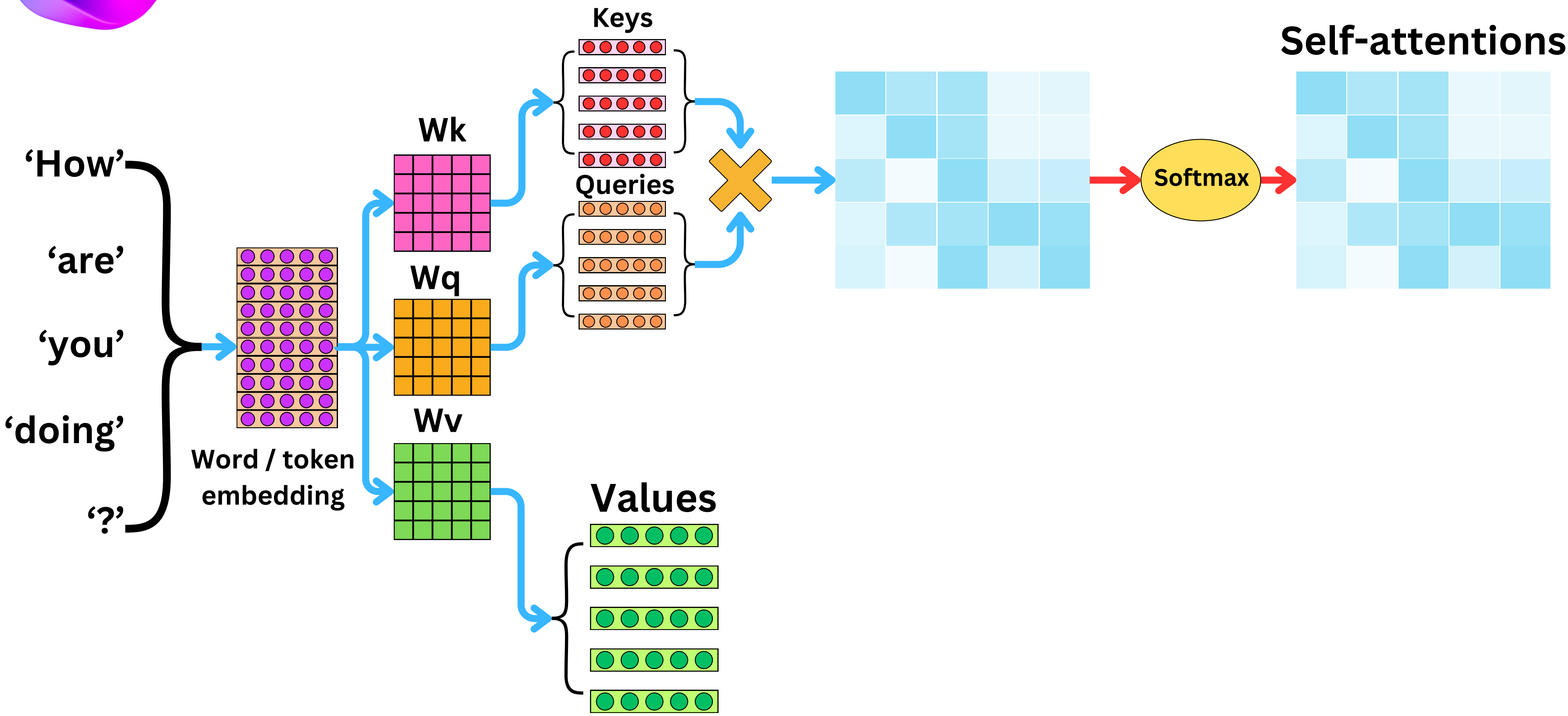


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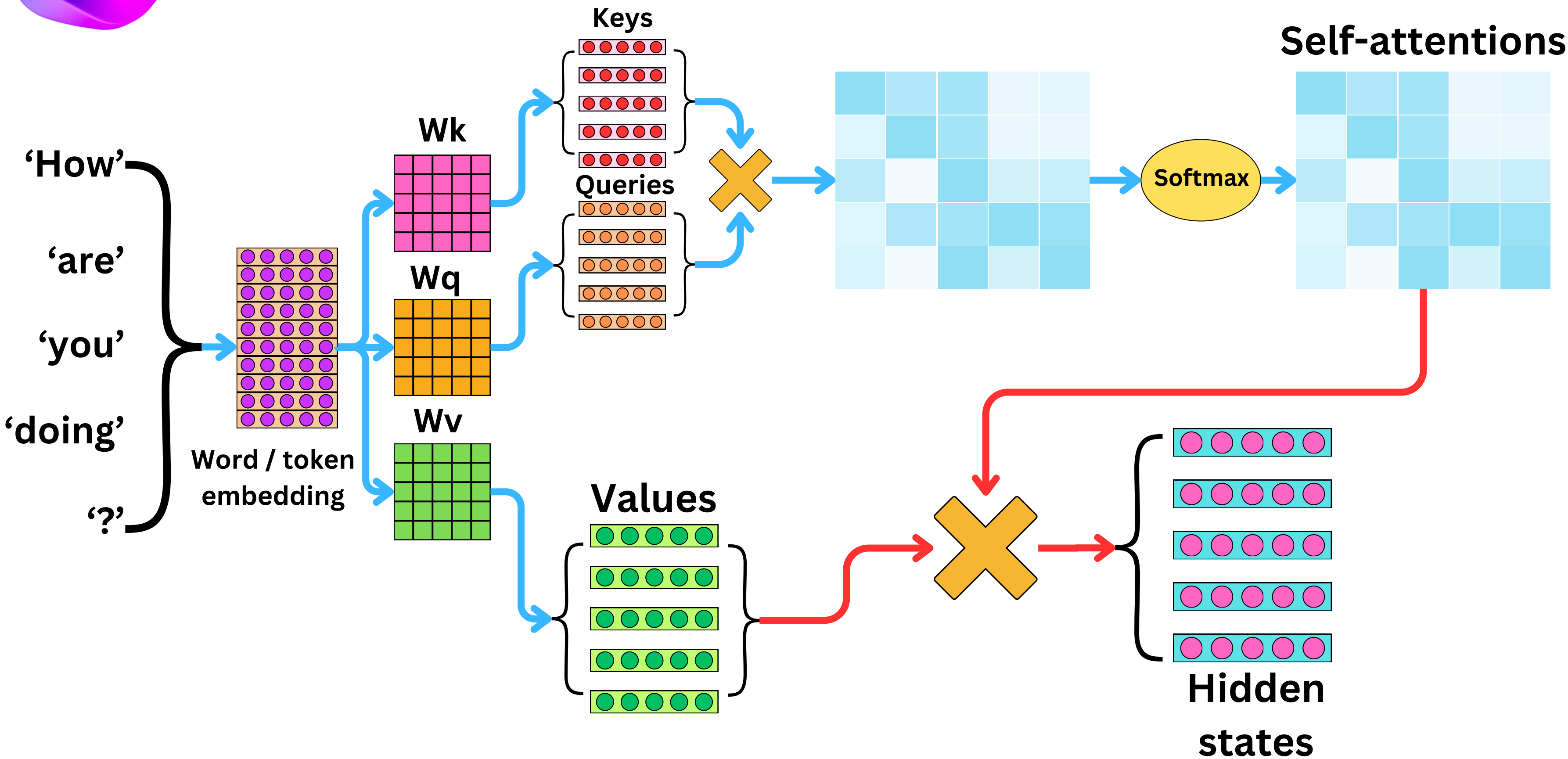


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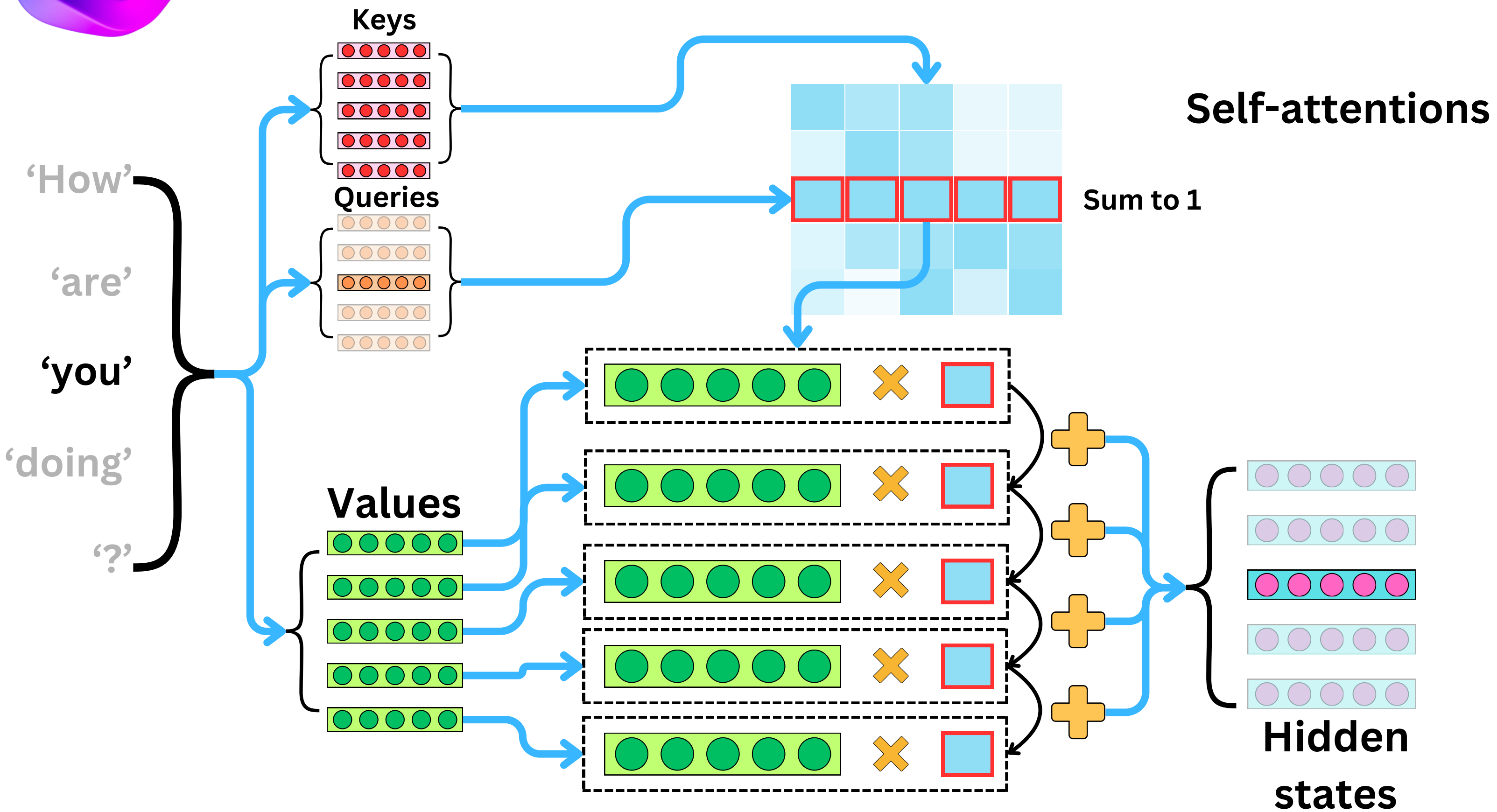


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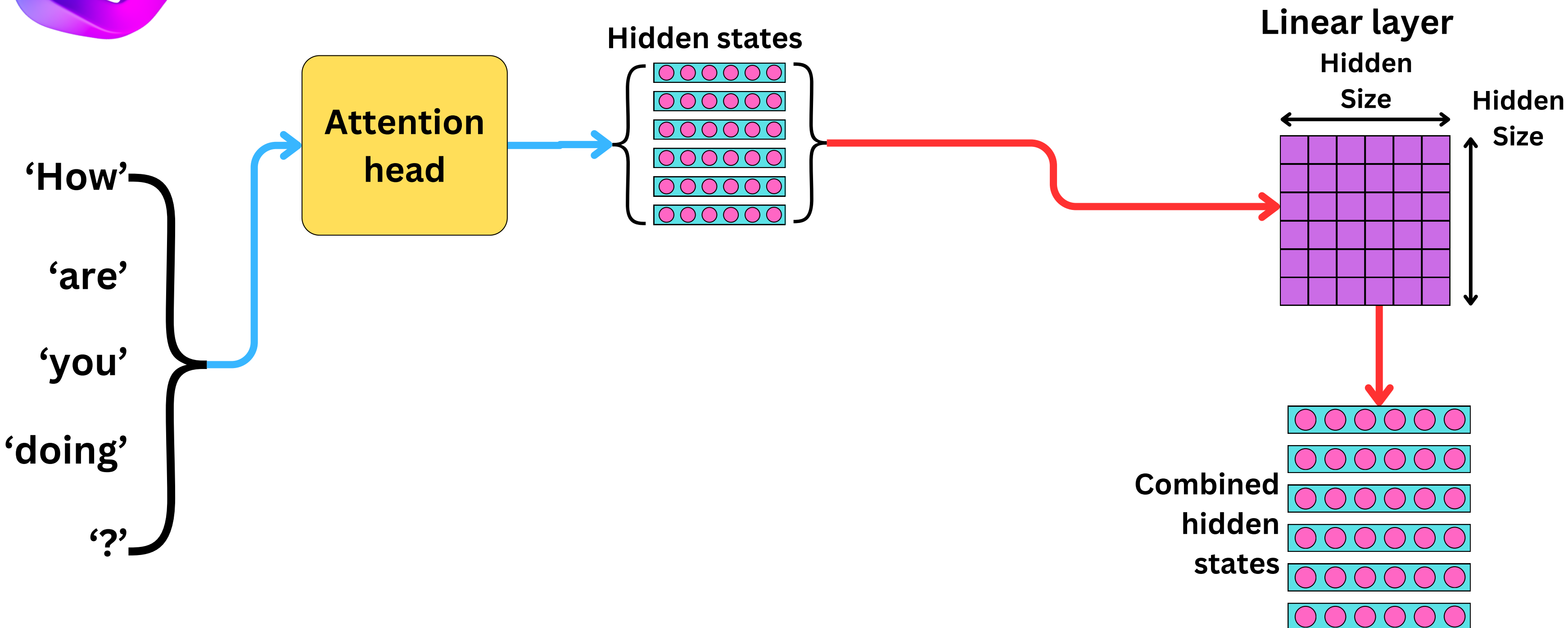


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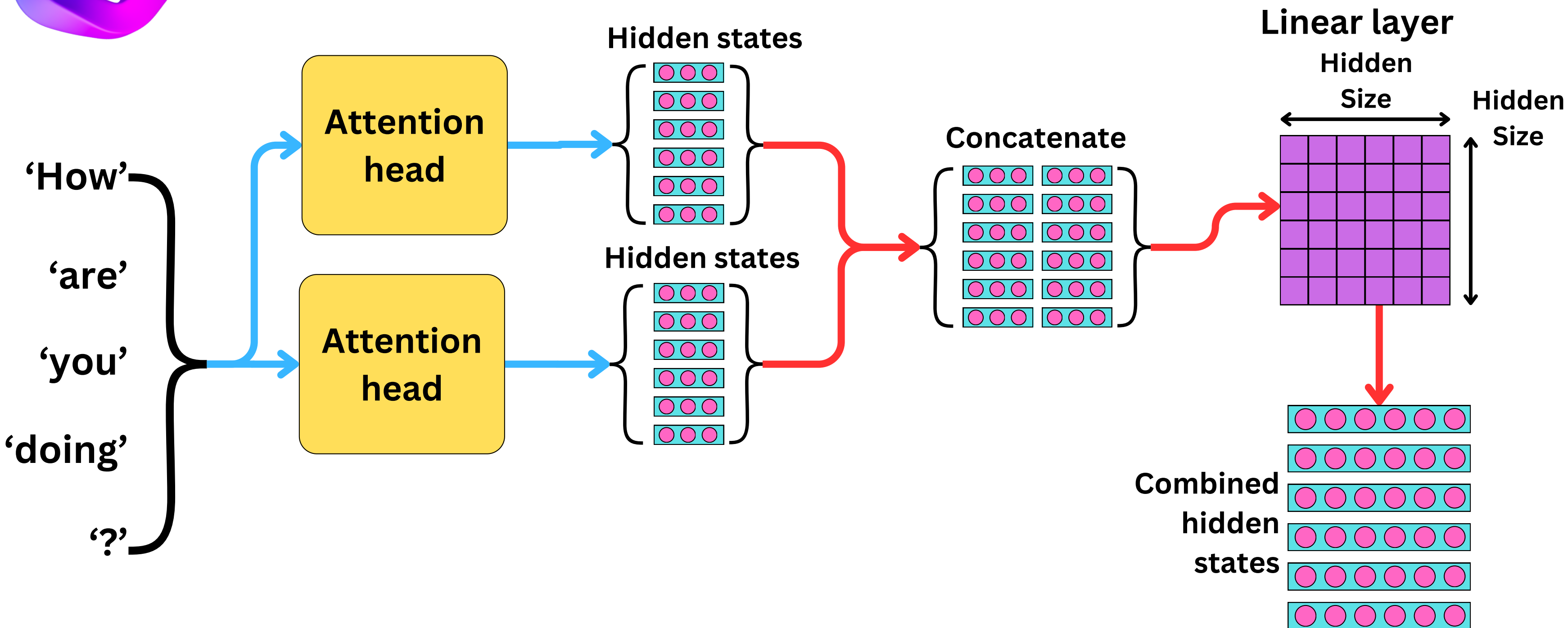


Multihead Attention Layer



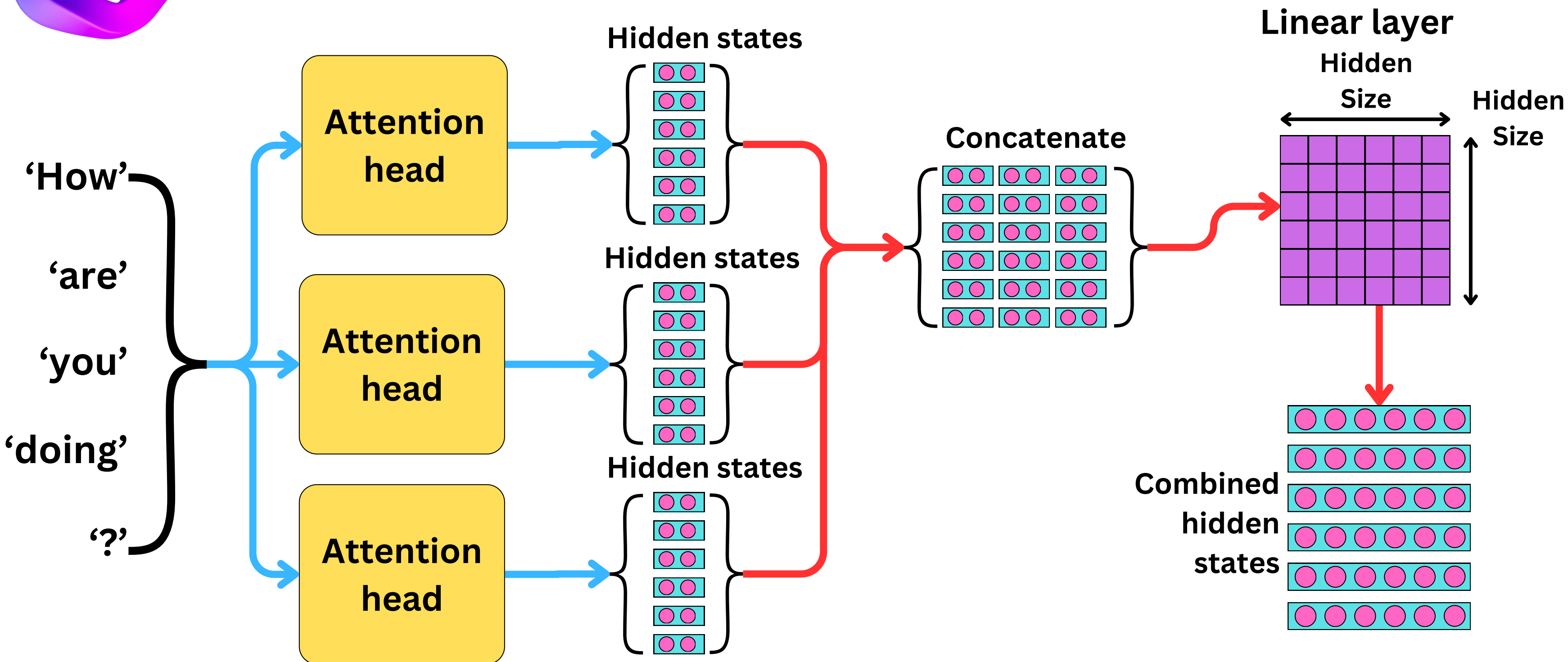


Multihead Attention Layer



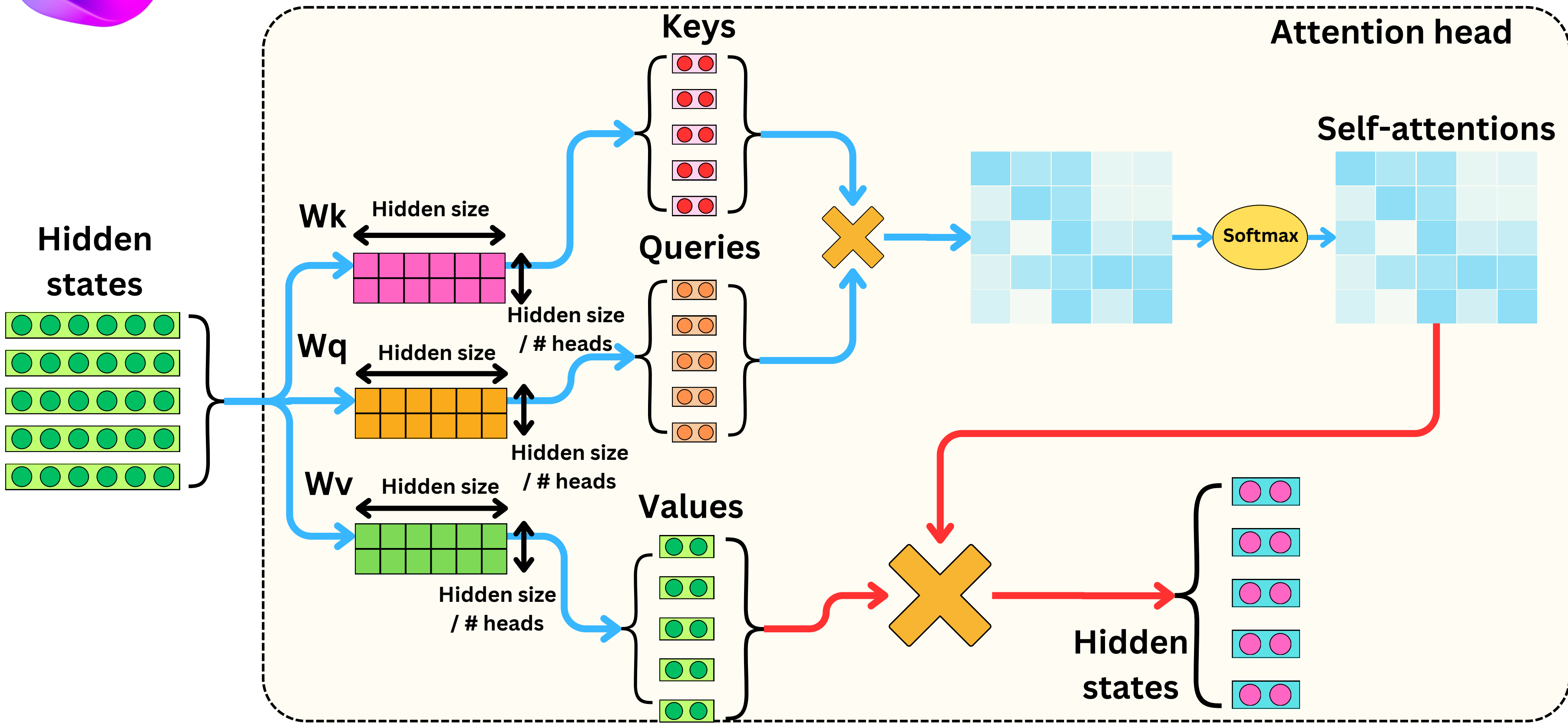


Multihead Attention Layer



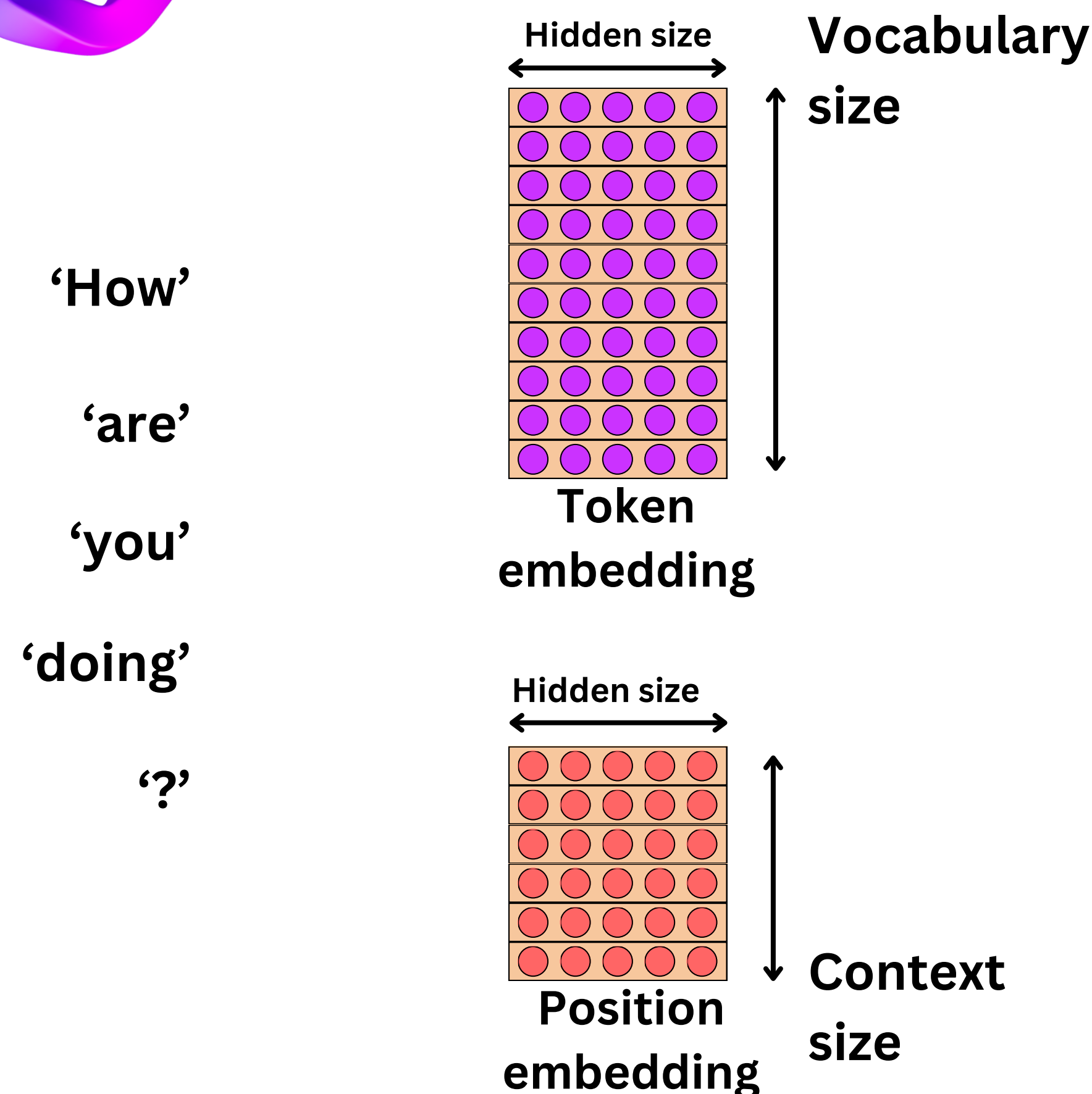


Multihead Attention Layer



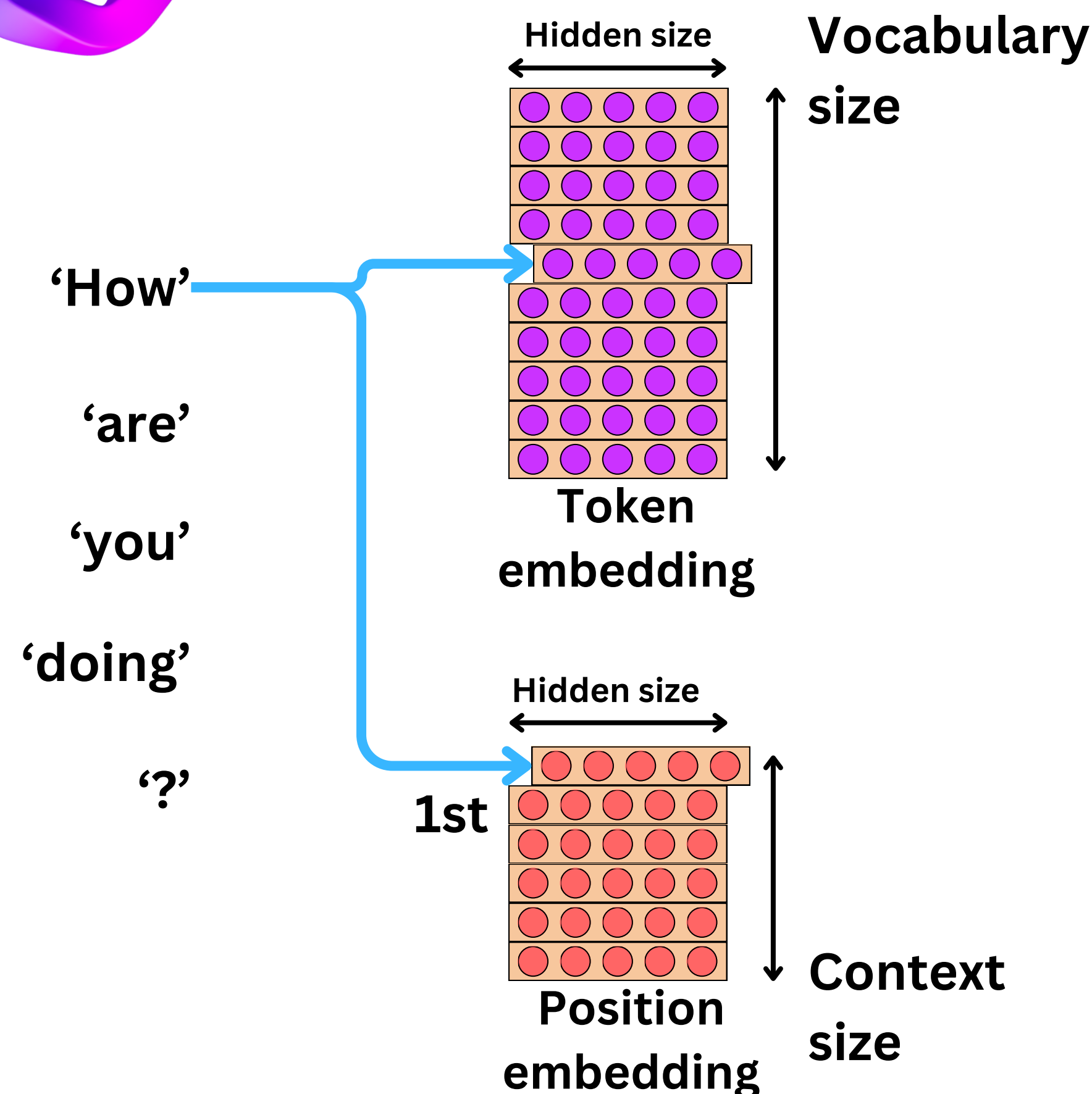


The Position Embedding



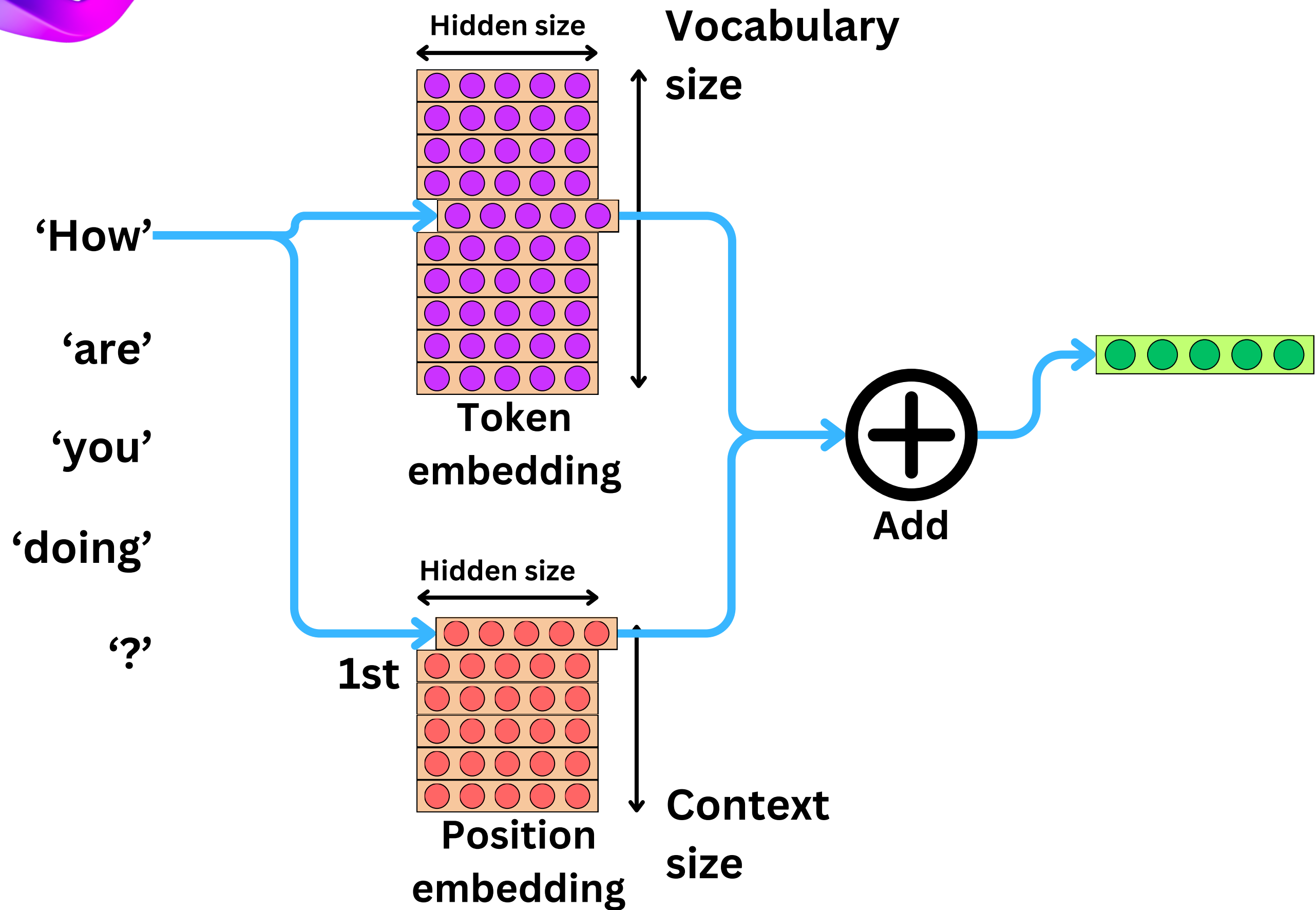


The Position Embedding



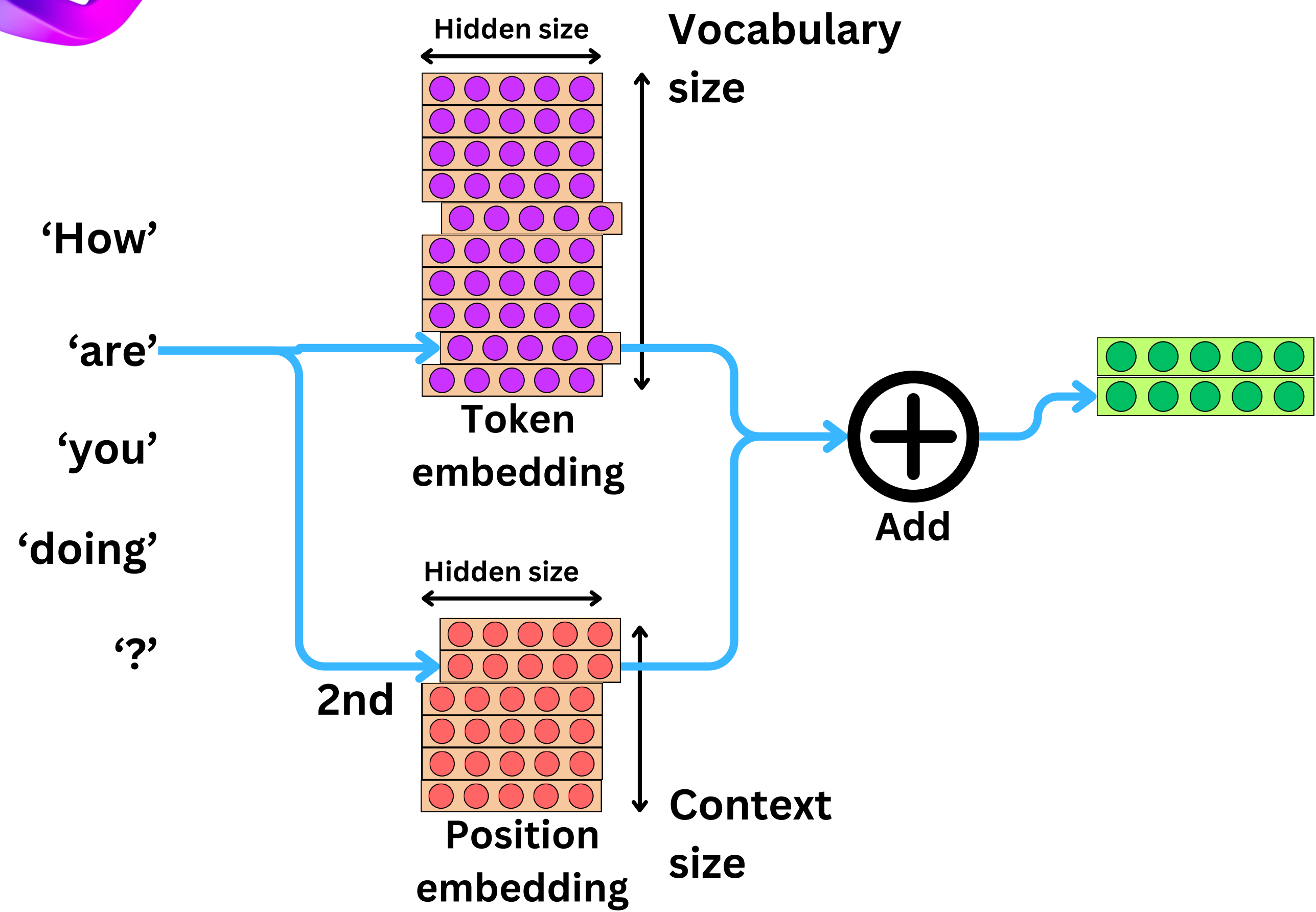


The Position Embedding



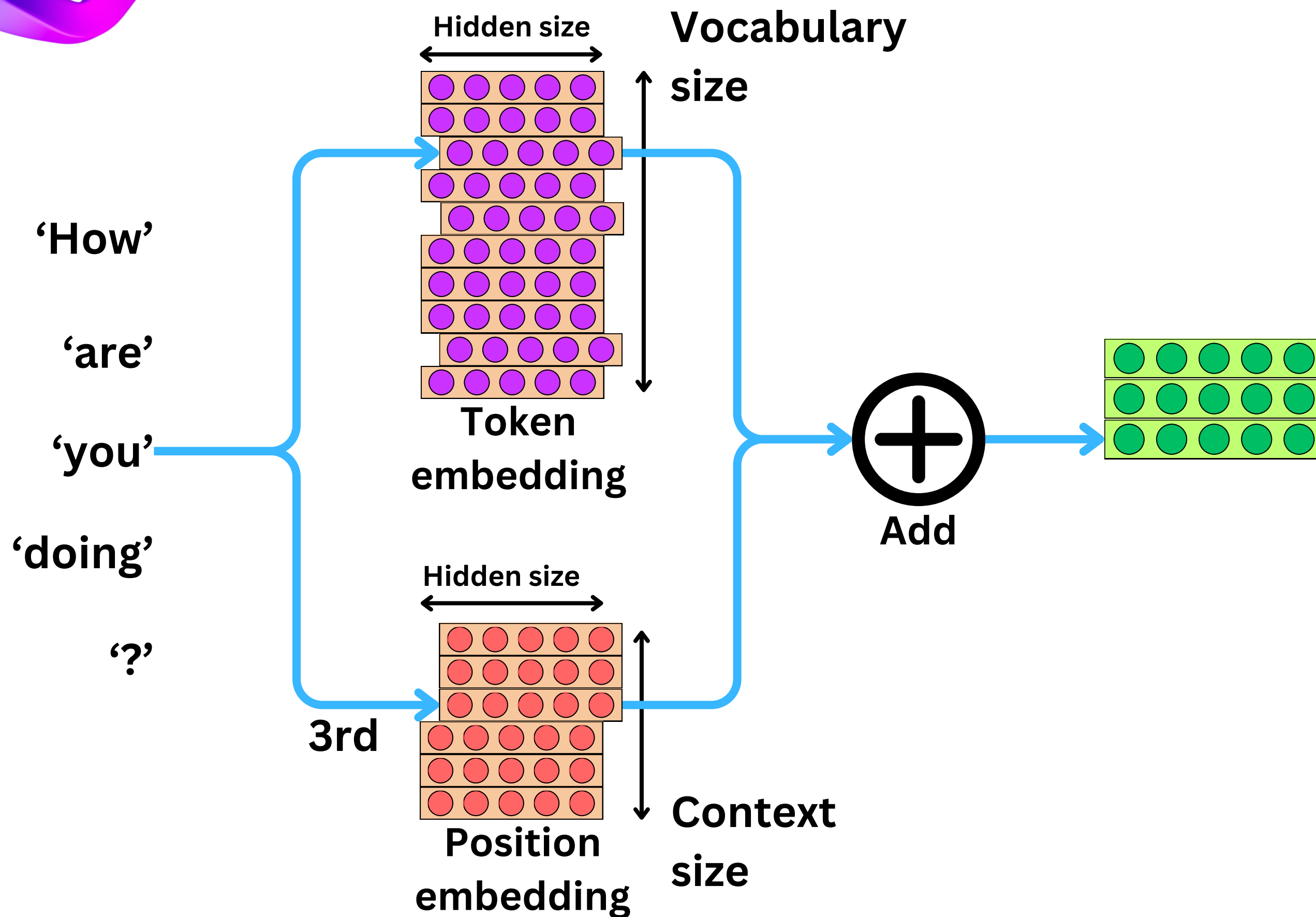


The Position Embedding



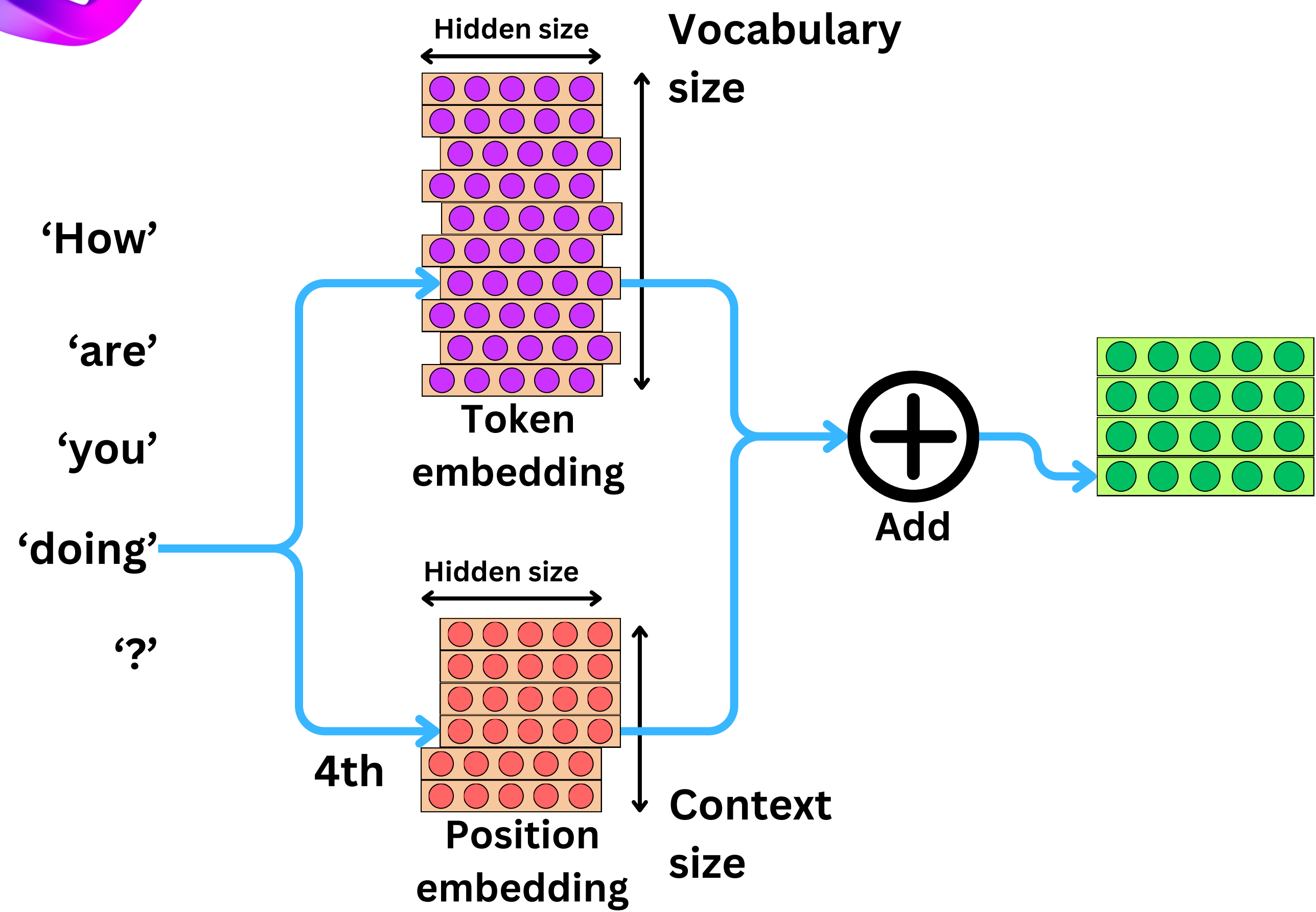


The Position Embedding



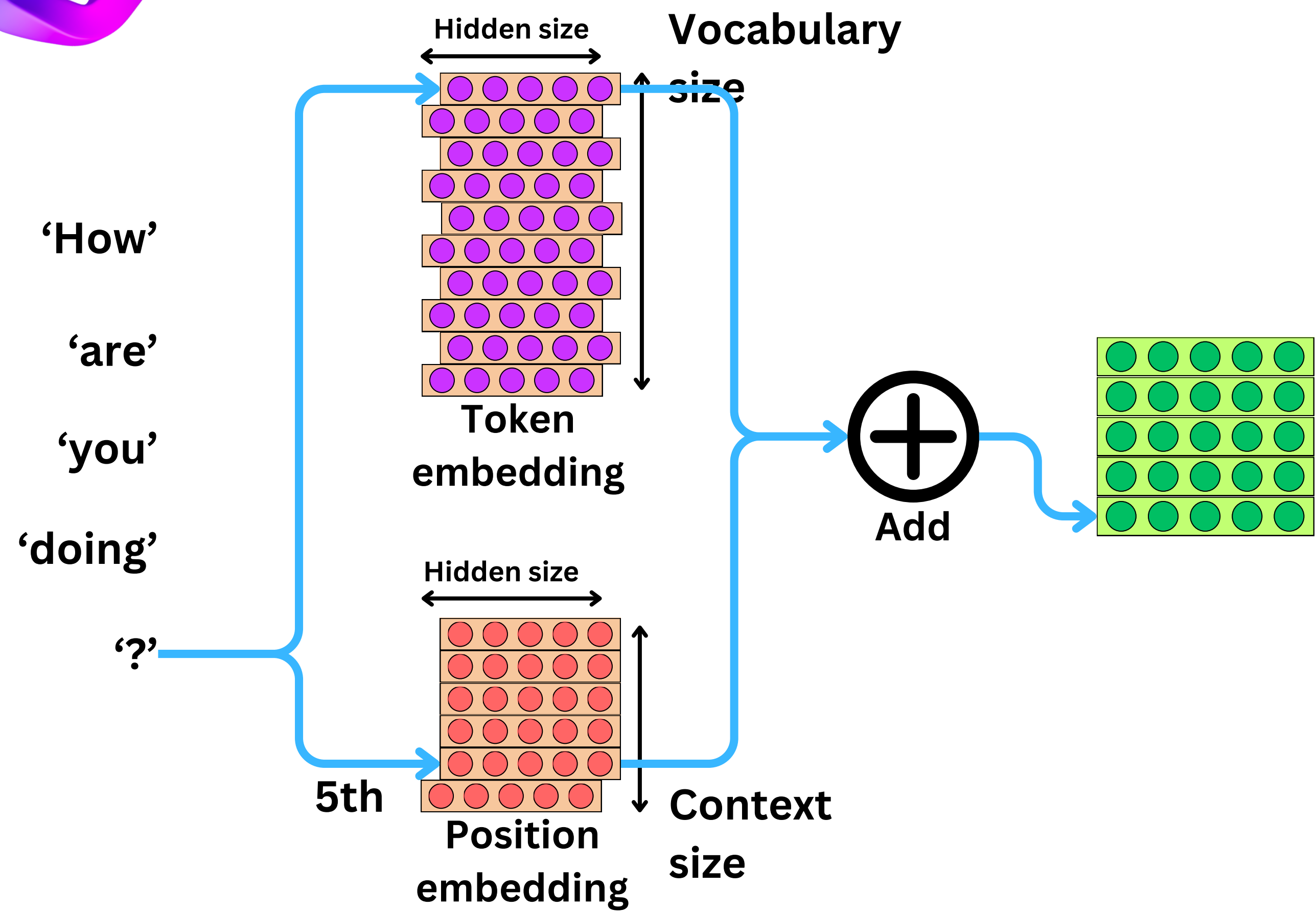


The Position Embedding





The Position Embedding





The Position Embedding

Correct
order

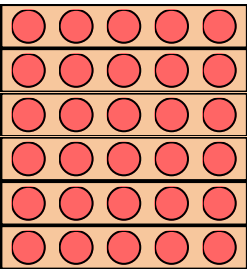
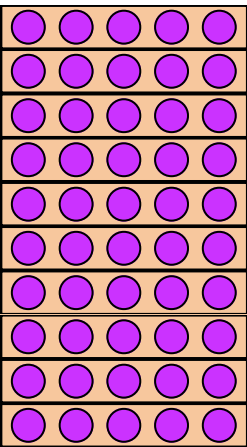
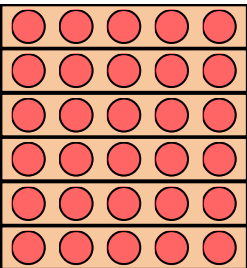
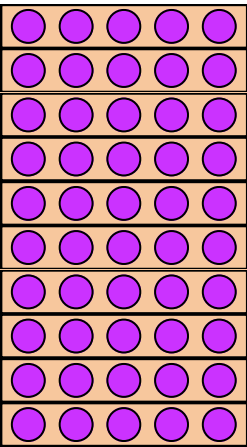
‘How’

‘are’

‘you’

‘doing’

‘?’



Wrong
order

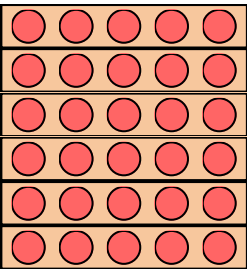
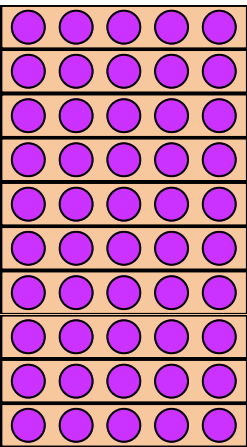
‘doing’

‘?’

‘are’

‘you’

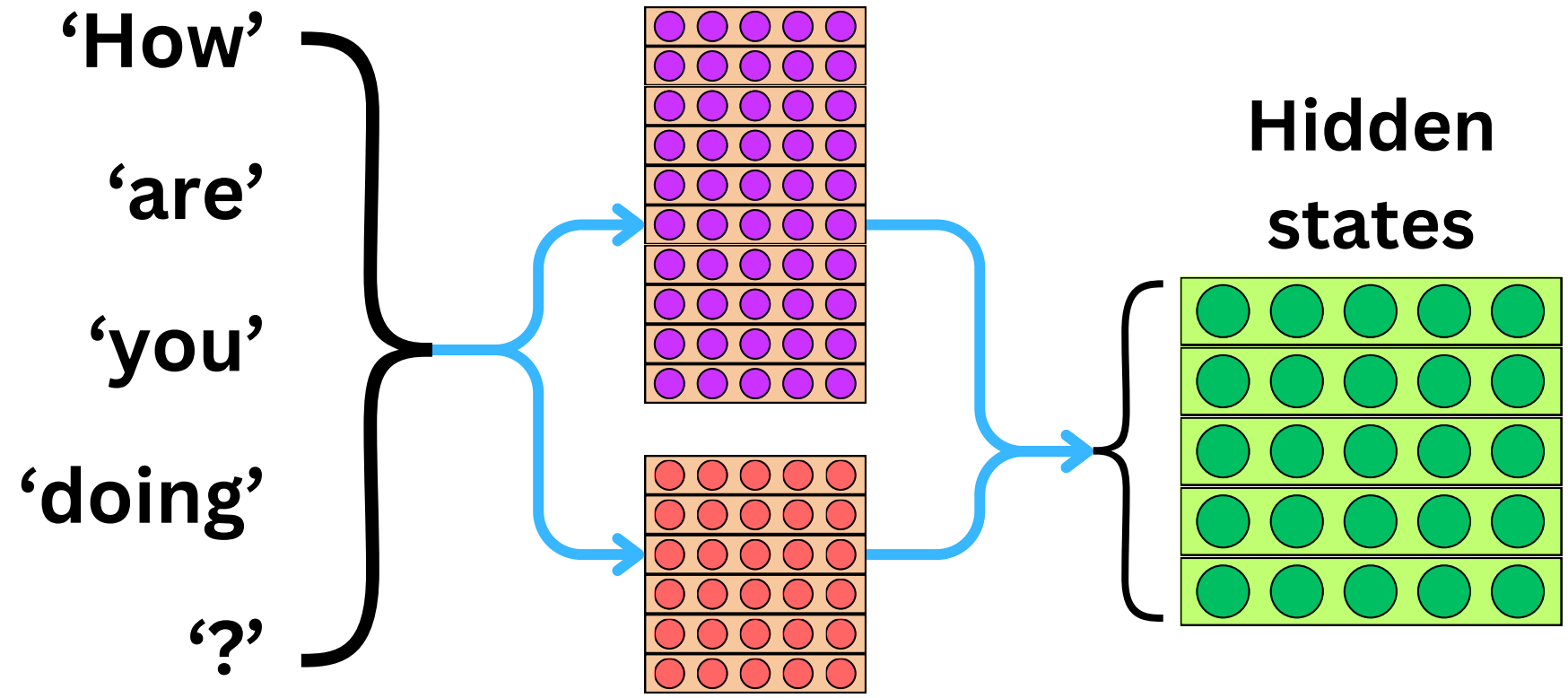
‘How’



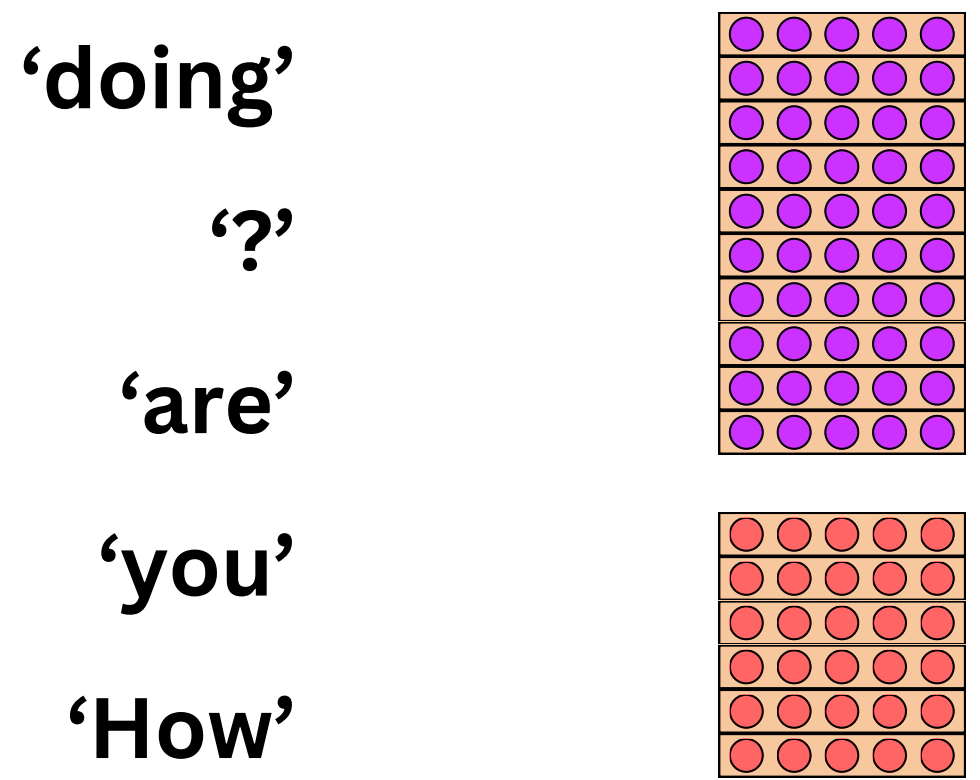


The Position Embedding

Correct
order



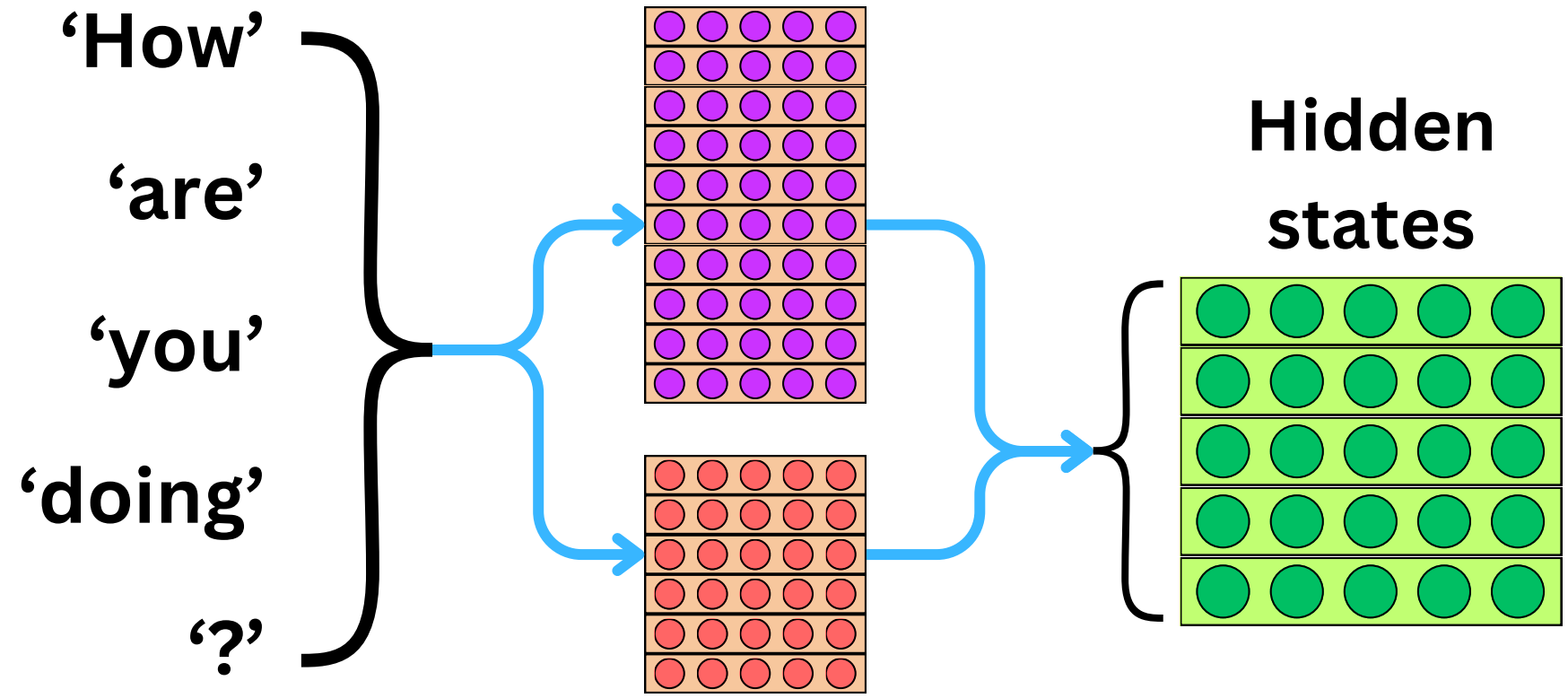
Wrong
order



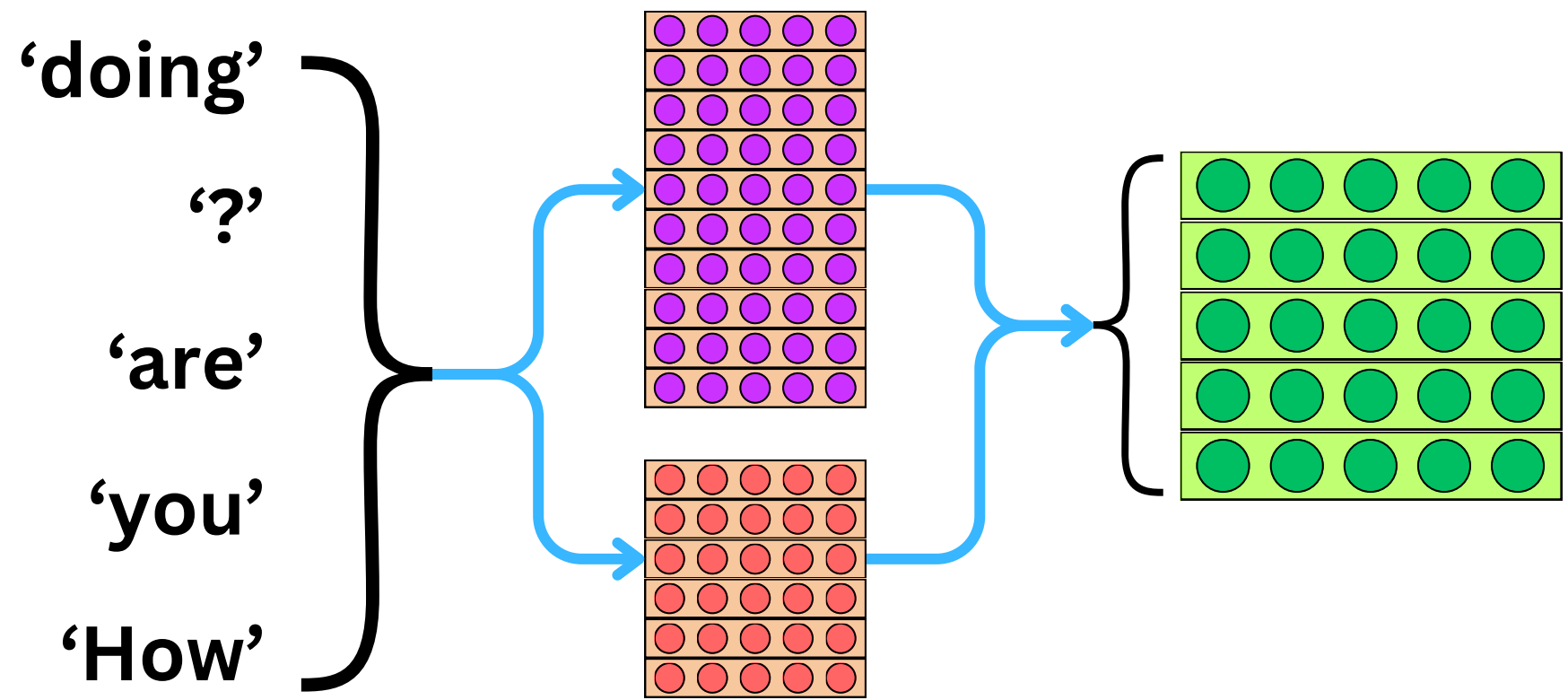


The Position Embedding

Correct
order



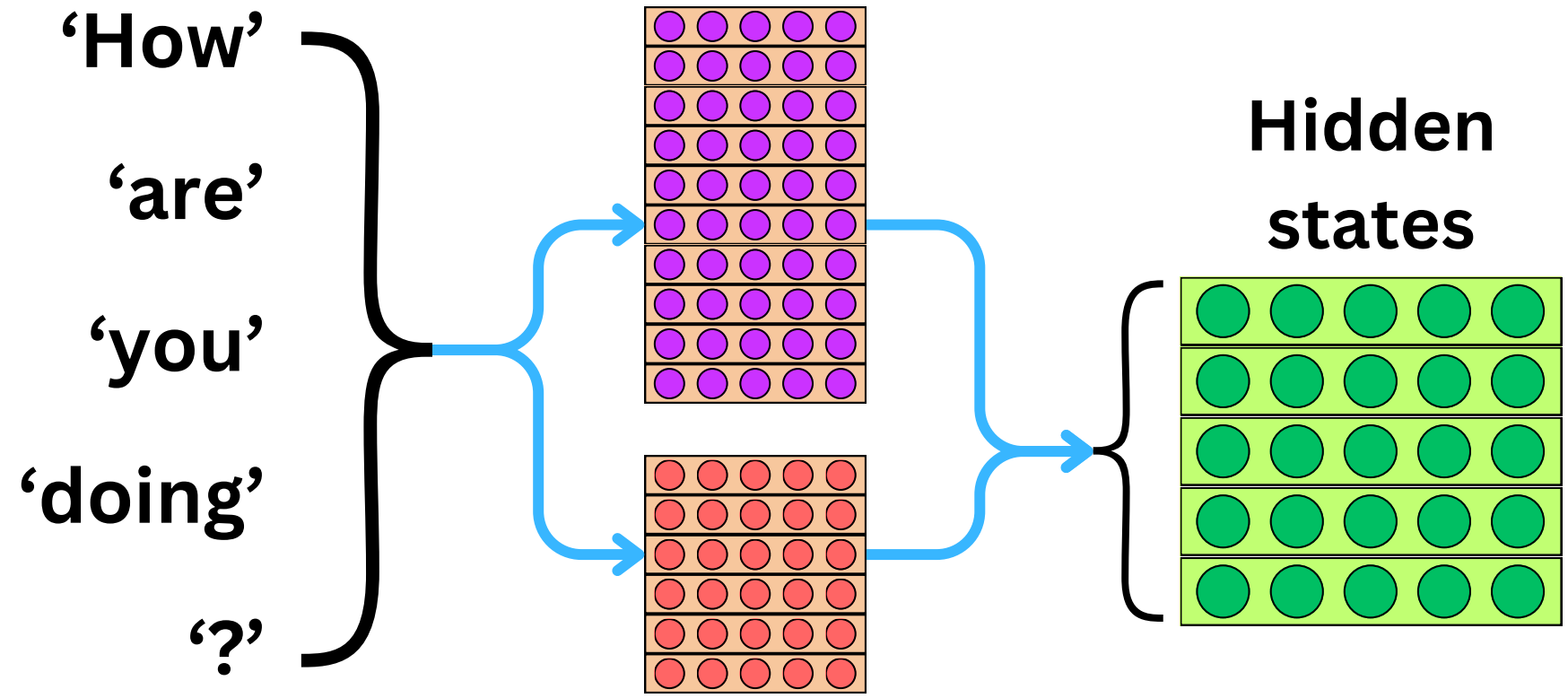
Wrong
order



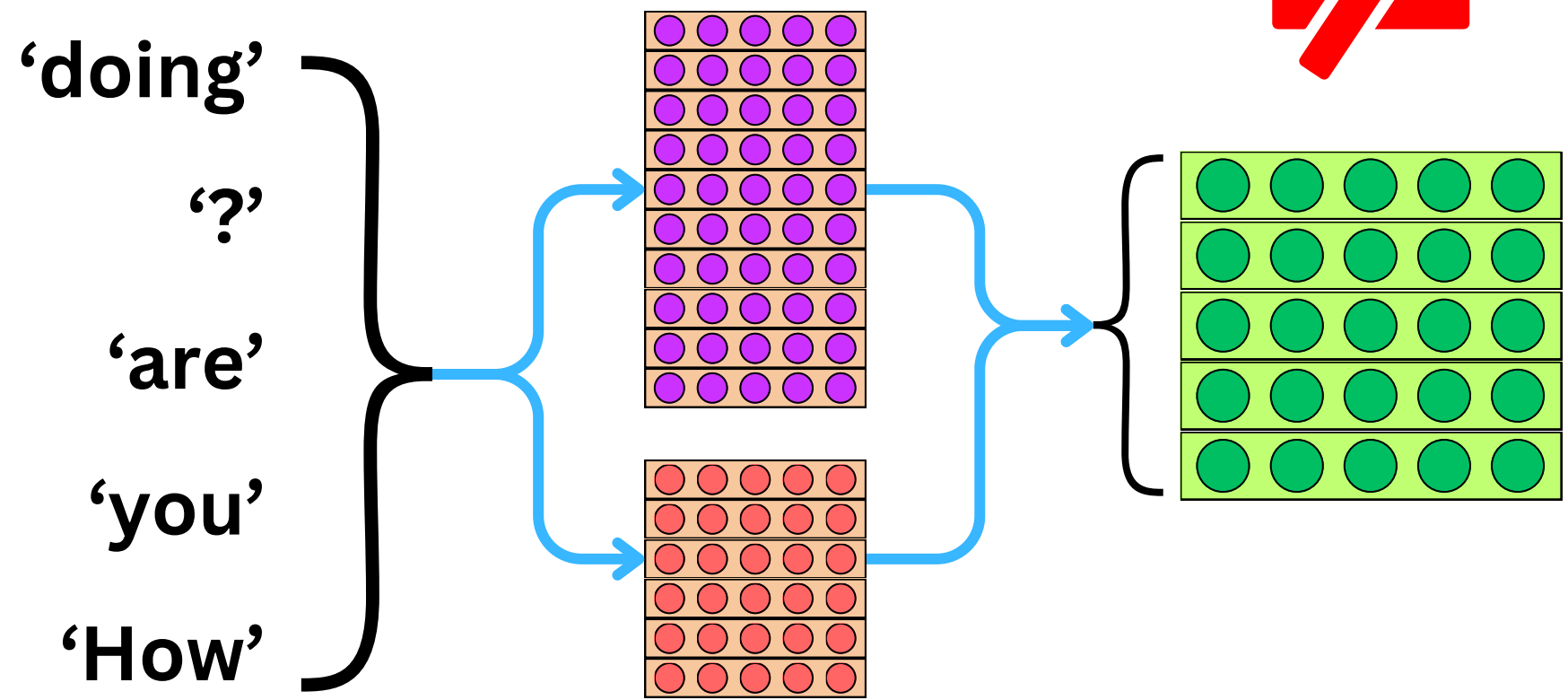


The Position Embedding

Correct
order



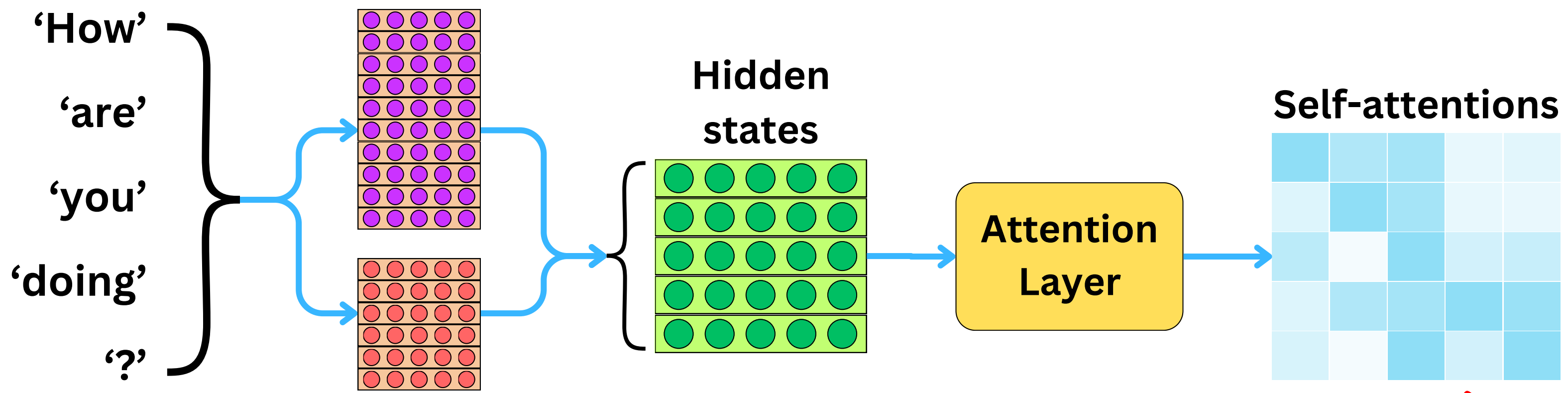
Wrong
order



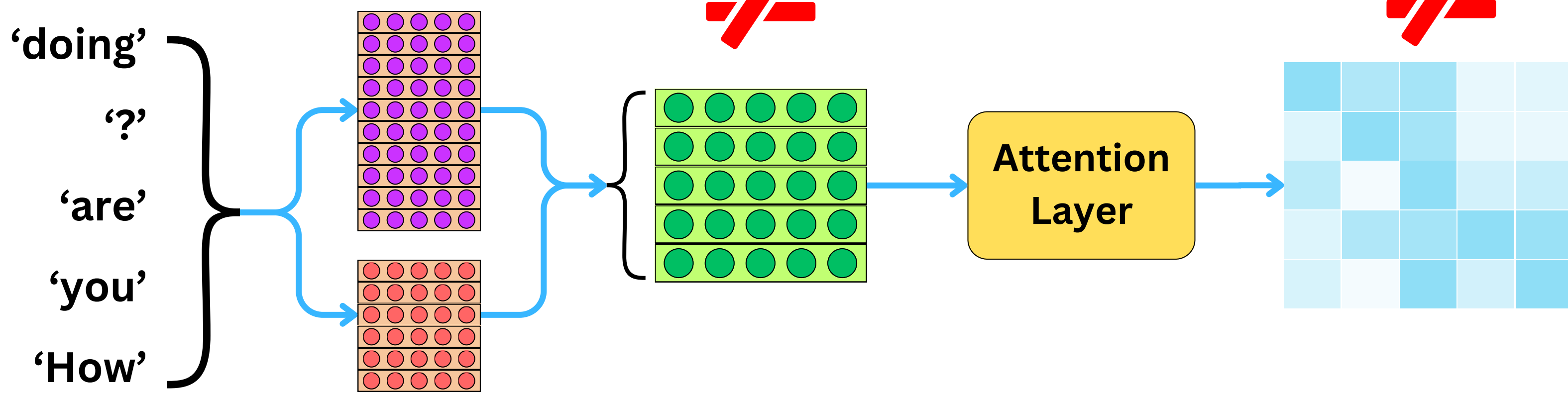


The Position Embedding

Correct
order



Wrong
order

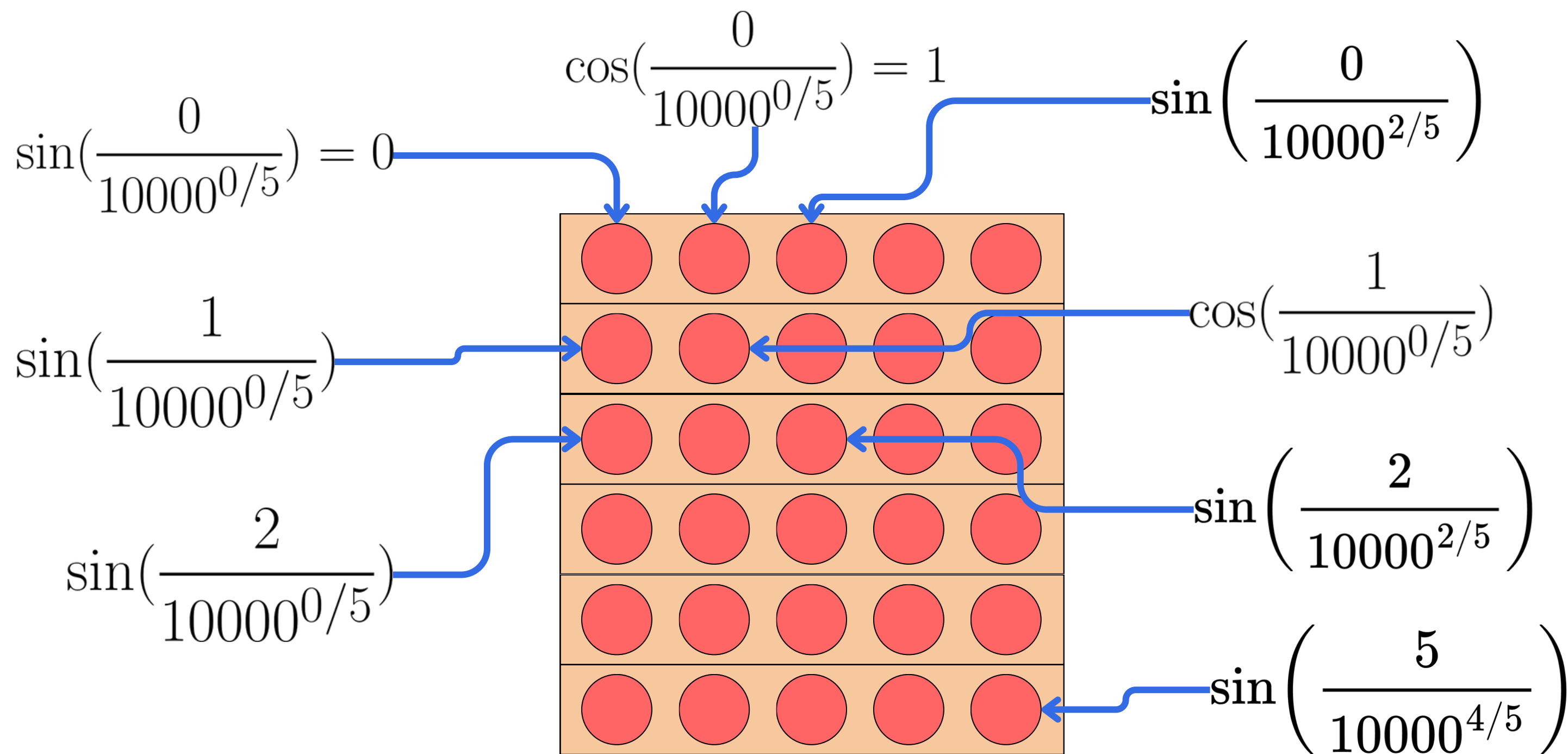




The Position Embedding

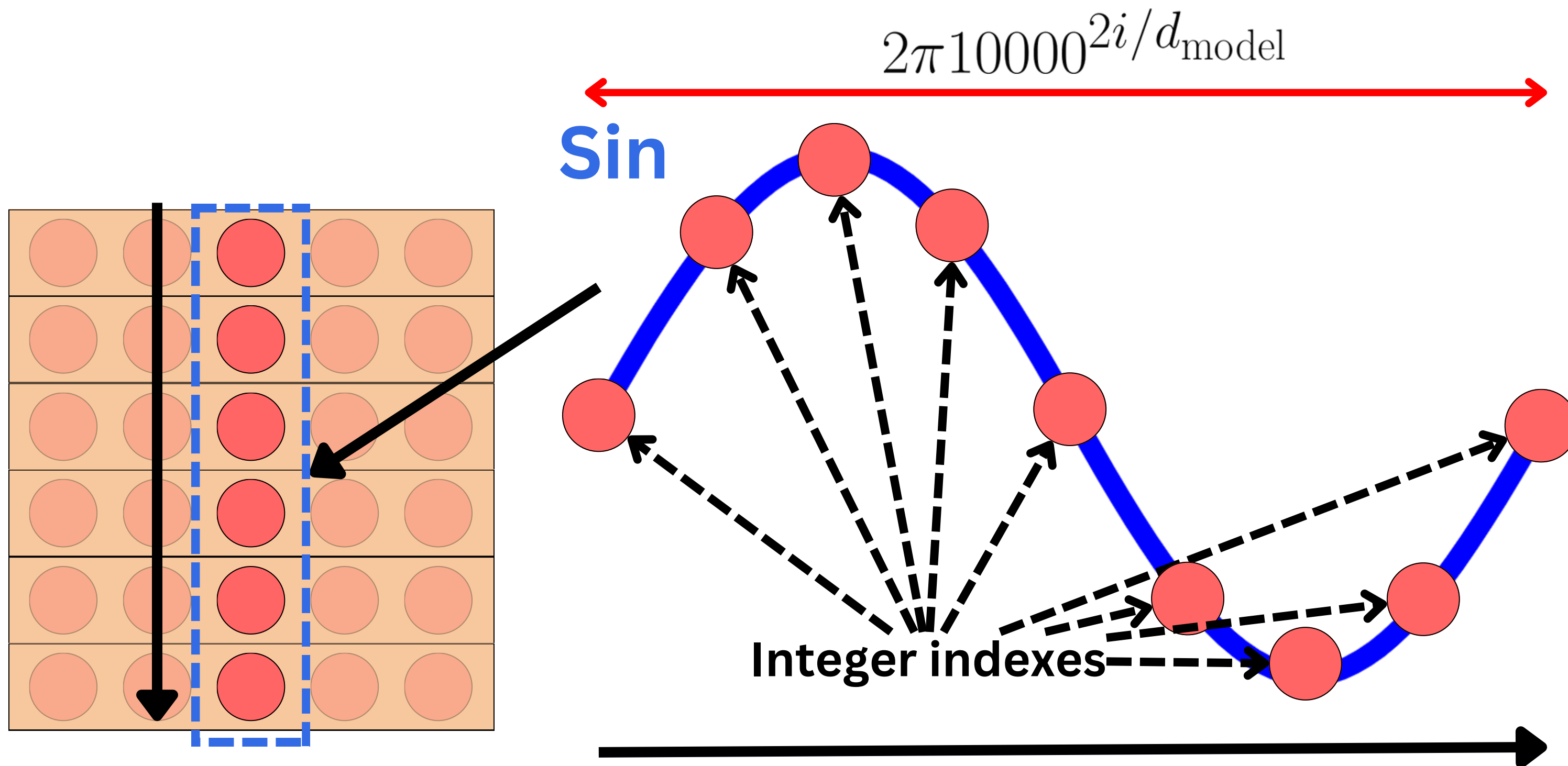
$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}}) \quad \text{even } i$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}}) \quad \text{odd } i$$



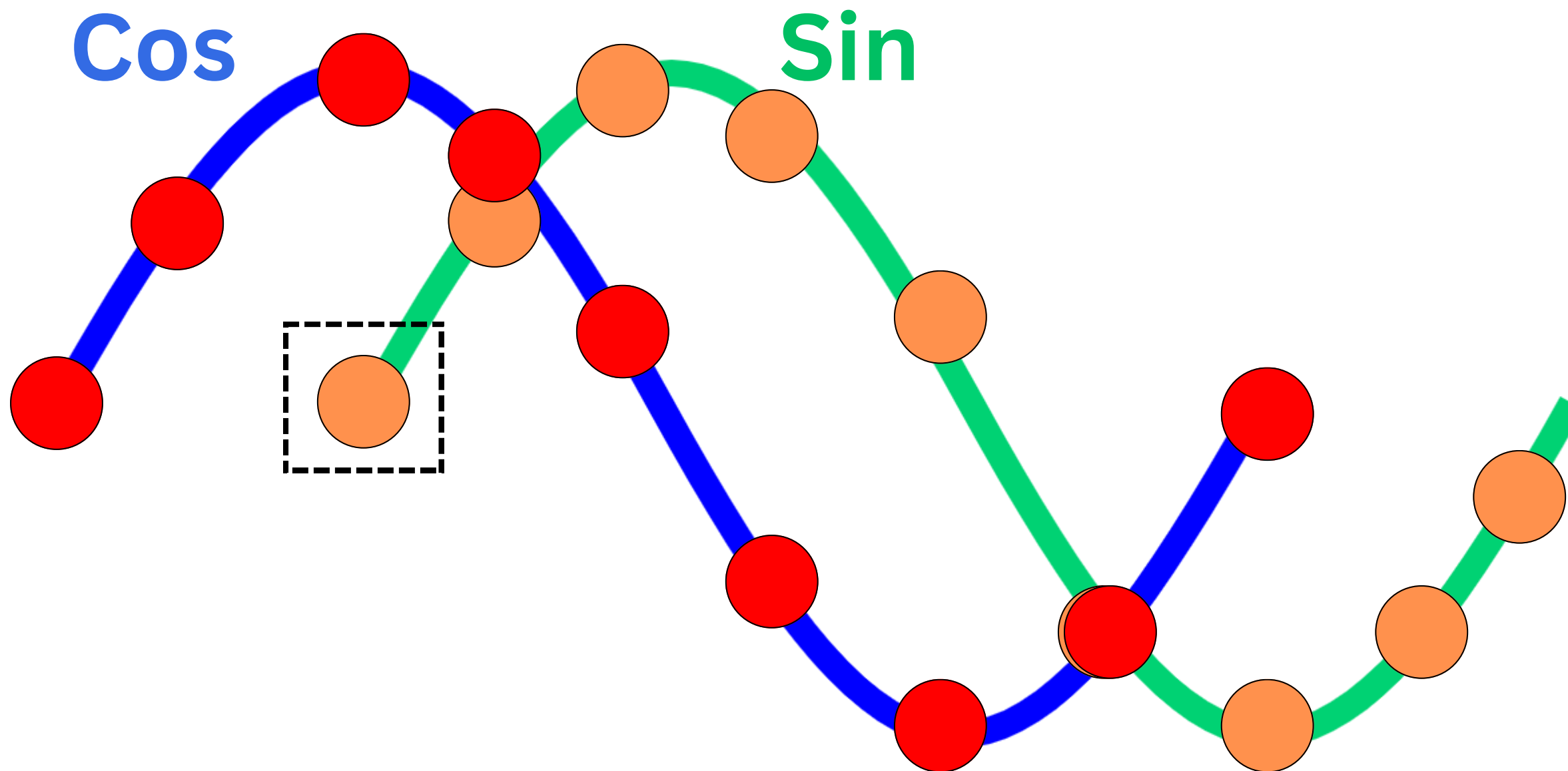
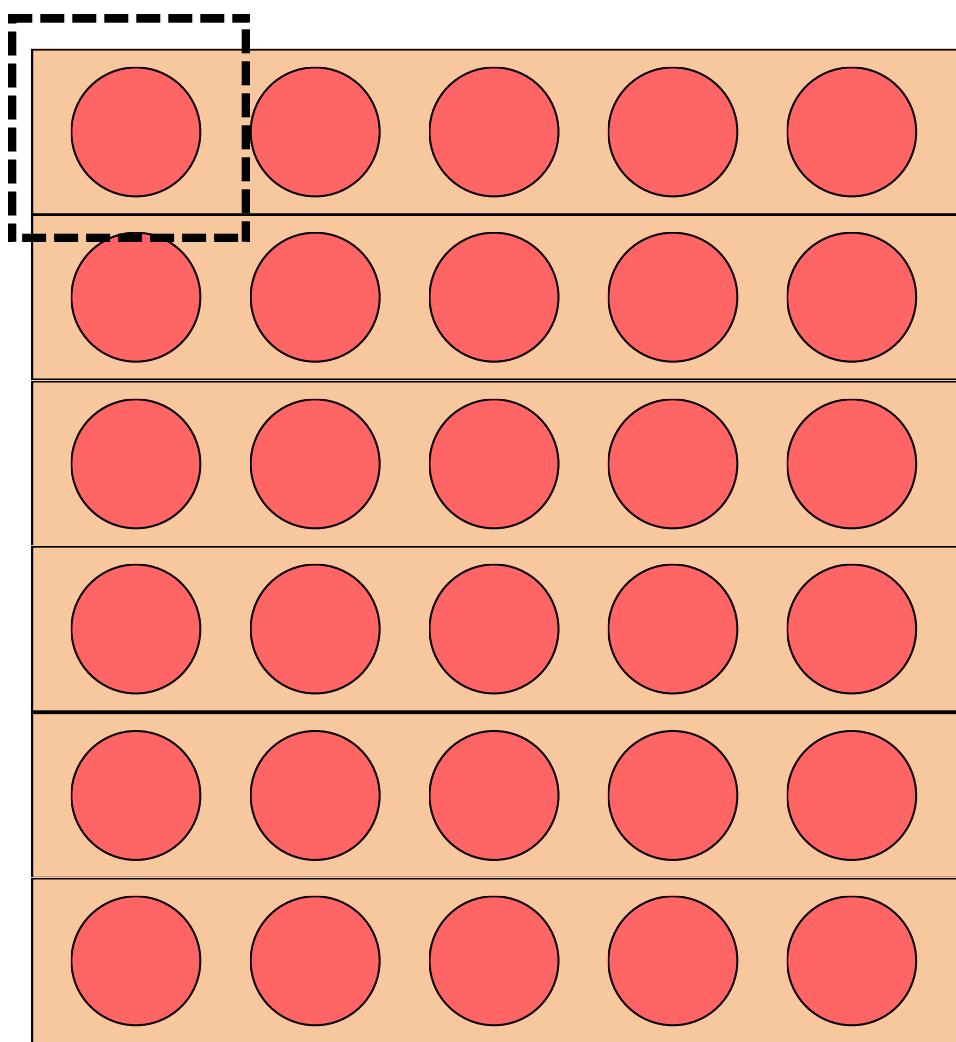


The Position Embedding



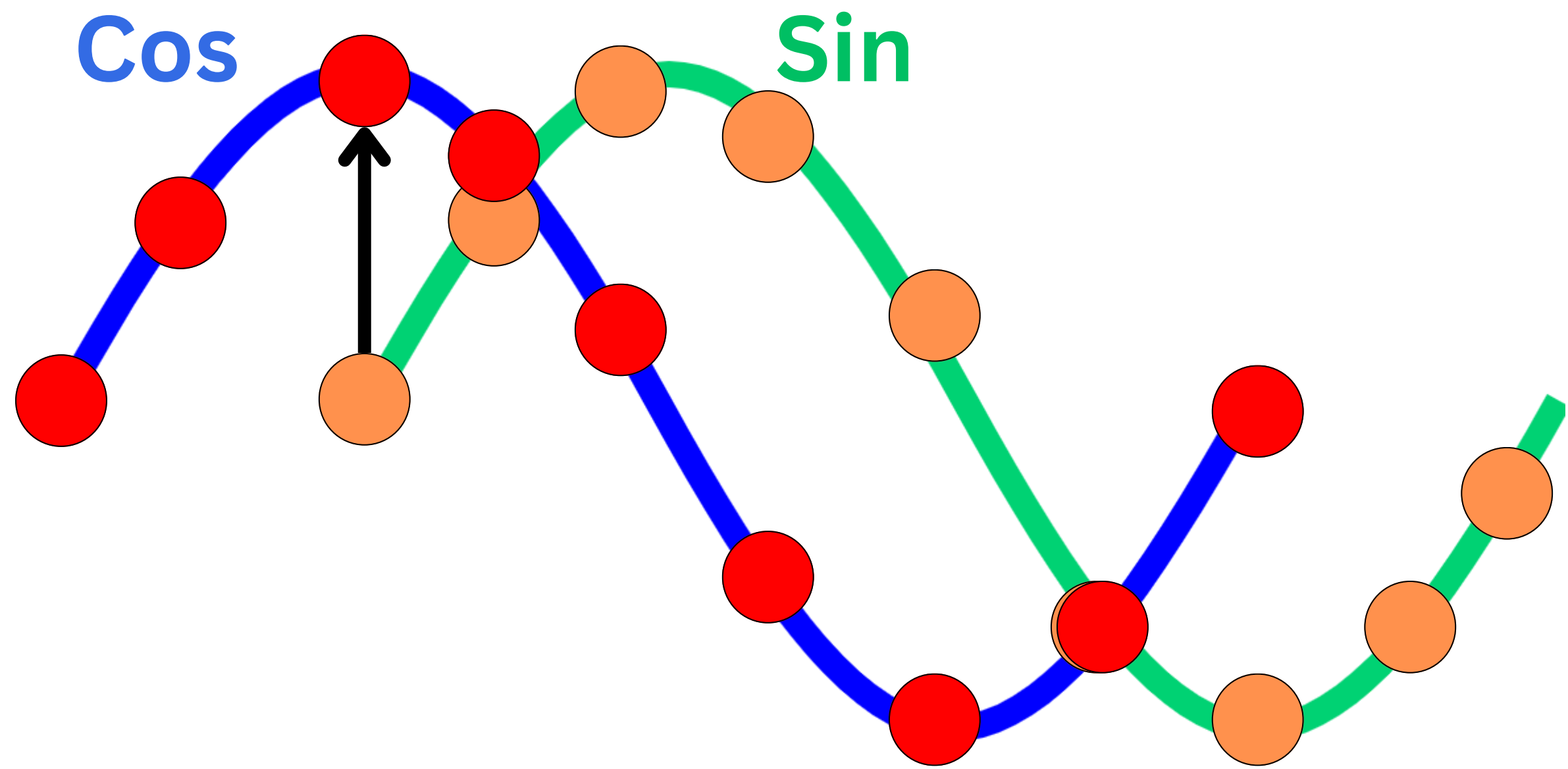
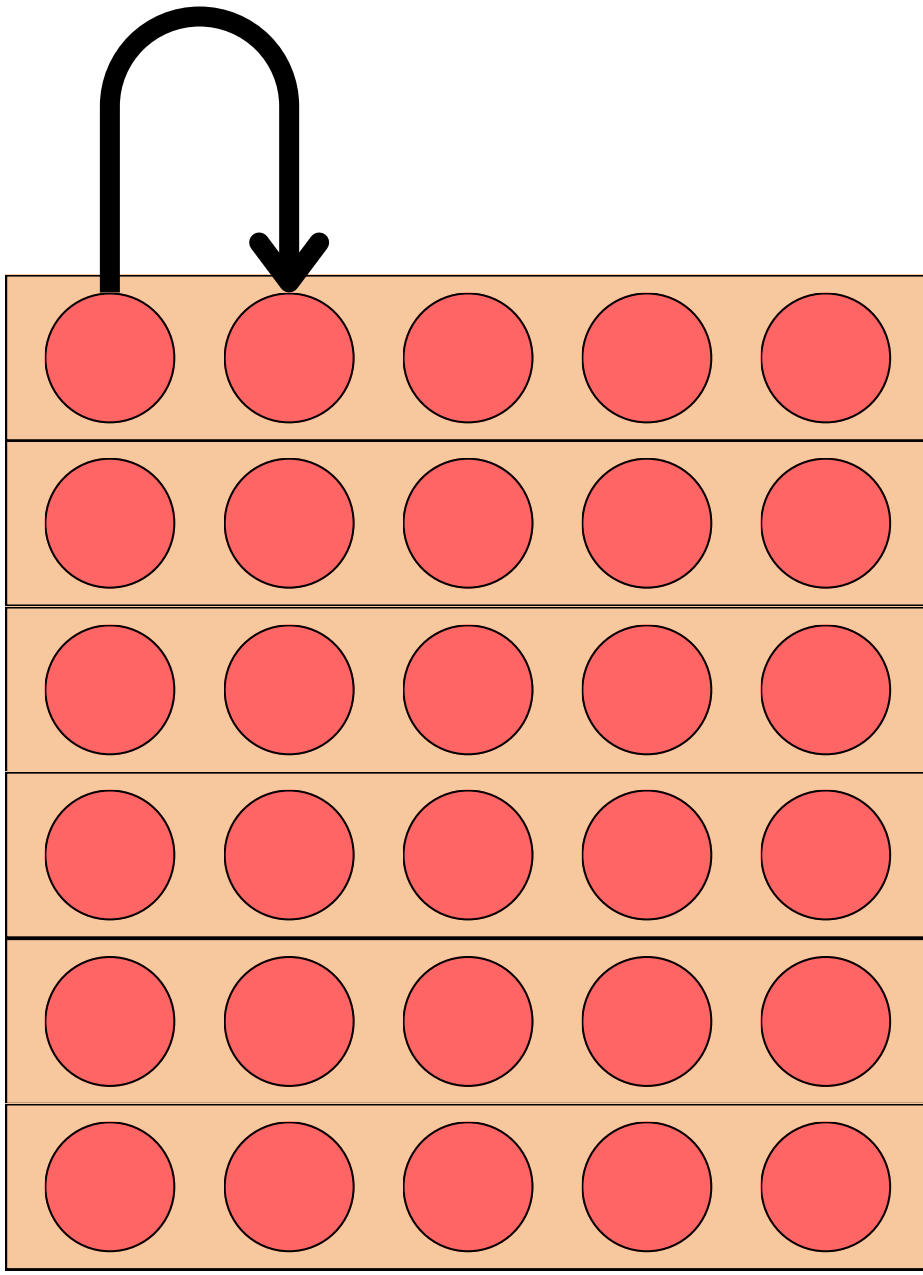


The Position Embedding



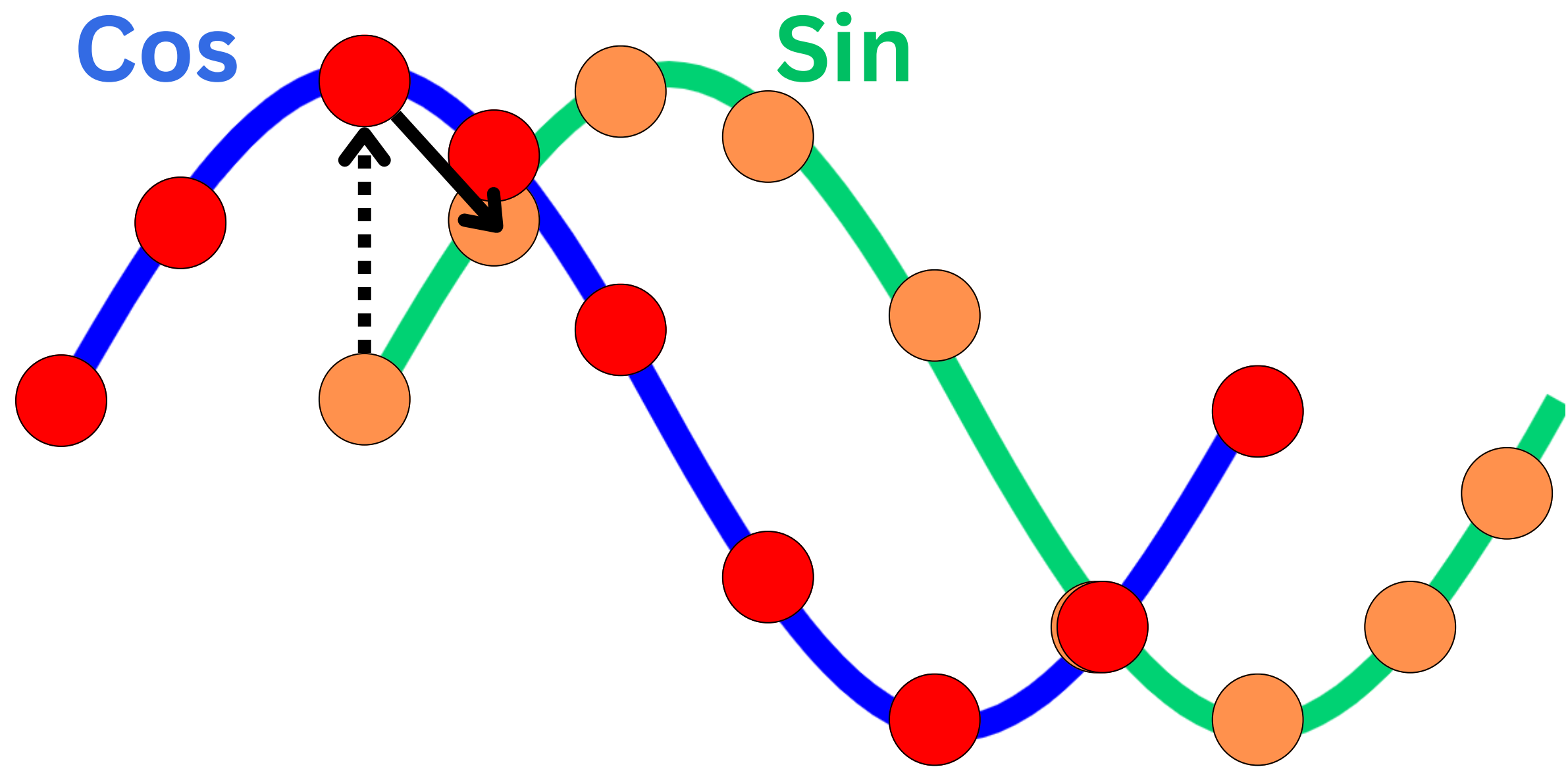
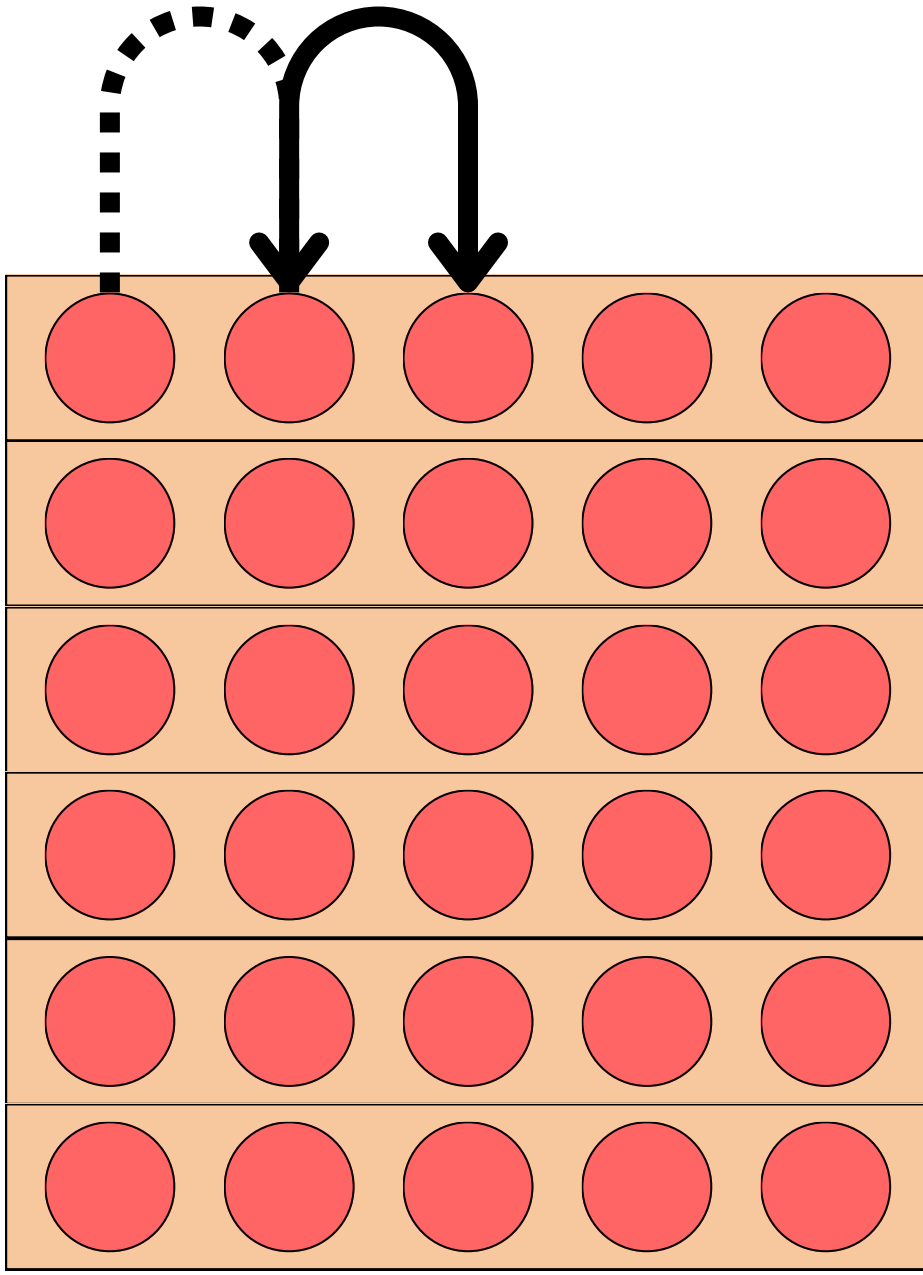


The Position Embedding



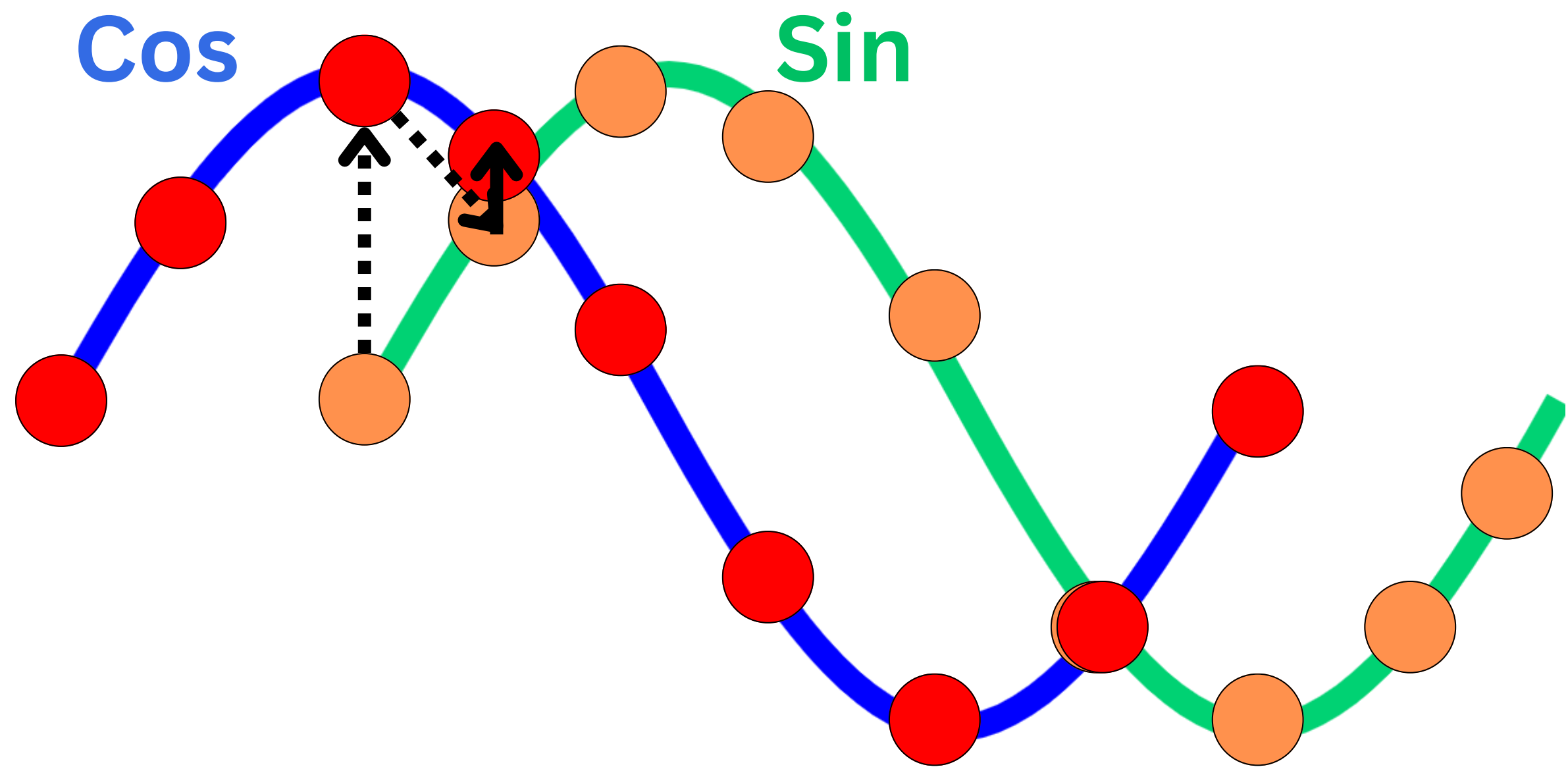
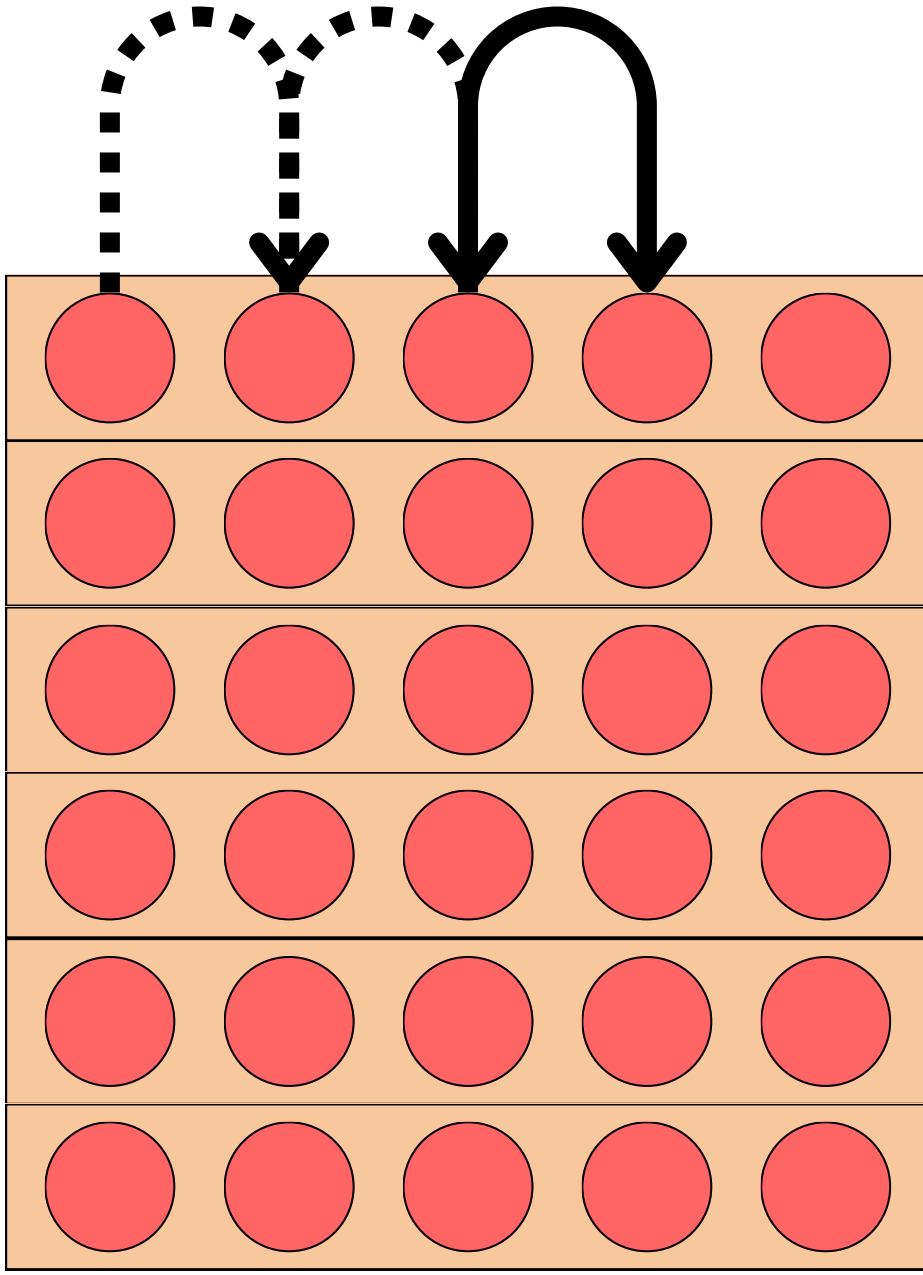


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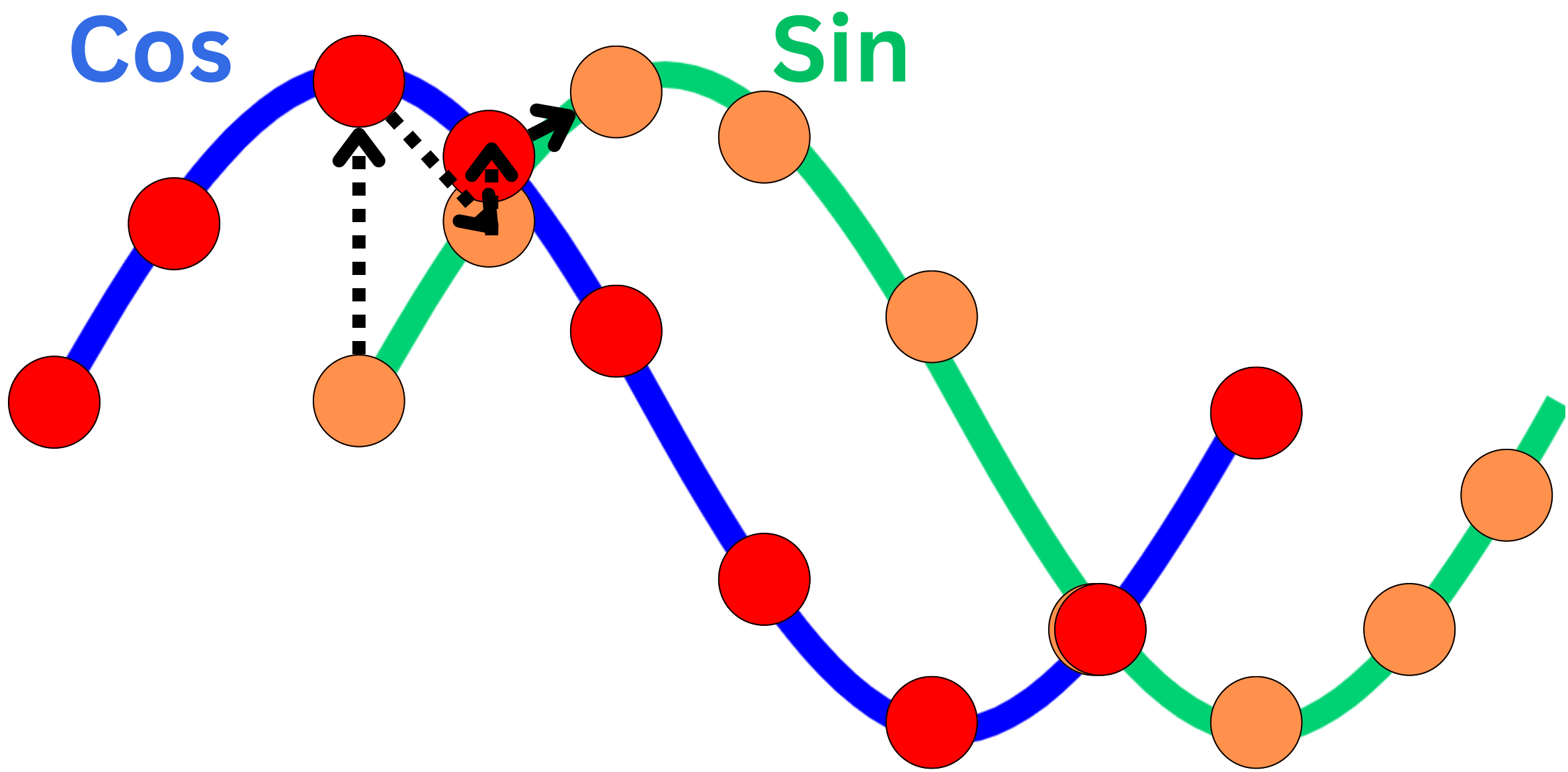
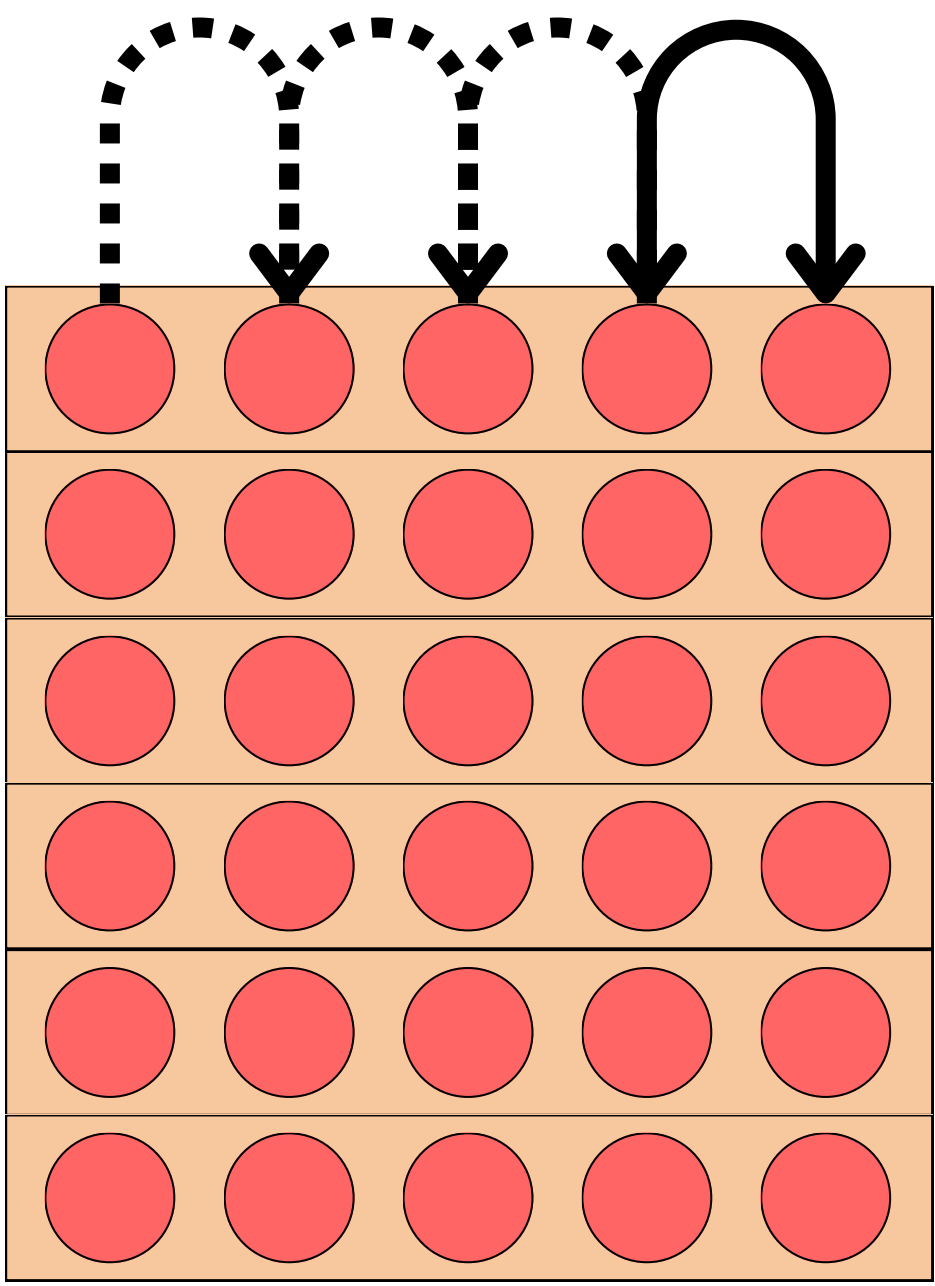


The Position Embedding



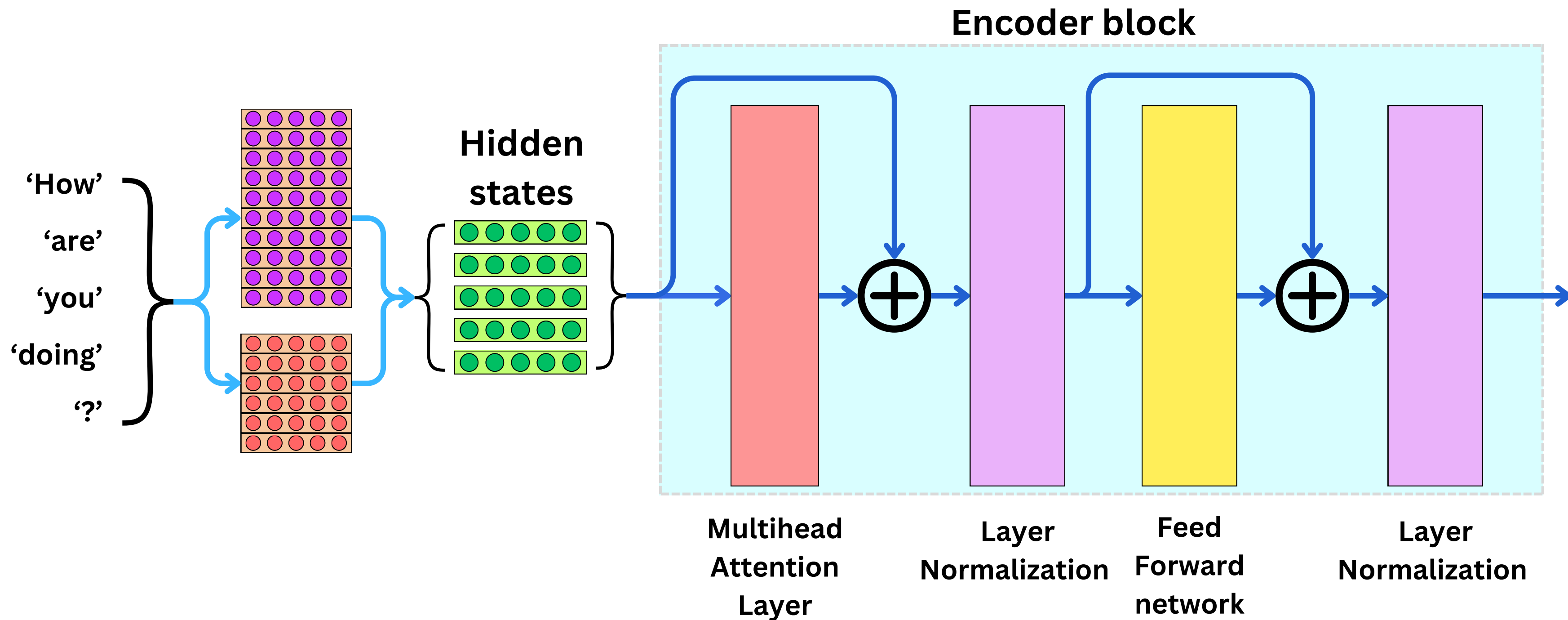


The Position Embedding



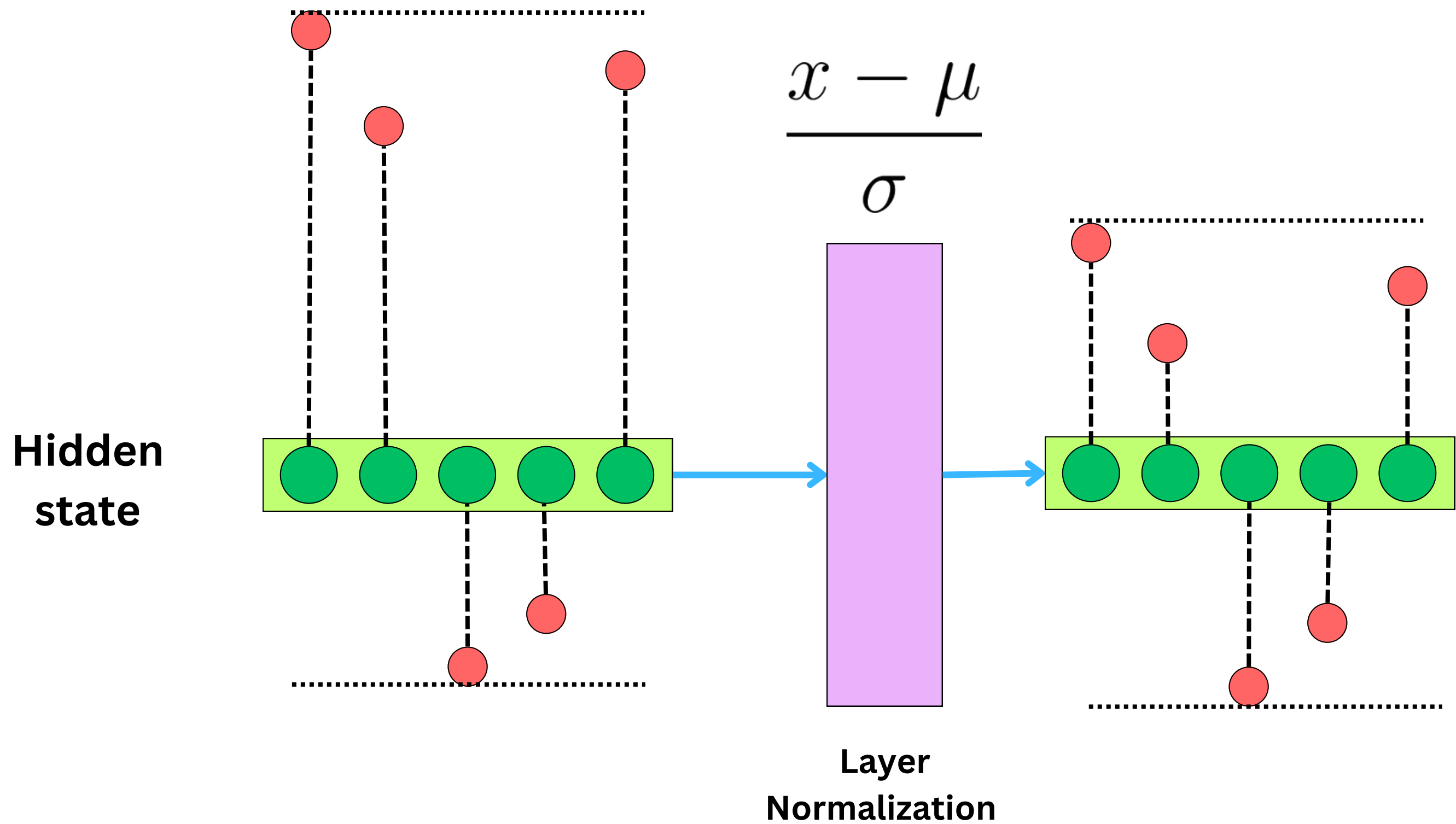


The Encoder Block



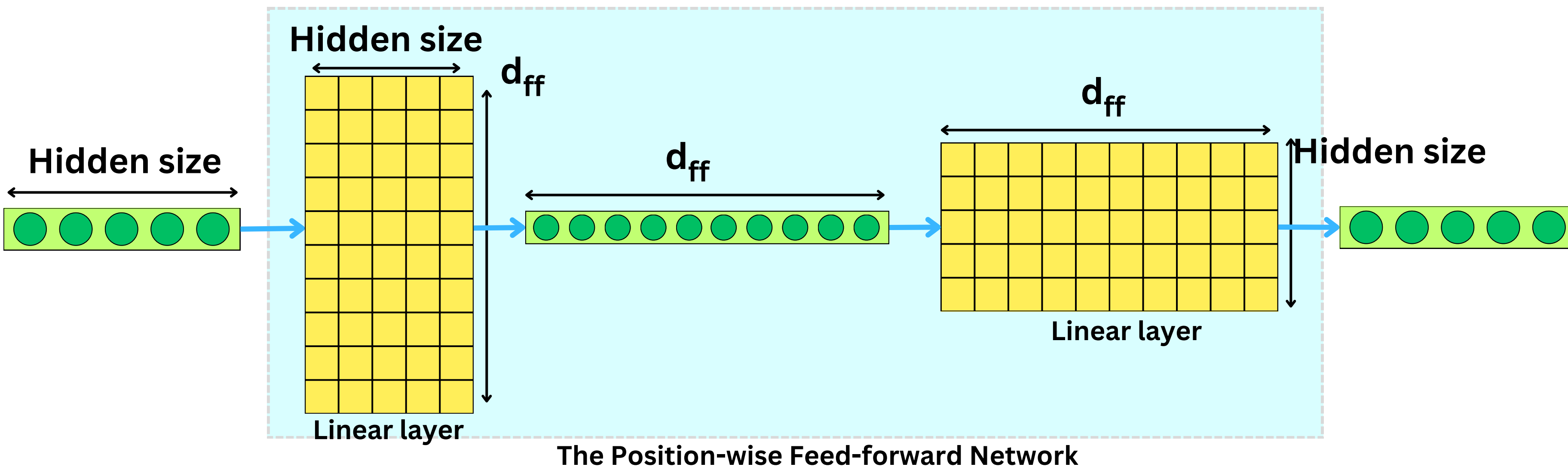


The Layer Normalization



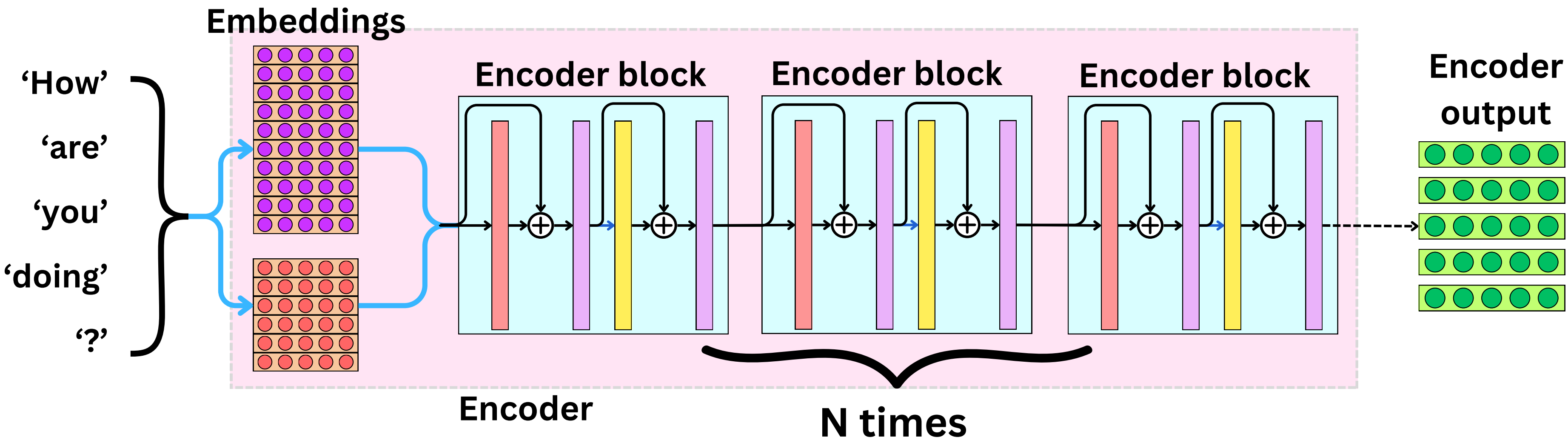


The Position-wise Feed-forward Network



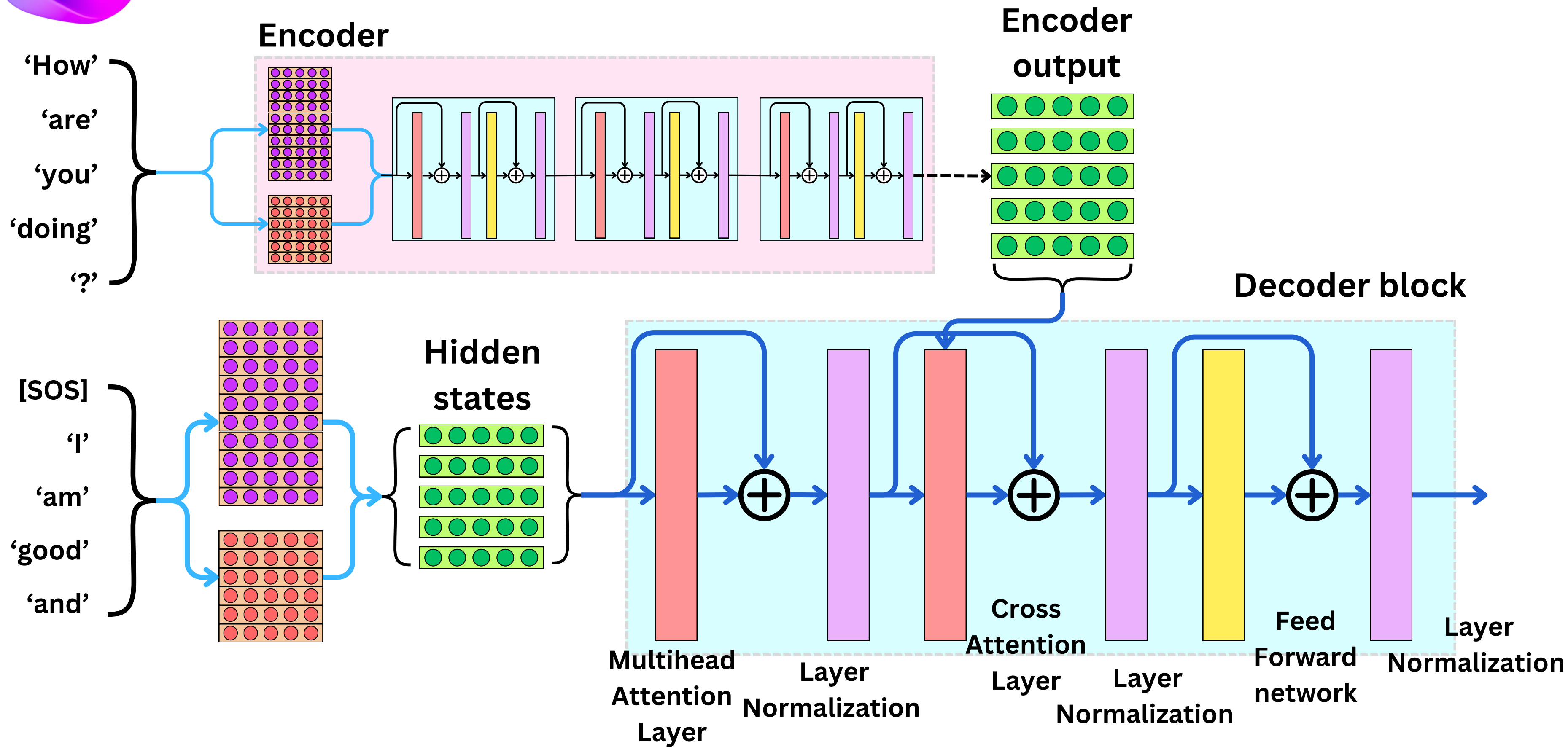


The Encoder



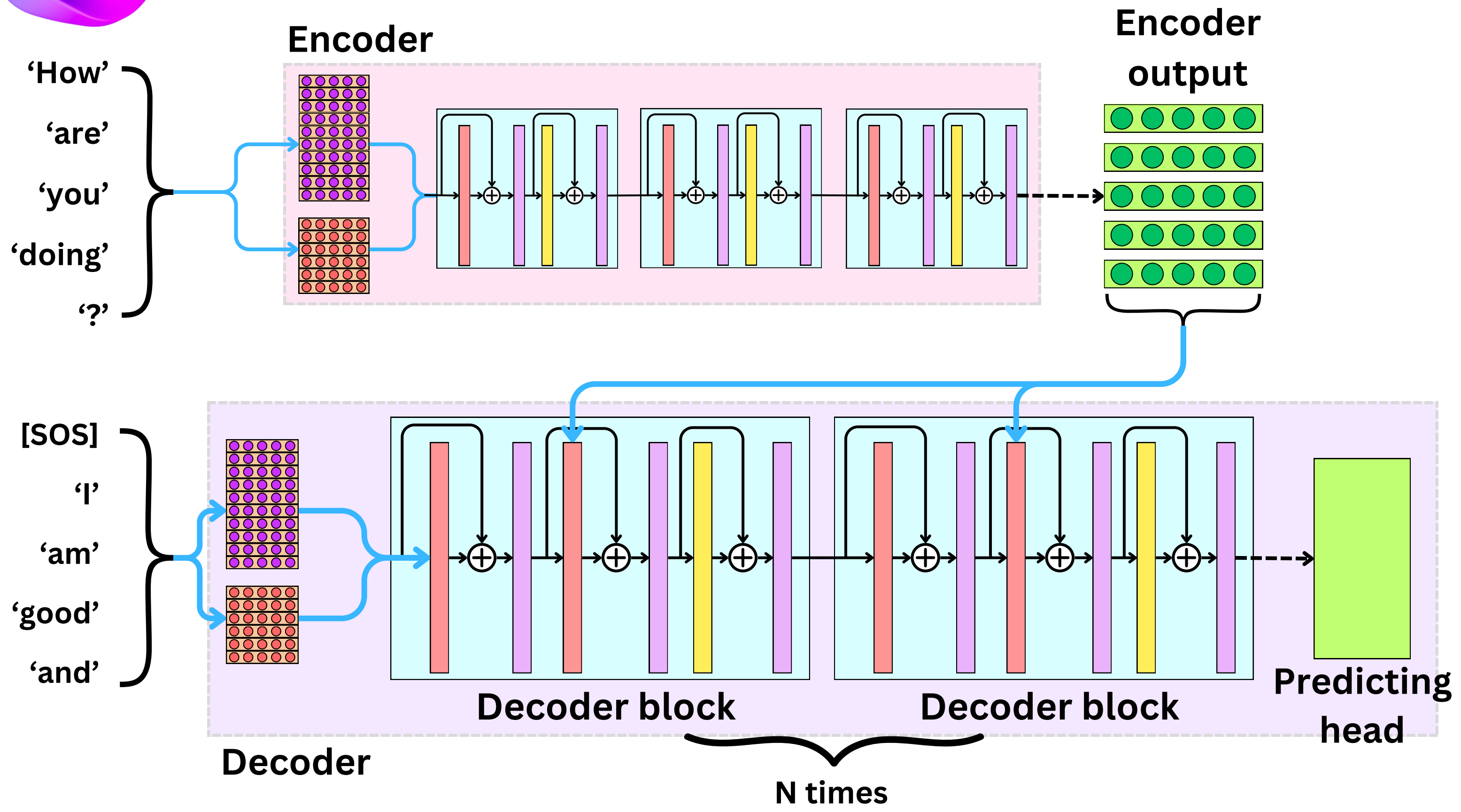


The Decoder Block



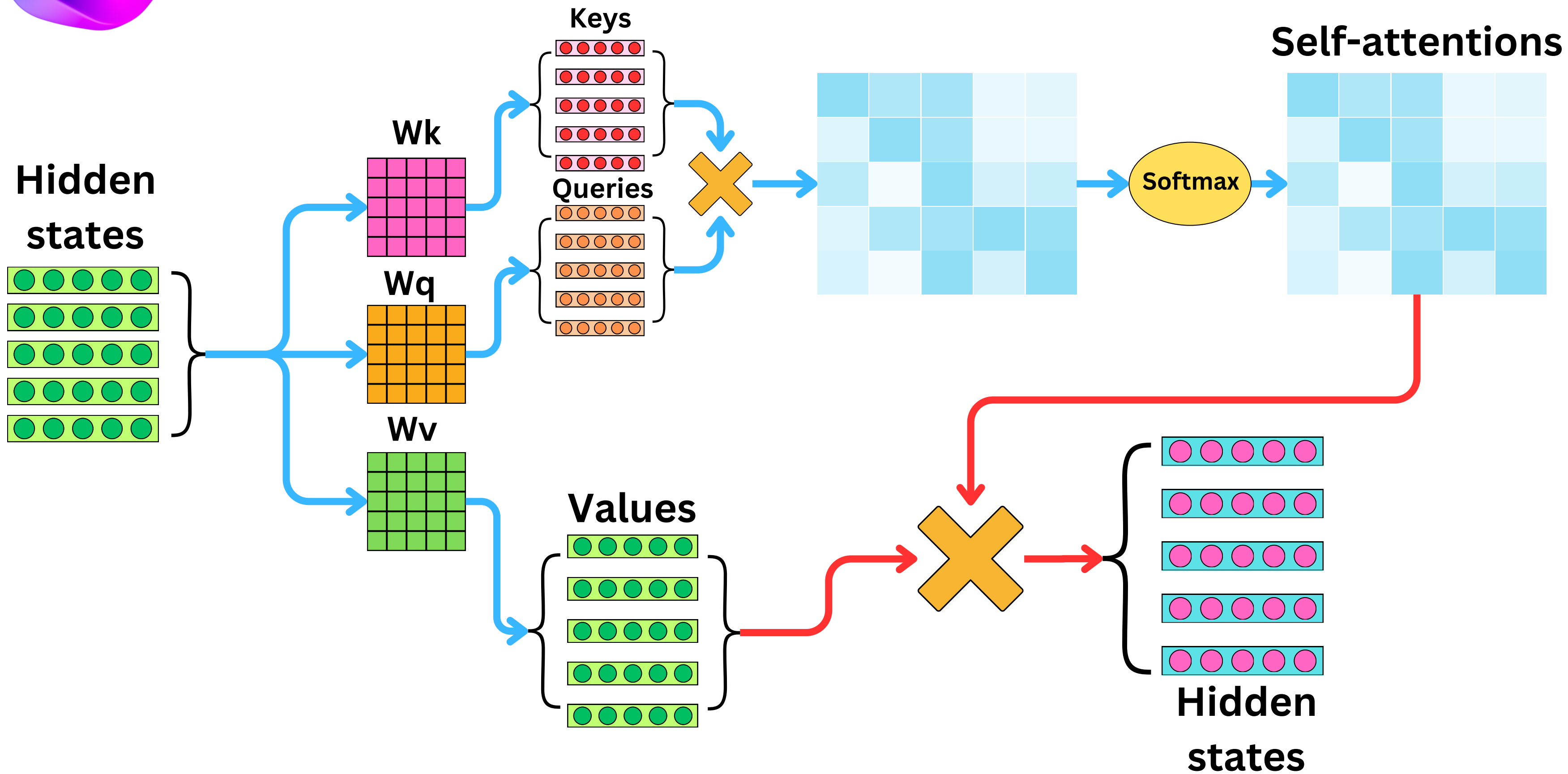


The Decoder



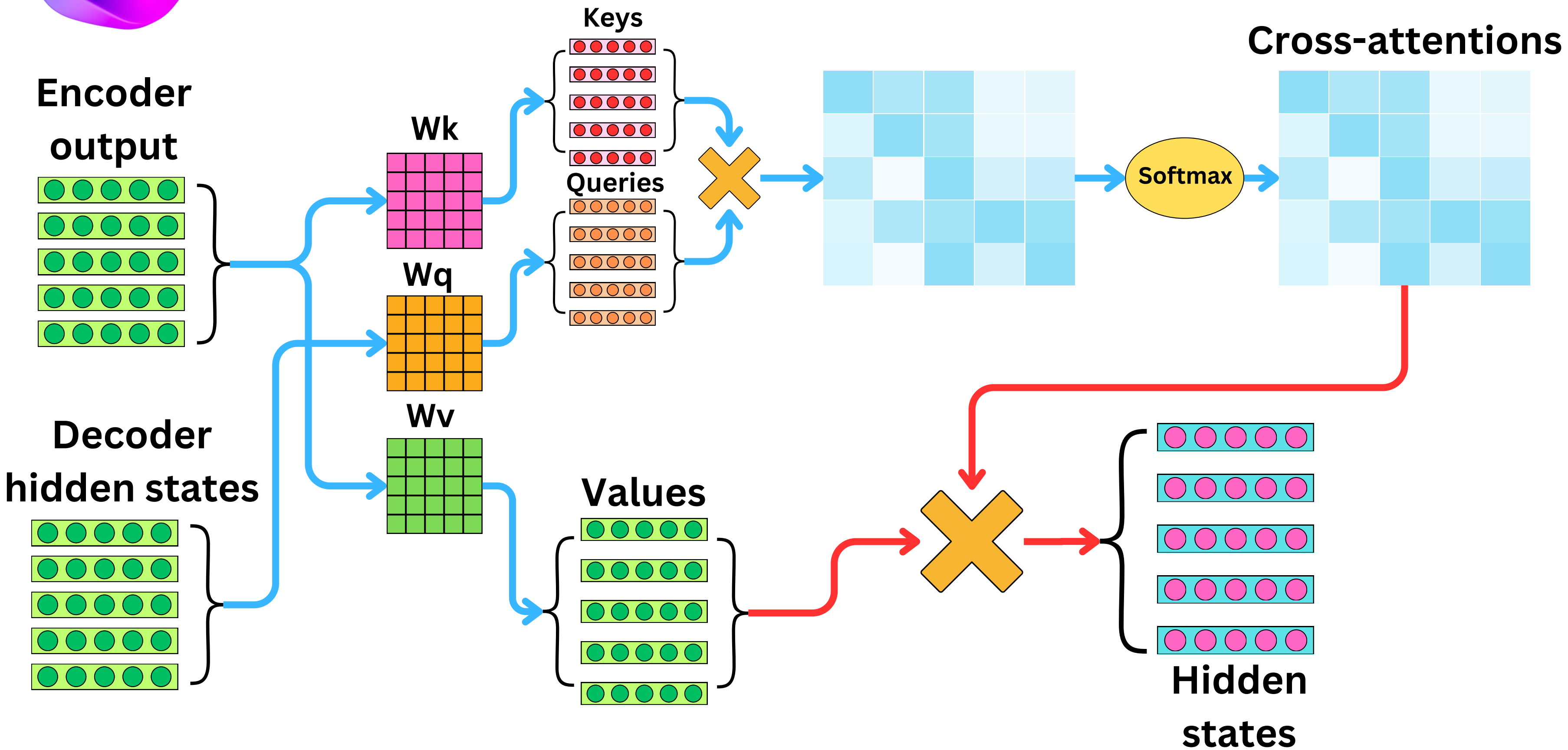


The Self-Attention Layer



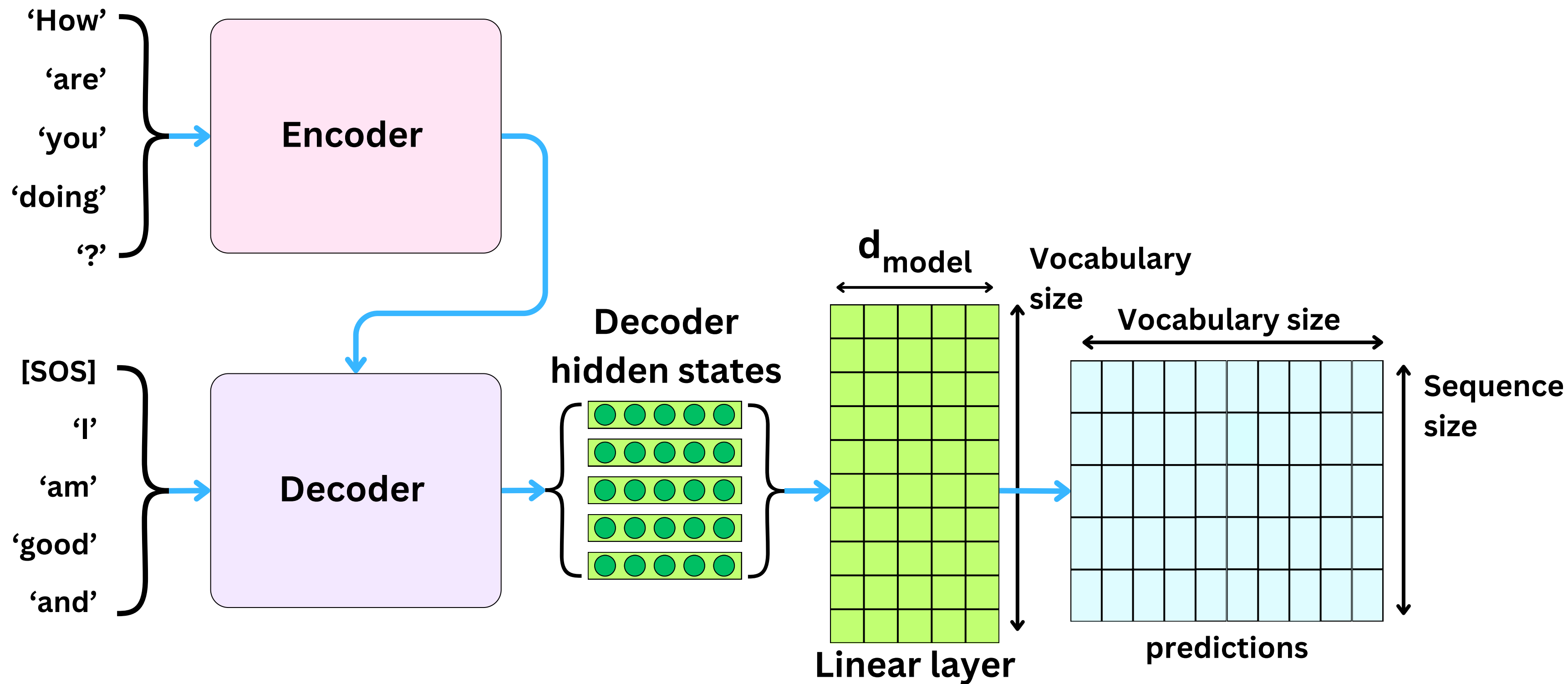


The Cross-Attention Layer





The Predicting Head





The Predicting Head

